

From Operational Prediction to Signed Renewal Memory

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Abstract

We investigate whether the mathematical structure used to describe emergent Lorentzian spacetime can itself arise from a more basic principle: prediction. We begin with composable operations and identify histories whenever no admissible future experiment can distinguish them. This removes redundant microscopic information and produces a minimal memory containing exactly what is needed to predict future outcomes. Under explicit operational assumptions, this reduced memory acquires the structure of finite-dimensional complex quantum dynamics and remains stable under small errors in predictive compression. We then show how resolved outcomes generate well-defined successor states and multiplicative path weights. Forward and retrodictive dynamics define a real measure of irreversibility, while stable-record transport produces an independent binary orientation class and a two-sheeted signed cover. Reversible transformations generate a Clifford algebra, pressure selects an intrinsic renewal scale, and spatial record statistics provide the remaining finite ingredients for a companion construction of Lorentzian geometry. The result is conditional: it identifies clear assumptions and phase conditions under which spacetime-like structure emerges, while allowing unoriented, subcritical, and torsional alternatives.

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1 Introduction

A companion work, *Renewal Spectral Geometry and the Emergence of Lorentzian Spacetime* [1], starts from a finite family of unital completely positive maps and studies the geometry encoded by histories of their composition. Under explicit regularity and phase conditions, that construction produces a positive spectral geometry, a minimal signed cover with a Krein fundamental symmetry, a primitive Clifford factor generated by reversible predictive transformations, and controlled flat and curved Lorentzian Dirac limits. Spacetime is therefore not assumed there, but one substantial object still is: the signed renewal memory from which the geometry is built.

The purpose of the present paper is to reconstruct that object from a more primitive operational level and to identify, with sharp logical boundaries, which later ingredients are canonical and which remain phase selections. The central question is not simply whether quantum channels can be used to model a pregeometric dynamics. It is why completely positive renewal maps, a predictive quotient, a retrodictive comparison, a signed orientation line, a modular scale, and a torsion-free translational sector should arise together at all. These structures are often packaged as if they had a common origin, but they do not. Positivity is weaker than complete positivity; a retrodictive channel is not an inverse; nonequilibrium affinity is real-valued whereas orientation is

mod two; a nontrivial double cover does not select one of its sheets; stationarity need not imply exchange symmetry; and detailed balance need not remove torsion. A continuum limit, moreover, requires analytic regularity that should not be disguised as a foundational axiom.

Our strategy is reconstructive. The primitive language contains composable operations, parallel composition, terminal tests, and real weights assigned to closed experiments. Histories are generated composites rather than trajectories in a pre-existing time. Two rooted histories are identified when every admissible continuation and terminal test gives the same result. This future-equivalence relation produces a universal reduced predictive memory: every other future-sufficient representation factors through it. The construction belongs to the broad operational and process-theoretic programme for reconstructing quantum theory [2, 3, 4, 5, 6, 7, 8, 9], but the organising object here is a quotient of histories rather than a single-shot state space. In particular, ordinary operational equivalence is compared with its complete ancillary extension. Their equality is isolated as a genuine no-holism condition, rather than being hidden inside a local-tomography axiom; the obstruction exhibited by real-vector-space quantum theory is correspondingly retained [10, 11, 12].

The first reconstruction theorem passes from this predictive quotient to finite quantum dynamics. A reduced sharp-reversible realisation supplies the purification and sharpness structure needed for operational diagonalisation and the Euclidean Jordan route [13, 14, 15]. Minimal continuous two-path interference then selects complex matrix type, while a compatible phase response fixes the associative orientation; this is the point at which the Jordan structure becomes a complex C^* -algebra [16]. Stability under arbitrary finite ancillas upgrades positive deterministic transformations to UCP maps. The paper also proves a quantitative perturbation theorem: a nearly idempotent, nearly future-sufficient compression with the correct rank and a transversality bound lies near an exact minimal predictive projection, and the induced compressed renewal dynamics is stable. Thus the finite UCP renewal alphabet is derived from operational prediction rather than postulated as primitive geometry.

A second reconstruction concerns resolved histories. Computational mechanics identifies stochastic pasts that induce the same conditional future law, and the resulting causal states form a minimal unifilar predictive model [17, 18]; quantum memory advantages for such models have also been studied [19, 20]. Here the same principle is lifted to resolved completely positive branches. Equality of conditional future profiles is proved to be stable under appending an observed outcome, so the conditional predictive quotient has a deterministic outcome-labelled successor map. In every reduced future-tomographic operator realisation, the normalised output of a branch is therefore forced to equal the state representing that successor class. This yields the branchwise reference-renewal identity and multiplicative scalar path weights as theorems, not hidden-state assumptions. A canonical successor refinement repairs under-resolved target graphs; Kraus changes within a fixed resolved outcome leave the weights invariant; and approximate future tomography gives an explicit error bound along every finite history.

Resolved outcomes themselves cannot, however, be recovered from an averaged channel without additional physical information. To make this boundary precise, we give an optional microscopic bridge. A minimal dilation together with a specified family of system–environment interactions determines a basis-independent environment coefficient algebra. In an interaction-complete monitoring phase, the associated dephasing semigroup converges to the trace-preserving conditional expectation onto its commutant, and the centre of the stable nondemolition algebra yields a canonical pointer instrument. This construction is related to einselection and decoherence [21, 22, 23], conditional expectations and fixed-point algebras [24, 25, 26, 27, 28], and the redundancy viewpoint of quantum Darwinism [29]. Its additional conclusion is a canonicity statement

relative to the specified interaction family: the stable pointer centre is fixed before the conditional predictive quotient is taken, while an averaged channel alone remains insufficient to select an instrument.

The stationary and retrodictive layers are reconstructed next. A primitive weighted transfer selects a unique faithful stationary state. When the elementary channels preserve that state, the symmetric KMS pairing determines their common completely positive retrodictive adjoints, in the lineage of the Accardi–Cecchini and Petz transpose maps [30, 31, 32] and KMS formulations of quantum detailed balance [33, 34, 35, 36]. In the commutative reduction, a stationary forward path has exactly the same weight as the reversed path under the canonical Petz chain. Consequently a genuine irreversibility observable must compare forward and retrodictive branch laws in the same path orientation, consistently with time-reversal and Bayesian readings of fluctuation relations [37, 38]. The resulting cycle affinity belongs to the established network-thermodynamic lineage of Schnakenberg and its extensions [39, 40, 41].

The Lorentzian sign has a different origin. Conditional future prediction reconstructs the minimal algebra of stable resolved records, and determinant transport on its contrast module defines a real line local system and a class in $H^1(G; \mathbb{Z}/2)$; the determinant-line viewpoint is standard in its natural geometric setting [42]. We extend the construction from reversible record permutations to positive noisy correspondences whose action on future-distinct contrasts remains invertible. The key result is a strict separation theorem: the real affinity class in $H^1(G; \mathbb{R})$ and the record-orientation class in $H^1(G; \mathbb{Z}/2)$ are logically and cohomologically independent. There is no coefficient homomorphism that reduces the former to the latter, and all four combinations of vanishing or nonvanishing affinity and orientation occur. To our knowledge, this theorem-level separation of network affinity from stable-record parity has not previously been formulated. It shows in particular that entropy production does not generate the signed cover and that orientability does not imply detailed balance.

The remaining steps are phase-theoretic. On a selected recurrent component, reversible predictive units act projectively on a finite factor. Their commutator bicharacter and radical quotient produce a primitive complex Clifford algebra, while the signed cover fixes its temporal row. Reciprocal branch capacities admit a unique decomposition into symmetric conductance and half-affinity. Perron–Frobenius pressure then selects a dimensionless renewal exponent, in the finite-graph thermodynamic-formalism tradition [43, 44, 45]. Its stationary law obeys the exact identity

$$\beta\mu_\ell = h_{\text{rate}} + \mathbb{E}[\log c_e] + \frac{1}{2}\sigma,$$

which ties modular depth, entropy rate, symmetric capacity, and entropy production in one formula. A graded path cover realises the corresponding affine modular clock, giving a renewal implementation of the thermal-time relation between modular flow and physical time [46] while avoiding the obstruction to a nonzero common exponent on a finite return cycle.

Spatial and Cartan data are obtained from record statistics rather than added as independent geometry. Covariance of positive spatial second moments derives orthogonal comparison between neighbouring frames. At the plaquette level, an exchange-covariant primitive law on microscopic two-step diamonds forces the exchange-odd translational defect to have zero stationary mean while leaving the exchange-even rotational defect unconstrained. The same condition is the zero-source Euler–Lagrange equation of a convex exchange pressure. Since torsion is precisely the translational obstruction separating Levi–Civita from general metric-compatible transport [47], this provides the finite input needed by the downstream Cartan reconstruction. An explicit Ising-type countermodel shows at the same time that one-step detailed balance and stationarity

can support both exchange-symmetric torsion-free states and symmetry-broken torsional states. Torsion freedom is therefore a genuine phase selection rather than an automatic consequence of reversibility.

The main contributions may be summarised as follows.

- (i) We construct the universal rooted and complete predictive quotients, reconstruct a finite complex C^* -algebra and UCP renewal dynamics from explicit operational assumptions, and prove quantitative stability of approximate predictive compression.
- (ii) We derive conditional unifilarity, branchwise reference renewal, and multiplicative path weights from the predictive quotient and future tomography, including canonical refinement and finite-error control.
- (iii) Relative to a specified interaction family, we extract a canonical stable pointer instrument from the centre of the nondemolition commutant, while proving why no averaged channel can determine such an instrument by itself.
- (iv) We separate the real affinity class from the mod-two record-orientation class, extend determinant orientation to noisy regular record transport, and classify the resulting signed cover and its amplitude lifts.
- (v) We derive the entropy–affinity–depth identity, the pressure-selected affine clock, orthogonal frame comparison, and exchange-selected translational closure, and we provide explicit countermodels showing which conclusions are phase dependent.

These statements are conditional reconstruction results, not a claim that every operational theory becomes Lorentzian spacetime. The predictive quotient, stationary reference, KMS–Petz dual, record algebra, determinant line, projective multiplier, and pressure root are canonical once their respective domains and hypotheses are present. By contrast, nontrivial sign holonomy, sufficient reversible rank, selection of one deck orientation, pressure supercriticality, spatial ellipticity, exchange-symmetric translation, and continuum regularity are genuine phase or scaling conditions. Their failure produces mathematically consistent holistic, unsigned, subcritical, unoriented, spatially degenerate, or torsional regimes. The aim is therefore not to replace one large geometric axiom by an equally opaque operational list, but to expose a short auditable dependency chain in which every indispensable premise has a distinct role and a distinct failure mode.

2 Predictive reduction and the emergence of UCP renewal memory

The first task is to construct the object whose geometry is studied downstream. We begin before vector spaces, probabilities, or operator algebras have been chosen. The primitive relation is operational indistinguishability under all admissible futures.

The argument proceeds in three conceptual stages. First, future equivalence is shown to define a canonical minimal memory, both for rooted histories and for processes placed in arbitrary ancillary contexts. Second, the exact quotient is related to practical finite-dimensional compressions and is proved stable under controlled approximation. Third, additional operational hypotheses reconstruct the algebraic form of the memory and identify its deterministic updates as UCP maps. The final subsection then supplies the tomographic and stationary interfaces needed for the retrodictive constructions of [Section 3](#).

2.1 Rooted predictive memory and process equivalence

Definition 2.1 (Finite operational process system). A finite operational process system consists of:

- (i) a small strict symmetric monoidal category $\mathbb{C}$ of typed processes, with tensor unit I ;
- (ii) for every object A , a deterministic discard effect $u_A : A \rightarrow I$, satisfying $u_{A \otimes B} = u_A \otimes u_B$;
- (iii) a finite generating family $\mathbb{G}$ of processes, together with the identities and symmetries needed for typed sequential and parallel composition;
- (iv) convex mixing of parallel processes and an evaluation $\Pr(\omega) \in [0, 1]$ for every closed experiment $\omega : I \rightarrow I$;
- (v) specified admissible continuations and terminal effects, closed under composition, classical postprocessing, and tensoring with identities.

A *rooted history ending at B* is a generated process $h : I \rightarrow B$. The word “future” means admissible postcomposition, not a pre-existing physical time.

Definition 2.2 (Rooted future equivalence). Let $h, h' : I \rightarrow B$ be rooted histories. Write $h \sim_+ h'$ when

$$\Pr(e \circ r \circ h) = \Pr(e \circ r \circ h')$$

for every admissible continuation $r : B \rightarrow C$ and terminal effect $e : C \rightarrow I$. Write

$$\mathbb{Q}_+(B) := \{h : I \rightarrow B \text{ generated}\} / \sim_+.$$

Proposition 2.3 (Rooted future congruence). *For each endpoint B , the relation $\sim_+$ is an equivalence relation. If $h \sim_+ h'$ and $k : B \rightarrow C$ is an admissible generated continuation, then*

$$k \circ h \sim_+ k \circ h'.$$

Consequently every continuation induces a well-defined map

$$\mathbb{Q}_+(k) : \mathbb{Q}_+(B) \longrightarrow \mathbb{Q}_+(C), \quad \mathbb{Q}_+(k)[h] = [k \circ h],$$

*and $\mathbb{Q}_+$ is a covariant functor from the generated continuation category to **Set** on the reachable component.*

Proof. Associate to $h : I \rightarrow B$ its future-profile function

$$\Pi_B(h)(r, e) := \Pr(e \circ r \circ h).$$

Then $h \sim_+ h'$ exactly when $\Pi_B(h) = \Pi_B(h')$, proving reflexivity, symmetry, and transitivity. If $h \sim_+ h'$ and $k : B \rightarrow C$, then for every $d : C \rightarrow D$ and $e : D \rightarrow I$,

$$\Pr(e \circ d \circ k \circ h) = \Pr(e \circ (d \circ k) \circ h) = \Pr(e \circ (d \circ k) \circ h') = \Pr(e \circ d \circ k \circ h'),$$

because $d \circ k$ is an admissible future continuation from B . Thus postcomposition descends. The identity and composition laws follow from those of $\mathbb{C}$. $\square$

Rooted future equivalence is intentionally one-sided. It is the correct object for predictive memory, but it is not by itself a congruence on all hom-sets. General process equality must also quantify over the admissible past.

The preceding proposition establishes the essential predictive fact: once two rooted histories have the same future profile, no later admissible operation can separate them. This gives a well-defined state-like memory. To compare general processes, however, one must also allow different admissible preparations before the process. The next definition therefore enlarges the comparison from rooted histories to arbitrary input–output transformations.

Definition 2.4 (Ordinary two-sided operational equivalence). For generated processes $p, q : A \rightarrow B$, write $p \equiv_1 q$ when

$$\Pr(e \circ r \circ p \circ a) = \Pr(e \circ r \circ q \circ a)$$

for every generated preparation $a : I \rightarrow A$, every admissible continuation $r : B \rightarrow C$, and every terminal effect $e : C \rightarrow I$.

Proposition 2.5 (Two-sided equivalence is a sequential congruence). *The relation $\equiv_1$ is stable under arbitrary typed precomposition and postcomposition. Hence the ordinary process quotients*

$$\mathbf{Q}_1(A, B) := \mathbf{C}_G(A, B) / \equiv_1$$

form a category under sequential composition.

Proof. Let $p \equiv_1 q : A \rightarrow B$, $u : A' \rightarrow A$, and $v : B \rightarrow B'$. Every rooted preparation before $p \circ u$ has the form $u \circ a'$ for an admissible $a' : I \rightarrow A'$, and every future continuation after $v \circ p$ is absorbed into $r \circ v$. The defining equality for $p \equiv_1 q$ therefore gives $p \circ u \equiv_1 q \circ u$ and $v \circ p \equiv_1 v \circ q$. Composition of classes is consequently well defined. $\square$

Definition 2.6 (Complete contextual equivalence). For $p, q : A \rightarrow B$, write $p \equiv_{\parallel} q$ when, for every ancillary object R , every joint preparation $\omega : I \rightarrow A \otimes R$, every continuation $c : B \otimes R \rightarrow C$, and every terminal effect $e : C \rightarrow I$,

$$\Pr(e \circ c \circ (p \otimes \text{id}_R) \circ \omega) = \Pr(e \circ c \circ (q \otimes \text{id}_R) \circ \omega).$$

Set

$$\mathbf{Q}_{\parallel}(A, B) := \mathbf{C}_G(A, B) / \equiv_{\parallel}.$$

Theorem 2.7 (Canonical parallel completion). *Complete contextual equivalence is a monoidal congruence and implies ordinary operational equivalence. Moreover it is the largest monoidal congruence contained in $\equiv_1$: if $\mathcal{R}$ is an equivalence relation on every process set such that*

- (a) $p \mathcal{R} q$ implies $p \equiv_1 q$;
- (b) $\mathcal{R}$ is stable under sequential composition;
- (c) $\mathcal{R}$ is stable under tensoring with arbitrary transformations,

then $p \mathcal{R} q$ implies $p \equiv_{\parallel} q$.

Proof. Equivalence and sequential stability follow by absorbing pre- and postprocessings into the joint preparation and terminal continuation. Tensor stability follows by regarding any further system as part of the ancillary object. Taking the trivial ancilla proves $\equiv_{\parallel} \subseteq \equiv_1$.

Now suppose $p \mathcal{R} q$. Tensor stability gives $p \otimes \text{id}_R \mathcal{R} q \otimes \text{id}_R$ for every R . By (a), the two extended processes are ordinarily equivalent on the composite input and output, so they have identical values on every joint preparation and every joint terminal test. This is exactly $p \equiv_{\parallel} q$. Since $\equiv_{\parallel}$ itself has properties (a)–(c), it is the largest such monoidal congruence. $\square$

The theorem identifies complete contextual equivalence as the correct process notion whenever parallel composition is physically meaningful. Ordinary experiments may miss distinctions that become visible only through an ancilla; the comparison map below measures exactly that possible loss of information.

Because $\equiv_{\parallel}$ is finer than $\equiv_1$, there is a canonical surjection

$$\kappa_{A,B} : \mathbf{Q}_{\parallel}(A, B) \longrightarrow \mathbf{Q}_1(A, B), \quad [p]_{\parallel} \longmapsto [p]_1.$$

Its fibres are the process distinctions invisible to all unassisted preparation–test experiments but visible in some ancillary context.

Definition 2.8 (Parallel completeness). The operational theory is *parallel complete* when every $\kappa_{A,B}$ is bijective, equivalently when

$$\equiv_1 = \equiv_{\parallel}$$

on every process set. This is the precise no-holism condition.

Theorem 2.9 (Universal complete predictive quotient). *The quotient*

$$\mathbf{Q}_{\text{pred}} := \mathbf{C}_G / \equiv_{\parallel}$$

is a symmetric monoidal category. Its quotient functor is universal among symmetric monoidal functors that are constant on complete contextual classes. Under parallel completeness it is canonically isomorphic to the ordinary two-sided operational quotient $\mathbf{C}_G / \equiv_1$.

Proof. The first assertion follows because $\equiv_{\parallel}$ is a monoidal congruence. For the universal property, if $F : \mathbf{C}_G \rightarrow \mathbf{D}$ is symmetric monoidal and $p \equiv_{\parallel} q$ implies $F(p) = F(q)$, define $\overline{F}([p]_{\parallel}) = F(p)$. This is well defined, symmetric monoidal, and unique because the quotient functor is surjective on morphism classes. If parallel completeness holds, the two kernel relations coincide, giving the canonical quotient isomorphism. $\square$

Proposition 2.10 (Real quantum theory has a strict parallel completion). *In finite-dimensional real quantum theory there are channels T_+, T_- such that $T_+ \equiv_1 T_-$ but $T_+ \not\equiv_{\parallel} T_-$. Thus ordinary process reduction need not be monoidal even in a theory with purification and sharp state–effect duality.*

Proof. Let

$$Y_{\mathbb{R}} := \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad Y_{\mathbb{R}}^T = -Y_{\mathbb{R}}, \quad Y_{\mathbb{R}}^2 = -I_2,$$

and define two real two-rebit density matrices

$$\rho_{\pm} := \frac{1}{4}(I_2 \otimes I_2 \pm Y_{\mathbb{R}} \otimes Y_{\mathbb{R}}).$$

The matrix $Y_{\mathbb{R}} \otimes Y_{\mathbb{R}}$ is real symmetric and squares to the identity, so $\rho_{\pm} \geq 0$ and $\text{Tr } \rho_{\pm} = 1$. Their first marginal is $I_2/2$ because $\text{Tr } Y_{\mathbb{R}} = 0$. Hence the real Choi correspondence associates to them trace-preserving completely positive maps $T_{\pm}$.

For real symmetric local effects A, B ,

$$\text{Tr}[(A \otimes B)(\rho_+ - \rho_-)] = \frac{1}{2} \text{Tr}(AY_{\mathbb{R}}) \text{Tr}(BY_{\mathbb{R}}) = 0,$$

because the trace pairing of a symmetric and an antisymmetric real matrix vanishes. Thus the two Choi states have identical product-effect statistics, and the corresponding channels are indistinguishable by every unassisted preparation–effect experiment. On the other hand

$$P_+ := \frac{1}{2}(I_4 + Y_{\mathbb{R}} \otimes Y_{\mathbb{R}})$$

is a valid real-symmetric projection and $\text{Tr}(P_+\rho_+) = 1$, $\text{Tr}(P_+\rho_-) = 0$. A joint ancillary experiment therefore distinguishes the channels. This proves the strictness of the comparison map κ ; compare the general real-quantum obstruction in [10]. $\square$

Remark 2.11 (Why no-holism is load-bearing). Parallel completeness does not follow from sequential prediction or purification. Finite-dimensional real quantum theory contains distinct composite states with identical product-effect statistics, and the associated faithful Choi processes are ordinarily indistinguishable but ancilla-distinct; see [10, 48]. Thus a faithful local-channel representation of all process distinctions requires either the explicit equality $\mathbf{Q}_{\parallel} \cong \mathbf{Q}_1$ or a richer carrier retaining the holistic fibres of κ .

Taken together, the quotient theorems and the real-quantum counterexample fix the logical starting point for renewal. Prediction canonically determines an equivalence class, but monoidal consistency may require the complete rather than the ordinary quotient. Having identified the information that must be retained, we next ask how that information can be represented as a minimal repeatedly updated memory.

2.2 Renewal as exact predictive compression

Fix an endpoint B and write $\mathbf{H}_B$ for the generated rooted histories $I \rightarrow B$. The canonical quotient map is

$$q_B : \mathbf{H}_B \twoheadrightarrow \mathbf{Q}_+(B).$$

The purpose of this subsection is not to choose a preferred microscopic history. It is to show that every future-sufficient representation contains the same reduced predictive content, and that any concrete renewal rule is merely a choice of representatives for that canonical content.

Definition 2.12 (Future-sufficient realization). A future-sufficient realization of the rooted histories consists of sets $X(B)$, surjective encodings $\eta_B : \mathbf{H}_B \rightarrow X(B)$, transition maps $\delta_k : X(B) \rightarrow X(C)$ for generated continuations $k : B \rightarrow C$, and output functions reproducing every terminal-test weight, such that

$$\delta_k \eta_B(h) = \eta_C(k \circ h).$$

It is *reduced* when distinct elements of $X(B)$ are separated by some future continuation and terminal effect.

Theorem 2.13 (Minimality and uniqueness of predictive memory). *The rooted quotient $\mathbf{Q}_+$ with the transition maps of Theorem 2.3 is future sufficient and reduced. For every future-sufficient realization X there is a unique surjection*

$$\pi_B^X : X(B) \longrightarrow \mathbf{Q}_+(B), \quad \pi_B^X(\eta_B(h)) = [h],$$

intertwining all transitions and outputs. The realization X is reduced if and only if every π_B^X is bijective. Hence $\mathbf{Q}_+$ is the unique reduced future-sufficient memory, up to the unique isomorphism fixing all encoded histories.

Proof. If $\eta_B(h) = \eta_B(h')$, transition compatibility and equality of all output functions imply that every future experiment has the same value on h and h' . Thus $h \sim_+ h'$, proving that the displayed formula for π_B^X is well defined. It is surjective because every class has a representative. Compatibility with continuations follows from

$$\pi_C^X(\delta_k \eta_B(h)) = \pi_C^X(\eta_C(k \circ h)) = [k \circ h] = Q_+(k)[h].$$

The quotient itself is sufficient by definition and is reduced because inequivalent classes differ on some future profile.

If X is reduced and $\pi_B^X(x) = \pi_B^X(y)$, choose histories representing x, y . Their quotient classes agree, so all future outputs agree; reducedness gives $x = y$. Conversely, if π_B^X is injective and two realized states have identical future outputs, their representing histories are future equivalent, so their images under π_B^X agree and injectivity gives equality. Thus reducedness is equivalent to bijectivity. Uniqueness follows from surjectivity of the encodings. $\square$

The theorem is the minimal-memory statement on which the rest of the paper rests. It says that predictive states are not introduced as hidden variables: they are forced by the equivalence relation, and every other sufficient memory maps onto them. The corollary translates this abstract quotient into the familiar language of an idempotent renewal operation on raw histories.

Corollary 2.14 (Exact renewal retract). *Suppose a type-preserving section $s_B : Q_+(B) \rightarrow H_B$ is chosen for every reachable endpoint. Then*

$$R_B := s_B q_B$$

is idempotent, satisfies $q_B R_B = q_B$, and replaces every history by a future-equivalent representative. Its image contains exactly one representative of every predictive class. Conversely every idempotent map with these properties is obtained from a unique section. The induced update

$$\Phi_k^{(s)} := R_C \circ (k \circ -) \big|_{\text{im } R_B}$$

forms an exact action of the continuation category and is conjugate to the canonical action on Q_+ .

Proof. Since $q_B s_B = \text{id}$, $R_B^2 = s_B q_B s_B q_B = R_B$ and $q_B R_B = q_B$. Conversely, an idempotent with one fixed point in each class defines the inverse of q_B restricted to its image. For updates,

$$q_C \Phi_k^{(s)} = Q_+(k) q_B.$$

The restricted quotient maps are bijective, so identity and composition of the canonical transition system imply the same laws for $\Phi^{(s)}$. $\square$

The quotient is canonical; a section is not. Renewal therefore means repeated return to minimal sufficient predictive content, not the selection of a preferred microscopic past.

Proposition 2.15 (No natural history-valued selector). *Let $q_B : H_B \twoheadrightarrow Q_+(B)$ be a rooted predictive quotient. Suppose one quotient class contains two distinct histories h_1, h_2 , and suppose there is a bijection*

$$\alpha_B : H_B \longrightarrow H_B$$

which preserves endpoint type and every future profile, satisfies $q_B \alpha_B = q_B$, and exchanges h_1 and h_2 . Then no section $s_B : Q_+(B) \rightarrow H_B$ can be invariant under all future-profile-preserving automorphisms of the presentation. Consequently the quotient data alone do not determine a natural exact renewal retract on raw histories.

Proof. Let $x = q_B(h_1) = q_B(h_2)$. If an invariant section existed, then $\alpha_B s_B = s_B$. Since $s_B(x)$ is one element of the fibre $q_B^{-1}(x)$, invariance would require it to be fixed by every profile-preserving automorphism. In particular, if $s_B(x) = h_1$, then

$$s_B(x) = \alpha_B s_B(x) = \alpha_B(h_1) = h_2,$$

contradicting $h_1 \neq h_2$; the same contradiction occurs if $s_B(x) = h_2$. More generally, any transitive automorphism orbit in a fibre contains no invariant choice of one representative. Since an exact minimal history-valued retract is equivalent to a section by [Theorem 2.14](#), no such retract is naturally determined by the quotient alone. $\square$

Remark 2.16 (Representative gauge). The obstruction is a naturality statement, not a failure of existence. Finite quotients admit sections, and arbitrary set-sized quotients admit them under the axiom of choice. What prediction fixes canonically is the quotient and its induced transition action; choosing a representative history requires extra structure, such as a normal form, metric projection, conditional expectation, or entropy criterion.

The exact theory therefore separates two levels. The quotient and its transition action are canonical, whereas a history-valued section is a gauge choice and in general cannot be selected naturally. Real implementations will moreover only approximate the exact quotient. The next subsection shows that this is not fatal: under explicit nondegeneracy conditions, an approximate compression determines a nearby exact minimal predictive memory.

2.3 Finite-dimensional stability of predictive compression

The quotient construction above is exact and set theoretic. A finite implementation may instead provide a linear compression that is only nearly idempotent and only nearly preserves all predictive profiles. We now show that, below the sharp spectral threshold and under explicit rank and transversality conditions, such a compression lies near an exact minimal predictive retract. No positivity is required in this subsection.

The strategy is deliberately modular. We first classify exact linear retracts, then turn a nearly idempotent map into a genuine projection, and finally correct that projection so that it preserves the predictive quotient exactly. This separates algebraic idempotence from the independent requirements of sufficiency and minimal rank.

Definition 2.17 (Finite linear predictive presentation). A finite linear predictive presentation is a surjective linear map

$$q : X \twoheadrightarrow Q$$

between finite-dimensional normed spaces. The quotient Q is the linear span of the retained predictive profiles, and

$$K_{\text{pred}} := \ker q, \quad r := \dim Q$$

is the space of observationally null directions. A linear map $P : X \rightarrow X$ is an *exact predictive projection* when

$$P^2 = P, \quad qP = q.$$

It is *minimal* when $\text{rank } P = r$.

Proposition 2.18 (Classification of exact linear predictive retracts). *Let P be an exact predictive projection for $q : X \rightarrow Q$. Then $\text{rank } P \geq r$ and $\ker P \subseteq K_{\text{pred}}$. Moreover the following are equivalent:*

$$\text{rank } P = r, \quad \ker P = K_{\text{pred}}, \quad q|_{\text{im } P} : \text{im } P \longrightarrow Q \text{ is an isomorphism.}$$

Consequently minimal exact predictive projections are in bijection with linear complements M of K_{pred} :

$$X = M \oplus K_{\text{pred}}, \quad P_M = (q|_M)^{-1}q.$$

Any two such carriers are related by the unique quotient-fixed linear isomorphism.

Proof. From $qP = q$ one obtains $\text{rank } q \leq \text{rank } P$. If $Px = 0$, then $qx = qPx = 0$, proving $\ker P \subseteq K_{\text{pred}}$. Since P is a projection,

$$\dim \ker P = \dim X - \text{rank } P, \quad \dim K_{\text{pred}} = \dim X - r.$$

Thus $\text{rank } P = r$ is equivalent to $\ker P = K_{\text{pred}}$. The restriction $q|_{\text{im } P}$ is surjective because $qP = q$; it is injective exactly when $\text{im } P \cap K_{\text{pred}} = 0$, which is equivalent to the same dimension condition.

If M complements K_{pred} , then $q|_M$ is an isomorphism and $P_M = (q|_M)^{-1}q$ is an idempotent with $qP_M = q$, image M , and kernel K_{pred} . Conversely the image of every minimal exact projection is such a complement. For two complements M, N , the map

$$U_{M,N} := (q|_N)^{-1}q|_M : M \longrightarrow N$$

is the unique isomorphism satisfying $qU_{M,N} = q|_M$. □

Lemma 2.19 (Idempotentization below the quarter threshold). *Let $E \in \text{End}(X)$ and assume*

$$\delta := \|E^2 - E\| < \frac{1}{4}.$$

Set

$$D := E^2 - E, \quad A := 2E - \mathbf{1},$$

and define, by the norm-convergent binomial series,

$$P_E := \frac{1}{2} \left[\mathbf{1} + A(\mathbf{1} + 4D)^{-1/2} \right].$$

Then

$$P_E^2 = P_E, \quad P_E E = E P_E,$$

and

$$\|P_E - E\| \leq \alpha_B(E, \delta) := \frac{\|2E - \mathbf{1}\|}{2} \left((1 - 4\delta)^{-1/2} - 1 \right).$$

If $E = E^*$ on a finite-dimensional Hilbert space, then

$$P_E = \mathbf{1}_{[1/2, \infty)}(E), \quad \|P_E - E\| \leq \alpha_H(\delta) := \frac{1 - \sqrt{1 - 4\delta}}{2},$$

and the constant α_H is sharp.

Proof. The defect D is a polynomial in E , hence commutes with A , and

$$A^2 = (2E - \mathbf{1})^2 = \mathbf{1} + 4D.$$

Because $4\|D\| < 1$, the inverse square root $(\mathbf{1} + 4D)^{-1/2}$ is defined by its absolutely norm-convergent binomial series. Therefore

$$S := A(\mathbf{1} + 4D)^{-1/2}$$

satisfies $S^2 = \mathbf{1}$, and $P_E = (\mathbf{1} + S)/2$ is an idempotent commuting with E . Furthermore,

$$P_E - E = \frac{1}{2}A \left[(\mathbf{1} + 4D)^{-1/2} - \mathbf{1} \right].$$

Majorizing the absolute binomial coefficients gives

$$\|(\mathbf{1} + 4D)^{-1/2} - \mathbf{1}\| \leq (1 - 4\|D\|)^{-1/2} - 1,$$

which proves the Banach-space estimate.

In the self-adjoint case, the scalar inequality $|\lambda^2 - \lambda| \leq \delta < 1/4$ separates the spectrum into two components lying near 0 and 1. Functional calculus therefore gives $P_E = \mathbf{1}_{[1/2, \infty)}(E)$. Solving $t(1 - t) = \delta$ on $0 \leq t \leq 1/2$ gives the maximal spectral distance

$$t = \frac{1 - \sqrt{1 - 4\delta}}{2}$$

to $\{0, 1\}$. Equality is attained by a scalar spectral value at an endpoint of the allowed component, proving sharpness. $\square$

Theorem 2.20 (Quantitative stability of minimal predictive memory). *Let $q : X \twoheadrightarrow Q$ be a finite linear predictive presentation. Choose a bounded right inverse $s : Q \rightarrow X$, $qs = \text{id}_Q$, and put $\kappa := \|s\|$. Suppose $E \in \text{End}(X)$ satisfies*

$$\|E^2 - E\| \leq \delta < \frac{1}{4}, \quad \|qE - q\| \leq \varepsilon.$$

Let $P = P_E$ be the projection of Theorem 2.19, and set

$$\alpha := \|P - E\|, \quad \eta := \varepsilon + \|q\| \alpha.$$

Assume

$$\text{rank } P = r = \dim Q, \quad \kappa\eta < 1.$$

Then $B := qPs \in \text{End}(Q)$ is invertible and

$$R := PsB^{-1}q$$

is an exact minimal predictive projection with

$$R^2 = R, \quad qR = q, \quad \text{rank } R = r, \quad \text{im } R = \text{im } P.$$

Moreover

$$\|R - E\| \leq \alpha + \frac{\|P\| \kappa \eta}{1 - \kappa \eta}.$$

Thus approximate idempotence and approximate future sufficiency determine a nearby exact minimal predictive memory whenever the near-one spectral subspace has the correct dimension and remains transverse to the observationally null directions.

Proof. First,

$$\|qP - q\| \leq \|q(P - E)\| + \|qE - q\| \leq \eta.$$

Hence

$$\|B - \text{id}_Q\| = \|(qP - q)s\| \leq \kappa\eta < 1.$$

The Neumann series makes B invertible and gives $\|B^{-1}\| \leq (1 - \kappa\eta)^{-1}$. The definition of R yields

$$qR = qPsB^{-1}q = BB^{-1}q = q$$

and

$$R^2 = PsB^{-1}qPsB^{-1}q = PsB^{-1}BB^{-1}q = R.$$

The range of R lies in $\text{im } P$. Since $qR = q$, $\text{rank } R \geq r$, whereas $\text{rank } R \leq \text{rank } P = r$; consequently $\text{rank } R = r$ and the two ranges agree.

The map $q|_{\text{im } P}$ is therefore an isomorphism with inverse PsB^{-1} . For every $x \in X$,

$$Px = PsB^{-1}qPx,$$

and hence

$$R - P = PsB^{-1}q(\mathbf{1} - P).$$

Because $q(\mathbf{1} - P) = q - qP$,

$$\|R - P\| \leq \|P\| \kappa \|B^{-1}\| \|q(\mathbf{1} - P)\| \leq \frac{\|P\| \kappa \eta}{1 - \kappa\eta}.$$

Adding $\|P - E\| = \alpha$ proves the estimate. $\square$

The theorem gives the central robustness conclusion: approximate predictive compression is close to an exact quotient carrier provided the retained spectral subspace has the correct dimension and is transverse to observationally null directions. The next corollary propagates this static estimate to the renewal updates themselves.

Corollary 2.21 (Stability of compressed renewal dynamics). *Let $U_\sigma : X \rightarrow X$ be bounded raw updates and let $M := \text{im } P = \text{im } R$. Define*

$$\tilde{\Phi}_\sigma := PU_\sigma P|_M, \quad \Phi_\sigma := RU_\sigma R|_M.$$

Then

$$\|\tilde{\Phi}_\sigma - \Phi_\sigma\| \leq \|P - R\| \|U_\sigma\|.$$

For every fixed word length, the corresponding products differ by the standard telescoping sum and are therefore norm stable.

Proof. Both P and R restrict to the identity on their common range M . Thus, for $x \in M$,

$$(\tilde{\Phi}_\sigma - \Phi_\sigma)x = (P - R)U_\sigma x,$$

which gives the displayed bound. For products, insert and subtract one factor at a time:

$$A_n \cdots A_1 - B_n \cdots B_1 = \sum_{j=1}^n A_n \cdots A_{j+1} (A_j - B_j) B_{j-1} \cdots B_1,$$

and take norms. $\square$

Remark 2.22 (Why the nondegeneracy hypotheses are necessary). Approximate idempotence and exact sufficiency do not imply minimality: if $X = Q \oplus K_{\text{pred}}$, q is the first projection, and $E = \mathbf{1}_X$, then $E^2 = E$ and $qE = q$, but E retains all redundant memory. Conversely, an exact rank- r projection may erase the predictive quotient. The rank and transversality clauses in [Theorem 2.20](#) therefore control genuinely independent failure modes.

Thus the predictive quotient is stable as a finite memory and its compressed dynamics is stable along every fixed finite history. Up to this point, however, the carrier is only a finite real linear space. We now add the operational principles that determine its order structure, its scalar field, and ultimately its associative operator algebra.

2.4 Finite sharp-complex reconstruction

To pass from predictive sets to operator algebras, additional operational structure is required. We isolate the hypotheses that produce the Jordan stage, the complex field, and the associative orientation.

The order of the reconstruction is important. Sharp purification first yields spectral decomposition and a Euclidean Jordan algebra. Two-path interference then eliminates the non-complex Jordan alternatives, while a compatible phase response chooses one of the two opposite associative products. Each step solves a distinct ambiguity and is therefore stated separately.

Assumption 2.1 (Finite operational reconstruction package). On each operationally irreducible reduced predictive sector the following hold.

- (O1) normalized predictive states form a finite-dimensional compact convex set, admissible effects separate states, and the resolved process theory is closed under sequential and parallel composition and convex mixing;
- (O2) *reduced sharp-reversible realisation*: there is a unique deterministic discard; every physical process admits a pure reversible dilation with a pure environment, unique up to reversible changes of a minimal environment; operationally pure processes with no retained environmental record are closed under sequential and parallel composition; and every normalized pure preparation has a normalized pure reverse effect certain on it;
- (O3) ordinary process reduction is parallel complete in the sense of [Theorem 2.8](#);
- (O4) for every nonclassical pair of primitive outcomes, the balanced two-path pure states form one effective transitive orbit of a compact connected one-dimensional phase group;
- (O5) pair phases have primitive period 2π and obey the linear, pair-local, permutation-covariant and triangle-compatible spectral response stated in [Assumption 2.2](#) below.

Definition 2.23 (Finite sharp theory with purification). A finite reduced operational theory satisfies the *sharp-purification package* when:

- (SP1) it is causal, meaning that every system has a unique deterministic effect;
- (SP2) sequential and parallel composites of pure processes are pure;
- (SP3) every system possesses a pure effect occurring with unit probability on some normalized pure state;

(SP4) every normalized state is the marginal of a pure bipartite state, and two purifications on the same purifying system differ by a reversible transformation of that system.

Proposition 2.24 (Sharp purification from reduced reversible realisation). *Item (O2) of Assumption 2.1 implies the sharp-purification package of Theorem 2.23.*

Proof. The unique discard in (O2) is precisely causality, giving (SP1). Apply the reversible-dilation clause to a preparation $\rho : I \rightarrow A$. Its dilation is a pure preparation $\Psi : I \rightarrow A \otimes E$ followed by discarding E , so every state has a purification. Essential uniqueness of minimal reversible dilations implies the usual uniqueness of purifications up to a reversible transformation on the purifying system; this is the reversible-realisation form of purification proved in [3]. Hence (SP4) holds.

The no-retained-record closure of pure dilations under sequential and parallel composition is exactly purity preservation, giving (SP2). Finally, if $\alpha : I \rightarrow A$ is normalized and pure, its normalized pure reverse effect $\alpha^\sharp : A \rightarrow I$ satisfies $\alpha^\sharp \circ \alpha = 1$. Thus a pure effect is certain on a pure state, which is (SP3). $\square$

Proposition 2.25 (Reversible dilation alone does not imply pure sharpness). *Unique discard, pure-environment reversible dilation, and essential uniqueness of minimal dilations do not imply pure sharpness. In particular, the pure-process reversal clause in item (O2) of Assumption 2.1 is logically independent of the transformation-level dilation clauses.*

Proof. Begin with finite-dimensional complex quantum theory, retaining its state spaces, channels, reversible dilations, tensor product, and deterministic discard. These data satisfy Stinespring dilation and essential uniqueness of minimal dilations. Modify only the admitted non-deterministic effects.

Fix $0 < \epsilon < 1/2$. On a qubit, for every rank-one projection P , let

$$e_{P,\epsilon}(\rho) := \epsilon + (1 - 2\epsilon) \text{Tr}(P\rho),$$

and admit the convex effect system generated by

$$0, \quad u, \quad e_{P,\epsilon}, \quad u - e_{P,\epsilon}.$$

The family separates states: equality of all $e_{P,\epsilon}$ -values implies equality of all rank-one projection probabilities because $1 - 2\epsilon \neq 0$, and rank-one projections separate density matrices. Thus the predictive state theory remains reduced.

Every non-deterministic generator takes values in $[\epsilon, 1 - \epsilon]$, so no such effect occurs with unit probability on any state. Convex combinations cannot increase this maximum unless the deterministic effect u is used. But u has a nontrivial refinement

$$u = e_{P,\epsilon} + (u - e_{P,\epsilon})$$

into two nonzero admitted effects and is therefore not pure in the operational, non-refinability sense. Hence the modified theory has no pure effect certain on a normalized pure state: pure sharpness fails.

The state spaces, channels, reversible dilations, minimal environments, and discard have not changed. Therefore the dilation assumptions alone cannot determine pure sharpness; the admitted effect theory is independent operational data. $\square$

Remark 2.26 (Exact boundary of the reconstruction package). The reversible-dilation clauses derive causality and purification, while the no-retained-record clause supplies purity preservation. Pure sharpness requires a separate preparation–observation reversal principle. The last clause of item (O2) is therefore not a hidden consequence of Stinespring-type realization.

Theorem 2.27 (Operational diagonalisation). *Under Theorem 2.23, every normalized predictive state has a finite spectral decomposition*

$$\rho = \sum_{j=1}^r p_j \alpha_j, \quad p_j > 0, \quad \sum_j p_j = 1,$$

where the α_j are perfectly distinguishable normalized pure states. Reversible transformations act transitively on normalized pure states, and each system has a unique invariant completely mixed state.

Proof. The diagonalisation statement is the finite operational diagonalisation theorem of Chiribella and Scandolo [13]. Its hypotheses are exactly causality, purity preservation, pure sharpness, and purification, which are (SP1)–(SP4). Pure-state transitivity and existence of the invariant completely mixed state are standard consequences of purification with essential uniqueness in the finite operational-probabilistic framework [3]. $\square$

Theorem 2.28 (Jordan reconstruction from sharp purification). *Under Theorem 2.23, the ordered state space of every operationally irreducible predictive sector is order-isomorphic to the positive cone of a finite-dimensional Euclidean Jordan algebra. The cone is self-dual for a canonical invariant inner product, and the decompositions of Theorem 2.27 are its Jordan spectral decompositions.*

Proof. Sharp theories with purification possess a canonical state–effect dagger, self-duality, pure face projectors, and the symmetry-of-transition-probability property. Barnum, Lee, Scandolo, and Selby prove that these structures identify each finite system with the state space of a Euclidean Jordan algebra [14]. The present theory is already reduced, so its states and effects satisfy the separation hypotheses of that theorem. The spectral decompositions agree with those of Theorem 2.27 by uniqueness of the operational dagger and the associated face projectors. $\square$

Remark 2.29 (What is now derived). Homogeneity, self-duality, and spectral decomposition are no longer primitive clauses of Assumption 2.1. They are consequences of the reduced sharp-reversible realisation principle through Theorems 2.24, 2.27 and 2.28. The remaining complex-field and associative-orientation inputs are genuinely composite and interferometric: parallel completeness and the two-path phase laws.

The preceding results explain what is already fixed before complex quantum structure is invoked: the predictive state cone is homogeneous, self-dual, and spectral. They do not yet distinguish real, complex, quaternionic, spin-factor, or exceptional Jordan systems. The next results use the geometry of balanced two-path interference to make precisely that distinction.

Let J be the Euclidean Jordan algebra supplied by Theorem 2.28, and let $(p_1, \dots, p_r)$ be a Jordan frame. We now show that the two-path phase condition selects complex matrix type rather than merely postulating a qubit. For $i \neq j$, write J_{ij} for the Peirce coherence space and

$$J[i, j] := \mathbb{R}p_i \oplus \mathbb{R}p_j \oplus J_{ij}$$

for the rank-two face generated by p_i, p_j . With the invariant Euclidean norm, the balanced pure states in this face are canonically a sphere in J_{ij} ; see the rank-two classification in [49, 50].

Theorem 2.30 (Two-path complex face). *Under item (O4), every nonclassical Peirce space has*

$$\dim_{\mathbb{R}} J_{ij} = 2,$$

and every nonclassical rank-two face is Jordan-isomorphic to $M_2(\mathbb{C})_{\text{sa}}$. The connected phase group is isomorphic to $\mathbb{T}$ and acts on $J_{ij} \cong \mathbb{R}^2$ by ordinary planar rotations, up to orientation.

Proof. Let $m = \dim_{\mathbb{R}} J_{ij}$. After rescaling the invariant norm, the balanced pure states form the sphere S^{m-1} . A transitive orbit of a one-dimensional Lie group has manifold dimension at most one, so $m - 1 \leq 1$. The case $m = 1$ gives S^0 , on which a connected group cannot act transitively; nonclassicality excludes $m = 0$. Hence $m = 2$. The unique rank-two Euclidean Jordan algebra with a two-dimensional Peirce coherence space is $M_2(\mathbb{C})_{\text{sa}}$. A compact connected one-dimensional Lie group is a circle, and every effective transitive orthogonal circle action on S^1 is conjugate to the standard rotation action, with the two conjugacy choices corresponding to orientation. $\square$

Corollary 2.31 (Pairwise complex matrix type). *Every nonclassical irreducible predictive Jordan sector is*

$$J \cong M_r(\mathbb{C})_{\text{sa}}$$

for some $r \geq 2$; the remaining irreducible sectors are one-dimensional classical sectors.

Proof. For rank two this is [Theorem 2.30](#). For rank at least three, the simple Euclidean Jordan classification gives real, complex, quaternionic Hermitian matrix algebras and the exceptional Albert algebra; their pairwise Peirce dimensions are respectively 1, 2, 4, 8. Spin factors have rank two. Since every nonclassical pair has Peirce dimension two, only the complex Hermitian matrix algebra remains [\[49\]](#). Operational irreducibility excludes a nontrivial direct sum inside one sector. $\square$

Pairwise two-path interference has now selected complex Hermitian matrix type, but the Jordan product alone still forgets the sign of the commutator and hence the orientation of the associative multiplication. The following phase-response assumption and rigidity lemma recover this remaining binary choice consistently across all pairs.

Assumption 2.2 (Compatible spectral phase response). For $a = \sum_k a_k p_k$ diagonal in a Jordan frame, let $\ell_{ij}(a)$ be the oriented angular velocity induced on J_{ij} . Assume:

- (P1) $a \mapsto \ell_{ij}(a)$ is real linear and is unchanged by adding a scalar multiple of $\mathbf{1}$;
- (P2) $\ell_{ij}(a)$ depends only on a_i, a_j and is covariant under frame permutations;
- (P3) the contrast $p_i - p_j$ has primitive unit angular speed;
- (P4) for every triple of distinct indices, $\ell_{ij}(a) + \ell_{jk}(a) + \ell_{ki}(a) = 0$.

Lemma 2.32 (Rigidity of pairwise phase velocity). *Under [Assumption 2.2](#), there is one sign $s \in \{\pm 1\}$ on each connected nonclassical block such that*

$$\ell_{ij}(a) = \frac{s}{2}(a_i - a_j)$$

for every frame and every pair.

Proof. Linearity and pair locality give $\ell_{ij}(a) = \alpha_{ij}a_i + \beta_{ij}a_j$. Scalar invariance gives $\alpha_{ij} + \beta_{ij} = 0$, so $\ell_{ij}(a) = c_{ij}(a_i - a_j)$. Since the contrast $p_i - p_j$ has eigenvalues 1, -1 on the pair, primitive unit speed gives $2|c_{ij}| = 1$. Write $c_{ij} = s_{ij}/2$. Substituting the three formulas into the triangle identity and comparing the independent coefficients of a_i, a_j, a_k gives $s_{ij} = s_{jk} = s_{ik}$. Connectivity of a Jordan frame propagates one sign through the irreducible block, and frame covariance makes it independent of the chosen frame. $\square$

Theorem 2.33 (Finite complex predictive algebra). *Under [Assumption 2.1](#), the predictive observable algebra is*

$$\mathcal{A}_{\text{pred}} \cong \bigoplus_{r=1}^s M_{n_r}(\mathbb{C}).$$

On each nonclassical connected block there is one sign $s = \pm 1$ such that the operational phase flow generated by $a = a^$ is*

$$\tau_a^t(b) = e^{sita/2} b e^{-sita/2}, \quad \left. \frac{d}{dt} \tau_a^t(b) \right|_{t=0} = s \frac{i}{2} [a, b].$$

Thus the ordered Jordan structure determines complex matrix type, while the composed phase response selects one of the two opposite associative products. The remaining freedom is blockwise reversal of this associative orientation.

Proof. By [Theorems 2.24](#) and [2.28](#), items (O1)–(O2) supply the finite Euclidean Jordan decomposition without separately assuming homogeneity, self-duality, or spectrality. Item (O4) and [Theorem 2.31](#) then force every nonclassical irreducible Jordan block to be complex Hermitian matrix type.

In a frame diagonalizing a , item (O5) and [Theorem 2.32](#) give the angular velocity $s(a_i - a_j)/2$ on every Peirce coherence plane. The unitary $e^{sita/2}$ fixes each p_i and has exactly those velocities, so the operational and matrix phase flows agree on every Peirce summand and hence on the whole algebra. Degenerate spectral subspaces cause no ambiguity because the exponential is scalar there. Differentiation yields $s \frac{i}{2} [a, \cdot]$. Replacing s by $-s$ reverses the associative multiplication while leaving the Jordan product, order, and all single-system probabilities unchanged; compare [\[51, 52\]](#).

Item (O3) is not used to select the single-system complex field. It ensures that independently prepared systems have no hidden ancillary fibres and supplies the faithful composite matrix ordering used in the UCP step below. Finite-dimensionality gives the stated direct sum of full matrix algebras. $\square$

Remark 2.34 (Sharp logical boundary). Real or quaternionic quantum systems, higher spin factors, and the exceptional Jordan system remain possible if the one-dimensional transitive two-path phase condition is removed. Conversely, single-system complex interference does not imply parallel completeness: a composite may contain globally visible degrees of freedom invisible to all product tests. The complex selector and the no-holism axiom are therefore independent.

The complex-algebra theorem completes the single-system reconstruction. Its conclusion is stronger than a choice of notation: it shows that complex matrix algebras and their commutator orientation arise from operational phase data. A separate question remains for transformations. Positivity on one system does not guarantee positivity when the system is embedded in a larger experiment, so the next subsection identifies ancillary stability with complete positivity.

2.5 Ancillary stability and complete positivity

Let $V_A = \mathcal{A}_A^{\text{sa}}$ be the observable order-unit space associated with a predictive object A . A deterministic process $T : A \rightarrow B$ acts in the Heisenberg picture by a linear map

$$\Phi_T : \mathcal{A}_B \longrightarrow \mathcal{A}_A.$$

Closed-experiment positivity makes Φ_T positive, and preservation of discard makes it unital. Complete positivity is the additional statement that this remains true in every matrix ancillary context.

This is the transformation-level counterpart of parallel completeness. The reconstructed algebra tells us what the predictive observables are; complete ancillary stability tells us which positive updates remain admissible in every composite context.

Definition 2.35 (Complete ancillary stability). A deterministic predictive transformation T is completely ancillary stable when, for every $k \geq 1$, its extension by the identity on a k -level complex ancilla induces a positive map

$$\text{id}_{M_k} \otimes \Phi_T : M_k(\mathbb{C}) \otimes \mathcal{A}_B \longrightarrow M_k(\mathbb{C}) \otimes \mathcal{A}_A.$$

Theorem 2.36 (Operational-to-UCP bridge). *Assume [Assumption 2.1](#). Every deterministic predictive transformation that is completely ancillary stable induces a unital completely positive map on $\mathcal{A}_{\text{pred}}$. Conversely every UCP map defines a deterministic transformation of the finite predictive operator system and remains positive under arbitrary finite ancillas. Thus the downstream renewal alphabet may be taken to be a finite family*

$$\Phi_\sigma : \mathcal{A}_{\text{pred}} \rightarrow \mathcal{A}_{\text{pred}}, \quad \sigma \in E,$$

of UCP maps, derived from operational prediction rather than imposed as primitive geometry.

Proof. Let T be deterministic. If $b \geq 0$ is an effect or positive observable in $\mathcal{A}_B$, then every state ω on $\mathcal{A}_A$ gives a nonnegative operational probability $\omega(\Phi_T(b))$; state separation implies $\Phi_T(b) \geq 0$. Determinism gives $\Phi_T(\mathbf{1}) = \mathbf{1}$, so Φ_T is positive and unital.

By complete ancillary stability, $\text{id}_{M_k} \otimes \Phi_T$ is positive for all k . This is the definition of complete positivity. In finite dimensions it is enough to test k equal to the largest matrix-block size, by Choi's theorem [53]. Hence Φ_T is UCP.

Conversely, a UCP map is unital and positive, so it preserves the order unit and sends effects to effects. Tensoring with an identity map preserves complete positivity, hence positivity, on every finite ancilla. Pairing with states therefore defines consistent probabilities in all parallel contexts, which is precisely a deterministic transformation of the reconstructed operator system. $\square$

Corollary 2.37 (Predictive units are automorphisms). *If a UCP predictive transformation Φ has a UCP inverse, then Φ is a $*$ -automorphism.*

Proof. Let Ψ be the UCP inverse. Kadison–Schwarz gives $\Phi(a)^* \Phi(a) \leq \Phi(a^* a)$. Applying Ψ and then Kadison–Schwarz to Ψ at $\Phi(a)$ yields

$$a^* a = \Psi(\Phi(a))^* \Psi(\Phi(a)) \leq \Psi(\Phi(a)^* \Phi(a)) \leq a^* a.$$

Thus equality holds and every a lies in the multiplicative domain of Φ . Hence Φ is a unital $*$ -homomorphism; bijectivity makes it a $*$ -automorphism. $\square$

Remark 2.38 (The first pillar). The output of this section is not yet spacetime, causality, or a signed geometry. It is the canonical finite object on which those structures may be built: a reduced predictive C^* -algebra and a monoid of UCP renewal updates. The construction has separated three issues that are often conflated: predictive quotienting is categorical, complex algebra reconstruction is operational, and complete positivity is ancillary stability.

We have therefore reached the first finite dynamical object promised in the introduction: a reduced complex predictive algebra acted on by UCP renewal maps. The remaining issue in this section is one of identification and reference. We must know when operationally indistinguishable histories determine the same full channel, and we must select a faithful stationary state relative to which forward dynamics can later be compared with retrodiction.

2.6 Tomographic identification and a canonical stationary reference

The predictive quotient is defined operationally. To identify a class of histories with a full accumulated channel, rather than only with its action on one prepared orbit, one needs a process-tomography interface.

The distinction is subtle but essential for the later branchwise arguments. A rooted history only probes the orbit generated from the preparations actually available, whereas a channel is a transformation on the entire state space. The following assumption makes the passage from the first notion to the second explicit rather than tacit.

Assumption 2.3 (Preparation–effect process tomography). For every reconstructed predictive object, the admissible preparations span its real ordered state space and the admissible effects separate that space. Two transformations are operationally equal when all pairings

$$a(Tv) = a(T'v)$$

agree for admissible spanning preparations v and separating effects a .

Proposition 2.39 (Operational profiles determine accumulated channels). *Under [Assumption 2.3](#), two generated histories have identical transformation-level operational profiles if and only if their accumulated linear state maps agree. Equivalently, after the reconstruction of [Theorems 2.33](#) and [2.36](#), their accumulated Heisenberg UCP maps agree.*

Proof. Equality of the accumulated maps plainly implies equality of every preparation–effect pairing. Conversely, suppose all such pairings agree. For each admissible spanning preparation v , separation by admissible effects gives $Tv = T'v$. Linearity and spanning then imply $T = T'$ on the entire state space. Taking duals gives equality of the Heisenberg maps. $\square$

Example 2.40 (A rooted orbit does not determine a channel). Let $V = \mathbb{R}^2$, suppose the only admissible initial preparation is $v_0 = e_1$, and let the terminal effects separate vectors in V . Define

$$T = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad T' = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}.$$

Then $Tv_0 = T'v_0 = e_1$. If every admissible future continuation is generated from the actual rooted output and never supplies an independent preparation with a component along e_2 , all rooted future statistics of T and T' agree. Nevertheless

$$Te_2 = e_2, \quad T'e_2 = e_1,$$

so $T \neq T'$. Thus equality on one reachable rooted orbit does not imply equality of accumulated transformations. The spanning-preparation clause in [Assumption 2.3](#) is therefore an independent interface condition, not a consequence of arbitrarily many future tests on the given orbit.

The proposition and counterexample isolate the exact role of tomography: future profiles determine accumulated channels only when preparations span and effects separate. Once this interface is present, the UCP alphabet may be treated as a genuine family of channels. We can then ask which state provides the common reference needed for stationarity and retrodictive duality.

Let a selected simple recurrent block be $M_n(\mathbb{C})$ and let $\{\Phi_\sigma\}_{\sigma \in E}$ be its UCP renewal alphabet. A stationary reference cannot be selected from the unweighted alphabet alone: one must also specify how elementary renewals are sampled.

Definition 2.41 (Weighted averaged renewal transfer). Choose $p_\sigma > 0$ with $\sum_\sigma p_\sigma = 1$ and set

$$\bar{\Phi} := \sum_{\sigma \in E} p_\sigma \Phi_\sigma, \quad T := \bar{\Phi}_*.$$

The trace-preserving CP map T is *primitive* when some $m \geq 1$ satisfies

$$X \geq 0, X \neq 0 \implies T^m(X) > 0.$$

Theorem 2.42 (Primitive transfer selects a unique faithful state). *If the weighted averaged renewal transfer of [Theorem 2.41](#) is primitive, then there is a unique density matrix $\rho_* > 0$ satisfying*

$$T(\rho_*) = \rho_*.$$

Moreover, for some $m \geq 1$ and $0 < c < 1$,

$$\|T^{km}(\rho) - \rho_*\|_1 \leq c^k \|\rho - \rho_*\|_1$$

for every density matrix ρ and every $k \geq 0$.

Proof. The compact convex set of density matrices is invariant under the continuous affine map T , so it has a fixed point ρ_* by Brouwer's theorem. Primitivity gives $\rho_* = T^m(\rho_*) > 0$.

Let

$$\mathcal{K}_0 := \{X = X^* : \text{Tr } X = 0, \|X\|_1 = 1\}.$$

This set is compact. For $X \in \mathcal{K}_0$, write its Jordan decomposition $X = X_+ - X_-$; then $\text{Tr } X_+ = \text{Tr } X_- = 1/2$. Primitivity makes $A = T^m(X_+)$ and $B = T^m(X_-)$ positive definite. Hence for some $\delta_X > 0$ one has $A, B \geq \delta_X \mathbf{1}$, and therefore

$$\|T^m(X)\|_1 = \|A - B\|_1 \leq \text{Tr}(A - \delta_X \mathbf{1}) + \text{Tr}(B - \delta_X \mathbf{1}) = 1 - 2n\delta_X < 1.$$

The continuous function $X \mapsto \|T^m(X)\|_1$ has a maximum $c < 1$ on $\mathcal{K}_0$. Homogeneity gives $\|T^m(X)\|_1 \leq c\|X\|_1$ for every traceless Hermitian X . Applying this to $\rho - \rho_*$ and iterating proves the estimate. If two stationary densities existed, the same strict contraction applied to their difference would force them to coincide. $\square$

Corollary 2.43 (Common KMS–Petz family). *Under [Theorem 2.42](#), the averaged channel $\bar{\Phi}$ has a canonical UCP KMS–Petz dual relative to ρ_* . If, in addition,*

$$(\Phi_\sigma)_*(\rho_*) = \rho_* \quad (\sigma \in E),$$

then every elementary channel has a UCP KMS–Petz dual relative to the same faithful state. Without channelwise stationarity the elementary Petz transposes remain CP, but need not be unital.

Proof. Apply [Theorem 3.3](#) to $\bar{\Phi}$ and to each channel that preserves ρ_* . From the defining formula, $\Phi_{\rho_*}^{\sharp}(\mathbf{1}) = \mathbf{1}$ is equivalent to $\Phi_*(\rho_*) = \rho_*$. $\square$

The weighted primitive transfer completes the predictive stage by selecting a unique faithful reference state and, under channelwise stationarity, a common family of KMS–Petz duals. Section [efsec:predictive](#) has therefore supplied three ingredients: the minimal predictive memory, its UCP renewal dynamics, and the stationary reference against which opposite temporal comparisons can be made. The next section shows that two such comparisons—real affinity and record orientation—must nevertheless be kept distinct.

3 Retrodictive duality and the signed orientation line

The positive predictive algebra distinguishes what future tests can retain. A Lorentzian construction additionally needs an oppositely oriented comparison. There are two different comparisons available: a real forward/backward weight, measured by affinity, and a parity of record transport, measured by a determinant line. This section proves that they are mathematically independent and that only the second classifies the signed cover.

The section follows the operational order in which these structures become available. We first construct the canonical KMS–Petz dual. We then resolve individual outcomes and prove that conditional predictive successors enforce multiplicative branch weights. Those weights define real affinity, while stable future-distinct records define an independent determinant parity. The final results compare the two classes and classify the additional amplitude lifts of the signed cover.

3.1 The canonical KMS–Petz retrodictive dual

Let $\mathcal{A} = M_n(\mathbb{C})$ for the moment; direct sums are handled blockwise. Let $\rho > 0$ be a faithful density matrix and let $\Phi : \mathcal{A} \rightarrow \mathcal{A}$ be UCP. Write Φ_* for the trace adjoint, determined by

$$\mathrm{Tr}(\Phi_*(x)a) = \mathrm{Tr}(x\Phi(a)).$$

Assume ρ is stationary, $\Phi_*(\rho) = \rho$.

Definition 3.1 (Symmetric KMS pairing). Define

$$\langle a, b \rangle_{\rho, 1/2} := \mathrm{Tr}(\rho^{1/2} a^* \rho^{1/2} b), \quad \Gamma_{\rho}(a) := \rho^{1/2} a \rho^{1/2}.$$

Definition 3.2 (KMS–Petz anti-renewal dual). The retrodictive or anti-renewal dual of Φ relative to ρ is

$$\boxed{\Phi_{\rho}^{\sharp} := \Gamma_{\rho}^{-1} \circ \Phi_* \circ \Gamma_{\rho}, \quad \Phi_{\rho}^{\sharp}(a) = \rho^{-1/2} \Phi_*(\rho^{1/2} a \rho^{1/2}) \rho^{-1/2}.}$$

Theorem 3.3 (Canonical retrodictive adjoint). *Let Φ be UCP and let $\rho > 0$ satisfy $\Phi_*(\rho) = \rho$. Then:*

- (i) $\Phi_{\rho}^{\sharp}$ is UCP and preserves the same stationary state;
- (ii) it is the adjoint of Φ for the symmetric KMS pairing,

$$\langle a, \Phi(b) \rangle_{\rho, 1/2} = \langle \Phi_{\rho}^{\sharp}(a), b \rangle_{\rho, 1/2};$$

- (iii) $(\Phi_{\rho}^{\sharp})_{\rho}^{\sharp} = \Phi$;

(iv) if Φ and Ψ preserve ρ , then

$$(\Phi \circ \Psi)_\rho^\sharp = \Psi_\rho^\sharp \circ \Phi_\rho^\sharp.$$

Thus retrodiction reverses compositional order but is not an inverse channel in general.

This theorem gives the unique backward comparison compatible with the stationary KMS pairing. Its importance is conceptual: retrodiction is an adjoint operation, not an undoing of dissipation. The next proposition compares this choice with more familiar adjoints and explains why the symmetric modular placement is needed to retain complete positivity.

Proof. The maps $a \mapsto \rho^{1/2}a\rho^{1/2}$ and $x \mapsto \rho^{-1/2}x\rho^{-1/2}$ are completely positive congruences. The trace adjoint Φ_* of a CP map is CP. Hence the composition defining $\Phi_\rho^\sharp$ is CP. Stationarity gives

$$\Phi_\rho^\sharp(\mathbf{1}) = \rho^{-1/2}\Phi_*(\rho)\rho^{-1/2} = \mathbf{1},$$

so the dual is UCP. To verify that it preserves the same state, for every $a \in \mathcal{A}$ compute

$$\mathrm{Tr}(\rho \Phi_\rho^\sharp(a)) = \mathrm{Tr}(\Phi_*(\rho^{1/2}a\rho^{1/2})) = \mathrm{Tr}(\rho^{1/2}a\rho^{1/2}\Phi(\mathbf{1})) = \mathrm{Tr}(\rho a).$$

Thus $(\Phi_\rho^\sharp)_*(\rho) = \rho$.

For the KMS adjoint identity,

$$\begin{aligned} \langle a, \Phi(b) \rangle_{\rho, 1/2} &= \mathrm{Tr}(\rho^{1/2}a^*\rho^{1/2}\Phi(b)) \\ &= \mathrm{Tr}(\Phi_*(\rho^{1/2}a^*\rho^{1/2})b) \\ &= \mathrm{Tr}(\rho^{1/2}(\Phi_\rho^\sharp(a))^*\rho^{1/2}b), \end{aligned}$$

using $*$ -preservation of positive maps. This proves (ii). The adjoint of an adjoint for a nondegenerate inner product is the original map, giving (iii). Alternatively it follows by direct substitution and the relation between the trace adjoints. Finally,

$$(\Phi\Psi)_* = \Psi_*\Phi_*,$$

so conjugation by Γ_ρ gives (iv). □

Proposition 3.4 (Why the symmetric KMS adjoint is selected). *Besides $\Phi_\rho^\sharp$, two common alternatives are the Hilbert–Schmidt adjoint $\Phi^\dagger := \Phi_*$ and the right-GNS adjoint*

$$\Phi_\rho^G(a) := \Phi_*(a\rho)\rho^{-1}.$$

The Hilbert–Schmidt adjoint is CP and trace preserving but is unital only when Φ is bistochastic. The right-GNS adjoint satisfies

$$\Phi_\rho^G = \mathcal{M}_\rho^{-1} \circ \Phi_\rho^\sharp \circ \mathcal{M}_\rho, \quad \mathcal{M}_\rho(a) = \rho^{-1/2}a\rho^{1/2},$$

and need not preserve Hermiticity because $\mathcal{M}_\rho$ is generally not $$ -preserving. If Φ commutes with the modular group of ρ , then $\Phi_\rho^G = \Phi_\rho^\sharp$. Thus the symmetric placement of the two factors $\rho^{1/2}$ is precisely what gives an unconditional CP adjoint relative to a faithful reference state.*

Proof. The Hilbert–Schmidt assertions follow from trace duality, with unitality $\Phi_*(\mathbf{1}) = \mathbf{1}$ equivalent to trace preservation of the Heisenberg map. Let $R_\rho(a) = a\rho$. By definition $\Phi_\rho^G = R_\rho^{-1}\Phi_*R_\rho$, while $R_\rho = \Gamma_\rho\mathcal{M}_\rho$. Substitution gives the displayed similarity relation. If Φ is modular covariant, then so are Φ_* and $\Phi_\rho^\sharp$; finite-dimensional analytic continuation to the imaginary half-step $\sigma_{i/2}^\rho = \mathcal{M}_\rho$ makes the similarity trivial. $\square$

Example 3.5 (The right-GNS adjoint can fail positivity). Let

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad \tau = \begin{pmatrix} 4/5 & 0 \\ 0 & 1/5 \end{pmatrix},$$

and define the Schrödinger channel

$$T(x) = \frac{1}{2}HxH^* + \frac{1}{2}\text{Tr}(x)\tau.$$

It has faithful stationary state

$$\rho = \begin{pmatrix} 7/10 & 1/10 \\ 1/10 & 3/10 \end{pmatrix}.$$

For the Heisenberg channel $\Phi = T^*$,

$$\Phi_\rho^G(E_{00}) = T(E_{00}\rho)\rho^{-1} = \begin{pmatrix} 129/200 & 57/200 \\ 19/100 & 67/100 \end{pmatrix},$$

which is not Hermitian. Hence the right-GNS adjoint is not positive, whereas $\Phi_\rho^\sharp$ is UCP by [Theorem 3.3](#).

Definition 3.6 (KMS detailed balance and cancellation current). The channel satisfies symmetric KMS detailed balance when $\Phi = \Phi_\rho^\sharp$. Its oriented cancellation current is

$$\mathcal{J}_\rho(\Phi) := \Phi - \Phi_\rho^\sharp.$$

Proposition 3.7 (Exact cancellation decomposition). *Every stationary UCP channel has the unique decomposition*

$$\Phi = \Phi_{\text{sym}} + \Phi_{\text{asym}}, \quad \Phi_{\text{sym}} := \frac{1}{2}(\Phi + \Phi_\rho^\sharp), \quad \Phi_{\text{asym}} := \frac{1}{2}\mathcal{J}_\rho(\Phi),$$

where Φ_{sym} is KMS self-adjoint and Φ_{asym} is KMS skew-adjoint. Detailed balance is equivalent to $\mathcal{J}_\rho(\Phi) = 0$.

Proof. Apply the involution $\sharp$ of [Theorem 3.3](#) to the two terms. One obtains $\Phi_{\text{sym}}^\sharp = \Phi_{\text{sym}}$ and $\Phi_{\text{asym}}^\sharp = -\Phi_{\text{asym}}$. Their sum is Φ and the decomposition into the ± 1 eigenspaces of an involution is unique. The last assertion is immediate. $\square$

Theorem 3.8 (Composition law for the cancellation current). *Let Φ and Ψ preserve the same faithful state ρ . Then*

$$\begin{aligned} \mathcal{J}_\rho(\Phi \circ \Psi) &= \mathcal{J}_\rho(\Phi) \circ \Psi + \Phi_\rho^\sharp \circ \mathcal{J}_\rho(\Psi) + [\Phi_\rho^\sharp, \Psi_\rho^\sharp] \\ &= \Phi \circ \mathcal{J}_\rho(\Psi) + \mathcal{J}_\rho(\Phi) \circ \Psi_\rho^\sharp + [\Phi_\rho^\sharp, \Psi_\rho^\sharp]. \end{aligned}$$

Consequently the current is not an ordinary derivation on a noncommutative renewal monoid: reversal of compositional order produces the commutator term.

Proof. Since $(\Phi\Psi)_\rho^\sharp = \Psi_\rho^\sharp\Phi_\rho^\sharp$,

$$\mathcal{J}_\rho(\Phi\Psi) = \Phi\Psi - \Psi_\rho^\sharp\Phi_\rho^\sharp.$$

For the first identity, insert and subtract $\Phi_\rho^\sharp\Psi$ and $\Phi_\rho^\sharp\Psi_\rho^\sharp$; for the second, insert and subtract $\Phi\Psi_\rho^\sharp$ and $\Phi_\rho^\sharp\Psi_\rho^\sharp$. $\square$

Corollary 3.9 (Balanced factors can have an unbalanced product). *If $\Phi = \Phi_\rho^\sharp$ and $\Psi = \Psi_\rho^\sharp$, then*

$$\mathcal{J}_\rho(\Phi \circ \Psi) = [\Phi, \Psi].$$

Thus the composite of two individually KMS-detailed-balanced channels is KMS-detailed-balanced if and only if they commute.

Proof. Substitute the two vanishing individual currents into [Theorem 3.8](#). $\square$

Remark 3.10 (Anti-renewal terminology). In this paper *anti-renewal* means the KMS–Petz retrodictive dual. It is not the opposite sheet of the later signed cover, not microscopic time reversal, and not the inverse of a dissipative channel. The opposite sheet is called the *counter-oriented renewal realisation*.

At the channel level, the KMS–Petz dual is now fixed and its failure of detailed balance is measured by the cancellation current. These statements concern averaged transformations. To obtain path weights, cycle affinities, and later a renewal graph, we must descend to resolved outcomes and identify the predictive state reached after each successful branch.

3.2 Conditional predictive unifilarity and derived reference renewal

The scalar edge law needed below is not fundamentally a consequence of stationarity. It is forced when resolved outcomes are represented at the same operational resolution as their conditional predictive successors.

The key point is that scalar renewal is not assumed at the microscopic level. It will emerge because conditioning removes the probability of the past, and future equivalence makes the observed outcome determine a unique successor class. A future-tomographic realization then has no freedom to represent that successor by a different normalized state.

Definition 3.11 (Conditional future profile). Let $h : I \rightarrow B$ be a successful resolved history and put

$$p(h) := \Pr(u_B \circ h) > 0.$$

A future experiment after h is an admissible resolved continuation followed by a terminal effect, written as a single process $F : B \rightarrow I$. Define

$$\text{Pred}^{\text{cond}}(h)(F) := \frac{\Pr(F \circ h)}{p(h)}.$$

Two successful histories are *conditionally predictively equivalent*, written $h \sim_{\text{cond}} h'$, when their conditional future profiles agree for every admissible future experiment.

The normalization removes the probability that the past itself occurred. Probabilities of later resolved outcomes remain part of the profile because such outcomes may be included at the beginning of F .

Lemma 3.12 (Resolved-outcome successor congruence). *Let $h \sim_{\text{cond}} h'$ and let $e : B \rightarrow C$ be a resolved continuation admissible after both histories. If*

$$p(e \circ h) > 0, \quad p(e \circ h') > 0,$$

then

$$e \circ h \sim_{\text{cond}} e \circ h'.$$

Proof. Let $F : C \rightarrow I$ be any future experiment after e . Conditional equivalence of h and h' applied first to $F \circ e$ and then to the discard experiment $u_C \circ e$ gives

$$\frac{\Pr(F \circ e \circ h)}{p(h)} = \frac{\Pr(F \circ e \circ h')}{p(h')}, \quad \frac{p(e \circ h)}{p(h)} = \frac{p(e \circ h')}{p(h')}.$$

The second common value is positive. Dividing the first equality by the second yields

$$\frac{\Pr(F \circ e \circ h)}{p(e \circ h)} = \frac{\Pr(F \circ e \circ h')}{p(e \circ h')},$$

which is precisely conditional predictive equivalence of the extended histories. $\square$

Definition 3.13 (Conditional predictive quotient and successor map). Let

$$\mathbf{Q}_{\text{cond}} := \{\text{successful resolved histories}\} / \sim_{\text{cond}}.$$

For a resolved outcome e and a class $x = [h]$ for which $p(e \circ h) > 0$, define

$$\delta_e(x) := [e \circ h].$$

By [Theorem 3.12](#), this is a well-defined partial map. The conditional predictive presentation is therefore *outcome unifilar*: the current predictive class and the observed resolved outcome determine one successor class.

Remark 3.14 (Predictive, not microscopic, unifilarity). No unifilar hidden-state model is assumed. Many microscopic states may realize one conditional predictive class, and their hidden dynamics may be nonunifilar. Deterministic successors appear only after quotienting pasts by all conditional future predictions.

Let X be a finite reachable component of $\mathbf{Q}_{\text{cond}}$. Represent every $x \in X$ by a normalized density matrix ρ_x on its finite-dimensional support algebra $\mathcal{A}_x$, so ρ_x is faithful on $\mathcal{A}_x$. A resolved edge $e : x \rightarrow y = \delta_e(x)$ is represented in the Schrödinger picture by a CP trace-nonincreasing map

$$\mathcal{E}_e : \mathcal{A}_x \longrightarrow \mathcal{A}_y.$$

Definition 3.15 (Reduced resolved operator realization). The family $(\rho_x, \mathcal{E}_e)$ is a *reduced resolved realization* when:

- (F1) the future statistics of ρ_x equal the conditional future profile of the class x ;
- (F2) for every positive-weight edge

$$w_e := \text{Tr } \mathcal{E}_e(\rho_x) > 0,$$

the normalized branch output

$$\hat{\rho}_e := \frac{\mathcal{E}_e(\rho_x)}{w_e}$$

represents the operational history obtained by appending the resolved outcome e ;

(F3) the represented admissible future effects separate the reachable normalized states in every target algebra.

The second clause is implementation consistency; the third is the future-tomographic reducedness condition.

Theorem 3.16 (Predictive derivation of reference renewal). *In every reduced resolved operator realization and for every positive-weight edge $e : x \rightarrow y = \delta_e(x)$,*

$$\boxed{\mathcal{E}_e(\rho_x) = w_e \rho_y, \quad w_e = \text{Tr } \mathcal{E}_e(\rho_x).}$$

Thus branchwise renewal on the canonical predictive orbit is derived without assuming stationarity, detailed balance, target atomicity, or invertibility of the branch.

Proof. By implementation consistency, $\hat{\rho}_e$ represents the successful extended history $e \circ h$ for any h in the class x . By [Theorem 3.13](#), this history belongs to the unique successor class $y = \delta_e(x)$. Hence $\hat{\rho}_e$ and ρ_y have the same value on every admissible future effect. Reducedness separates reachable states, so $\hat{\rho}_e = \rho_y$. Multiplication by w_e proves the identity. $\square$

The theorem converts quotient-level unifilarity into an operator identity. Every resolved branch carries the canonical reference state of its source to a scalar multiple of the canonical reference state of its successor. Iteration therefore reduces the full branch dynamics on this orbit to a product of one-edge probabilities, as stated next.

Corollary 3.17 (Exact scalar path law). *Let $\gamma = e_n \cdots e_1$ be a successful resolved path with $x_k = \delta_{e_k}(x_{k-1})$ and $W_k := \prod_{j=1}^k w_{e_j}$. Then*

$$\boxed{\mathcal{E}_{e_n} \cdots \mathcal{E}_{e_1}(\rho_{x_0}) = W_n \rho_{x_n}.}$$

In particular, the probability of the resolved path on the canonical predictive orbit is the product of its one-edge probabilities.

Proof. Induct on the path length using $\mathcal{E}_{e_k}(\rho_{x_{k-1}}) = w_{e_k} \rho_{x_k}$. $\square$

Proposition 3.18 (Branchwise Petz anti-renewal on the predictive orbit). *Let $e : x \rightarrow y$ satisfy the reference-renewal identity with faithful support states,*

$$\mathcal{E}_e(\rho_x) = w_e \rho_y, \quad 0 < w_e \leq 1.$$

Let $\mathcal{E}_e^$ be the trace adjoint and define*

$$\mathcal{E}_e^{\leftarrow} := \Gamma_{\rho_x} \circ \mathcal{E}_e^* \circ \Gamma_{\rho_y}^{-1}, \quad \Gamma_{\rho}(a) = \rho^{1/2} a \rho^{1/2}.$$

Then $\mathcal{E}_e^{\leftarrow} : \mathcal{A}_y \rightarrow \mathcal{A}_x$ is a CP trace-nonincreasing branch,

$$\text{Tr } \mathcal{E}_e^{\leftarrow}(\rho_y) = w_e, \quad (\mathcal{E}_e^{\leftarrow})^*(\mathbf{1}_x) = w_e \mathbf{1}_y,$$

and it renews the reverse reference state,

$$\mathcal{E}_e^{\leftarrow}(\rho_y) = w_e \rho_x$$

if and only if $\mathcal{E}_e^(\mathbf{1}_y) = w_e \mathbf{1}_x$. Reversal is involutive:*

$$(\mathcal{E}_e^{\leftarrow})^{\leftarrow} = \mathcal{E}_e.$$

For composable branches $\mathcal{E}_e : x \rightarrow y$ and $\mathcal{E}_f : y \rightarrow z$,

$$(\mathcal{E}_f \mathcal{E}_e)^{\leftarrow} = \mathcal{E}_e^{\leftarrow} \mathcal{E}_f^{\leftarrow}.$$

Proof. Complete positivity follows because the three factors in the definition are CP. Taking the trace adjoint gives

$$(\mathcal{E}_e^\leftarrow)^* = \Gamma_{\rho_y}^{-1} \circ \mathcal{E}_e \circ \Gamma_{\rho_x}.$$

Hence

$$(\mathcal{E}_e^\leftarrow)^*(\mathbf{1}_x) = \rho_y^{-1/2} \mathcal{E}_e(\rho_x) \rho_y^{-1/2} = w_e \mathbf{1}_y.$$

Since $w_e \leq 1$, this is the success effect of a trace-nonincreasing CP map. Moreover,

$$\mathrm{Tr} \mathcal{E}_e^\leftarrow(\rho_y) = \mathrm{Tr}(\rho_x \mathcal{E}_e^*(\mathbf{1}_y)) = \mathrm{Tr} \mathcal{E}_e(\rho_x) = w_e.$$

Directly from the definition,

$$\mathcal{E}_e^\leftarrow(\rho_y) = \Gamma_{\rho_x} \mathcal{E}_e^*(\mathbf{1}_y) = \rho_x^{1/2} \mathcal{E}_e^*(\mathbf{1}_y) \rho_x^{1/2},$$

which equals $w_e \rho_x$ exactly when $\mathcal{E}_e^*(\mathbf{1}_y) = w_e \mathbf{1}_x$, because ρ_x is faithful. Finally, inserting $\Gamma_{\rho_y}^{-1} \Gamma_{\rho_y} = \mathrm{id}$ proves contravariant composition, and applying the construction twice cancels the two modular congruences and returns $\mathcal{E}_e$. $\square$

Corollary 3.19 (Reciprocal weights from paired Petz branches). *For every pair of opposite resolved branches $e : x \rightarrow y$ and $\bar{e} : y \rightarrow x$, define the same-orientation retrodictive weight of e to be the reference-state weight of the Petz reverse of the opposite branch:*

$$w_e^- := \mathrm{Tr}(\mathcal{E}_{\bar{e}}^\leftarrow(\rho_x)).$$

Then

$$\boxed{w_e^- = w_e^+}.$$

If the forward branches are reference renewing, their path weights are multiplicative by [Theorem 3.17](#). If, in addition, each opposite branch satisfies

$$\mathcal{E}_{\bar{e}}^*(\mathbf{1}_x) = w_e^+ \mathbf{1}_y,$$

then $\mathcal{E}_{\bar{e}}^\leftarrow(\rho_x) = w_e^+ \rho_y$, so the forward branches and the reindexed Petz branches form a ray-homogeneous bi-instrument with $p_e^- = p_e^+$. Without this additional reverse-renewal condition, the Petz flux still supplies the reciprocal scalar weight, but not a state-independent retrodictive path law.

Proof. The equality is the reference-state weight identity of [Theorem 3.18](#) applied to the opposite branch. $\square$

Remark 3.20 (The exact operational boundary). A primitive stationary averaged channel does not force branchwise reference renewal. The decisive datum is that the target of a resolved branch is the actual conditional predictive successor represented in a reduced future-tomographic realization. If outcomes are forgotten, distinct successors are replaced by their mixture; if target labels are coarser than conditional predictive classes, the graph must first be refined.

The preceding results also locate the exact boundary of the scalar description. It is valid when target labels already coincide with conditional predictive successors; an averaged or under-resolved presentation can merge distinct future states. The following construction repairs only this labelling defect, without inventing any unobserved outcome decomposition.

Construction 3.21 (Canonical predictive successor refinement). Suppose a resolved instrument is presented on a graph whose target labels may be coarser than the conditional predictive quotient. For every reachable pair (x, e) with

$$w_e(x) := \text{Tr } \mathcal{E}_e(\rho_x) > 0,$$

form the normalized branch output

$$\hat{\rho}_{x,e} := \frac{\mathcal{E}_e(\rho_x)}{\text{Tr } \mathcal{E}_e(\rho_x)}.$$

Declare

$$(x, e) \approx_{\text{succ}} (x', e')$$

when the two normalized outputs have identical conditional future profiles. The refined target vertices are the $\approx_{\text{succ}}$ -classes of reachable pairs, and the branch (x, e) is redirected to its own class.

Proposition 3.22 (Minimality of successor refinement). *In a reduced future-tomographic realization, [Theorem 3.21](#) is the coarsest relabelling of the already resolved branches for which every positive-weight branch is reference renewing. It is finite whenever the reachable conditional predictive rank is finite.*

Proof. All normalized outputs assigned to one refined vertex have the same conditional future profile. Reduced future tomography therefore represents them by one and the same normalized target state. Consequently every redirected branch satisfies

$$\mathcal{E}_e(\rho_x) = w_e(x) \rho_{[(x,e)]},$$

so the refined graph is reference renewing.

Conversely, let r be any relabelling for which each positive-weight branch is reference renewing in a reduced realization. If $r(x, e) = r(x', e')$, then the two normalized outputs equal the same target reference state and hence have identical conditional future profiles. Thus $(x, e) \approx_{\text{succ}} (x', e')$. Every such relabelling therefore factors uniquely through the quotient of [Theorem 3.21](#), proving coarseness and minimality. Finiteness is immediate when only finitely many conditional predictive successor classes are reachable. $\square$

Remark 3.23 (Relabelling is not hidden outcome splitting). The construction refines labels of outcomes that are already operationally resolved. If only an averaged CP map is available, decomposing it into finer CP branches is additional instrument data and is not canonically determined by the channel. The refinement corrects under-resolved targets; it does not infer an unobserved Kraus decomposition.

Canonical refinement gives an exact reference-renewing graph whenever the conditional predictive quotient is available. For empirical or approximate models, exact equality of all future statistics is too strong. The next theorem replaces it with a finite separating family and quantifies how local tomographic errors accumulate along a path.

Exact future tomography may be weakened to a finite quantitative separating family.

Definition 3.24 (Tomographic frame constant). Let $\mathcal{Q}_y$ be a finite family of effects on the target support algebra. A number $\kappa_y < \infty$ is a *tomographic frame constant* when

$$\|\tau - \sigma\|_1 \leq \kappa_y \max_{Q \in \mathcal{Q}_y} |\text{Tr}((\tau - \sigma)Q)|$$

for all reachable normalized states τ, σ at y . Such a constant exists whenever $\mathcal{Q}_y$ separates the finite-dimensional affine span of the reachable state set.

Theorem 3.25 (Quantitative approximate reference renewal). *Let $e : x \rightarrow y$ have $w_e > 0$, and suppose its normalized output $\hat{\rho}_e = \mathcal{E}_e(\rho_x)/w_e$ obeys*

$$\max_{Q \in \mathcal{Q}_y} |\text{Tr}((\hat{\rho}_e - \rho_y)Q)| \leq \varepsilon_e.$$

Then

$$\|\mathcal{E}_e(\rho_x) - w_e \rho_y\|_1 \leq w_e \kappa_y \varepsilon_e.$$

For a path $\gamma = e_n \cdots e_1$, with $W_k = \prod_{j=1}^k w_{e_j}$ and $W_0 = 1$,

$$\|\mathcal{E}_{e_n} \cdots \mathcal{E}_{e_1}(\rho_{x_0}) - W_n \rho_{x_n}\|_1 \leq \sum_{k=1}^n W_k \kappa_{x_k} \varepsilon_{e_k}.$$

Consequently approximate conditional predictive sufficiency is stable along every fixed finite history.

Proof. The frame inequality gives $\|\hat{\rho}_e - \rho_y\|_1 \leq \kappa_y \varepsilon_e$; multiplication by w_e proves the one-edge estimate.

For the path estimate set

$$S_k := \mathcal{E}_{e_k} \cdots \mathcal{E}_{e_1}(\rho_{x_0}), \quad D_k := \|S_k - W_k \rho_{x_k}\|_1.$$

A positive trace-nonincreasing map is contractive on Hermitian inputs in trace norm. Indeed, if $a = a_+ - a_-$ is the Jordan decomposition, then

$$\|\mathcal{E}(a)\|_1 \leq \text{Tr } \mathcal{E}(a_+) + \text{Tr } \mathcal{E}(a_-) \leq \text{Tr } a_+ + \text{Tr } a_- = \|a\|_1.$$

Therefore

$$\begin{aligned} D_k &\leq \|\mathcal{E}_{e_k}(S_{k-1} - W_{k-1} \rho_{x_{k-1}})\|_1 \\ &\quad + W_{k-1} \|\mathcal{E}_{e_k}(\rho_{x_{k-1}}) - w_{e_k} \rho_{x_k}\|_1 \\ &\leq D_{k-1} + W_k \kappa_{x_k} \varepsilon_{e_k}. \end{aligned}$$

Iteration from $D_0 = 0$ gives the result. $\square$

Remark 3.26 (Rare resolved outcomes). The normalized conditional error is independent of w_e , whereas the subnormalized branch error is multiplied by w_e . A rare outcome can therefore have a poorly estimated conditional state without strongly changing the unconditioned path law. Relative statements conditioned on that rare outcome require separate statistical control.

We have thus derived both the exact scalar path law and its finite-error version from conditional prediction. The matrix formulation is the operational source of the construction, but later cohomological arguments use only the resulting one-dimensional action on reference rays. The next subsection records that minimal abstract structure.

3.3 Abstract ray-homogeneous formulation

[Theorem 3.16](#) derives ray homogeneity on the canonical predictive reference orbit. The following ordered-space formulation is useful when the forward and retrodictive branch families are specified abstractly. In matrix realizations the retrodictive branches may be taken to be the branchwise Petz maps of [Theorem 3.18](#), reindexed in the same path orientation.

This abstraction deliberately discards all information not needed for path weights. It preserves the source and target rays, the positive edge scalars, and the distinction between forward and retrodictive instruments. These are exactly the data from which affinity will be built.

A generic resolved quantum instrument does not assign state-independent numbers to arbitrary microscopic inputs. The reference-renewal theorem states that the necessary one-dimensional closure nevertheless holds on the reduced conditional predictive orbit. Abstractly, let G be a finite bidirected graph. At each vertex x , let X_x be an ordered real vector space with a fixed nonzero positive reference vector η_x .

Definition 3.27 (Ray-homogeneous bi-instrument). A ray-homogeneous bi-instrument consists of positive resolved branch maps

$$F_e, R_e : X_{s(e)} \longrightarrow X_{t(e)}$$

and scalars $p_e^+, p_e^- > 0$ such that

$$F_e \eta_{s(e)} = p_e^+ \eta_{t(e)}, \quad R_e \eta_{s(e)} = p_e^- \eta_{t(e)}.$$

The family F is the renewal instrument. The family R is the retrodictive instrument reindexed in the same path orientation. When these are probabilistic branches, their outgoing sums are required to be normalization preserving.

Proposition 3.28 (Multiplicative path weights). *For a path $\gamma = e_n \cdots e_1$, put $F_\gamma = F_{e_n} \cdots F_{e_1}$ and similarly for R_γ . Then*

$$F_\gamma \eta_{s(\gamma)} = W_+(\gamma) \eta_{t(\gamma)}, \quad R_\gamma \eta_{s(\gamma)} = W_-(\gamma) \eta_{t(\gamma)},$$

where

$$W_\pm(\gamma) = \prod_{k=1}^n p_{e_k}^\pm.$$

In particular the path weights are multiplicative under concatenation.

Proof. Induct on path length. If the claim holds through $e_{k-1} \cdots e_1$, then

$$F_{e_k} \left(\prod_{j < k} p_{e_j}^+ \eta_{s(e_j)} \right) = \left(\prod_{j \leq k} p_{e_j}^+ \right) \eta_{t(e_k)}.$$

The same argument applies to R . Splitting the product at a concatenation point gives multiplicativity. $\square$

Multiplicativity is the first payoff of ray homogeneity: a resolved history is assigned a scalar weight by multiplying local branch factors. Before those weights can be attached to predictive path classes, however, one must verify that they are unchanged by the path equivalences used in the quotient.

Theorem 3.29 (Descent of operational path weights). *Let $\equiv$ be any path congruence. The functions $W_\pm$ descend uniquely to the quotient path category if and only if they are constant on $\equiv$ -classes. A sufficient condition is*

$$\gamma \equiv \gamma' \implies F_\gamma \eta_{s(\gamma)} = F_{\gamma'} \eta_{s(\gamma)}, \quad R_\gamma \eta_{s(\gamma)} = R_{\gamma'} \eta_{s(\gamma)}.$$

In particular, equality of accumulated branch transformations is sufficient.

Proof. A scalar function factors through a quotient exactly when it is constant on quotient classes. Under the displayed hypothesis, the two images are equal nonzero multiples of the same terminal reference vector, so their scalar coefficients coincide. Equality of the full accumulated maps implies equality on the reference ray. $\square$

Proposition 3.30 (Kraus invariance and outcome dependence). *Let a resolved quantum branch be represented by the CP map*

$$\mathcal{F}_e(\rho) = \sum_a K_{e,a} \rho K_{e,a}^*,$$

and suppose that on the predictive reference orbit

$$\mathcal{F}_e(\rho_x) = p_e \rho_y, \quad p_e > 0.$$

Then $p_e = \text{Tr } \mathcal{F}_e(\rho_x)$ depends only on the CP branch map, not on its Kraus representation. Hence unitary or isometric Kraus mixing within a fixed resolved outcome leaves the scalar path weights and the resulting affinity unchanged. By contrast, splitting, merging, or physically relabelling outcomes changes the branch maps and possibly the resolved graph; no invariance under those operations follows without a separate coarse-graining theorem.

Proof. Two Kraus families related by an isometric change of Kraus section define the same completely positive map. Since ρ_y is normalized,

$$p_e = \text{Tr}(p_e \rho_y) = \text{Tr } \mathcal{F}_e(\rho_x),$$

which is determined solely by $\mathcal{F}_e$ and the fixed reference input. Therefore every product of such coefficients along a resolved path, and every logarithmic ratio formed from them, is Kraus-representation invariant. Outcome splitting replaces one CP branch by several CP summands, while merging replaces several branches by their sum. These operations alter the instrument rather than merely its Kraus section, so the preceding invariance argument does not apply. $\square$

Example 3.31 (A primitive averaged channel need not renew branchwise). Let $\rho_* = \mathbf{1}_2/2$ and

$$\sigma_0 = \begin{pmatrix} 3/4 & 0 \\ 0 & 1/4 \end{pmatrix}, \quad \sigma_1 = \begin{pmatrix} 1/4 & 0 \\ 0 & 3/4 \end{pmatrix}.$$

Define Schrödinger CP subchannels

$$\mathcal{E}_i(x) = \frac{1}{2} \text{Tr}(x) \sigma_i, \quad i = 0, 1.$$

Their sum is the one-step primitive replacement channel $x \mapsto \text{Tr}(x) \rho_*$, but $\mathcal{E}_i(\rho_*) = \sigma_i/2$ is not proportional to ρ_* . Thus stationarity and primitivity of the averaged channel do not imply ray-homogeneity of its resolved branches.

Remark 3.32 (Scope of the scalar reduction). For a generic instrument acting on arbitrary inputs, the probability of the next outcome depends on the entire conditional state left by preceding branches. Path probabilities then form a cocycle over the conditional-state bundle rather than a scalar one-cochain on G . The scalar reduction is exact on the canonical predictive orbit because [Theorem 3.16](#) returns every resolved successor to its uniquely represented predictive state. Outside that orbit, or when target labels are under-resolved, the state-bundle description remains necessary.

The abstract formulation therefore identifies both the strength and the scope of the scalar reduction. It is invariant under changes of Kraus representation for a fixed resolved branch, but not under physical changes of outcome resolution, and it need not hold away from the canonical predictive orbit. We next fix an important convention: which forward/backward path comparison actually measures irreversibility.

3.4 Classical Petz path reversal and the affinity convention

The canonical retrodictive chain is constructed to reproduce stationary forward experiments after reversing the path. Consequently that particular forward/reverse ratio cannot measure irreversibility.

The classical calculation makes this point transparently. Petz reversal is constructed so that a stationary forward path and the oppositely traversed Petz path have identical total weight. A nonzero affinity must therefore compare the forward and retrodictive laws after placing them in the same orientation.

Proposition 3.33 (Stationary Petz path reversal is exact). *Let $P = (p_{ij})$ be a finite row-stochastic matrix with faithful stationary law π , and let*

$$p_{ij}^\# := \frac{\pi_j p_{ji}}{\pi_i}$$

be its stationary reverse kernel, the commutative specialization of the KMS–Petz dual [31, 32]. For every path $\gamma = (i_0, i_1, \dots, i_m)$,

$$\boxed{\pi_{i_0} \prod_{k=1}^m p_{i_{k-1} i_k} = \pi_{i_m} \prod_{k=1}^m p_{i_k i_{k-1}}^\# .}$$

Therefore the logarithm of stationary forward-path weight divided by the weight of the reversed path under the Petz chain is identically zero, whether or not P satisfies detailed balance.

Proof. For every step,

$$p_{i_k i_{k-1}}^\# = \frac{\pi_{i_{k-1}} p_{i_{k-1} i_k}}{\pi_{i_k}} .$$

Multiplication over k telescopes all intermediate stationary weights:

$$\pi_{i_m} \prod_{k=1}^m p_{i_k i_{k-1}}^\# = \pi_{i_m} \prod_{k=1}^m \frac{\pi_{i_{k-1}} p_{i_{k-1} i_k}}{\pi_{i_k}} = \pi_{i_0} \prod_{k=1}^m p_{i_{k-1} i_k} .$$

□

Remark 3.34 (Why affinity uses the same orientation). The identity in [Theorem 3.33](#) is adjointness, not detailed balance. A genuine cycle obstruction must instead compare forward and retrodictive branch laws after both have been indexed in the same path orientation, or compare the two opposite directions under the same forward law. This is exactly the convention used in [Theorems 3.19](#) and [3.35](#).

With the convention fixed, the next subsection extracts the genuine continuous obstruction to reciprocity. It depends only on ratios of opposite oriented edge weights and is insensitive to how the record frames themselves are oriented.

3.5 Real affinity from forward and backward branch weights

Assume henceforth that the selected resolved branch law is ray homogeneous and instrument compatible in the sense of [Theorems 3.27](#) and [3.29](#). Let $G = (V, E_G)$ be a finite connected bidirected graph, with reverse edge $\bar{e} : y \rightarrow x$ for every $e : x \rightarrow y$, and let $w_e > 0$ denote the resulting positive scalar capacity of the oriented branch. In a classical model this may be a stationary edge flux; in a quantum reset instrument it is the scalar by which the branch carries the selected reference ray to its conditional successor ray.

Taking logarithms turns multiplicative path ratios into an additive one-cochain. Vertex rescalings of the reference rays change this cochain by a coboundary, so only its cycle class is physically invariant.

Definition 3.35 (Edge affinity). The edge affinity is the antisymmetric one-cochain

$$A(e) := \log \frac{w_e}{w_{\bar{e}}}, \quad A(\bar{e}) = -A(e).$$

A vertex gauge $f : V \rightarrow \mathbb{R}$ acts by

$$A^f(e : x \rightarrow y) = A(e) + f(x) - f(y).$$

The gauge-invariant class is

$$[A] \in H^1(G; \mathbb{R}).$$

Proposition 3.36 (Cycle affinity and detailed balance). *For every oriented cycle $c = e_k \cdots e_1$,*

$$\oint_c A := \sum_{j=1}^k A(e_j) = \log \frac{\prod_j w_{e_j}}{\prod_j w_{\bar{e}_j}}.$$

The following are equivalent:

- (i) $[A] = 0$ in $H^1(G; \mathbb{R})$;
- (ii) every cycle affinity vanishes;
- (iii) there is a positive vertex function π such that $\pi(x)w_e = \pi(y)w_{\bar{e}}$ for every $e : x \rightarrow y$.

Proof. The cycle identity follows by summing logarithms. If $[A] = 0$, then for some vertex function f one has $A(e : x \rightarrow y) = f(y) - f(x)$. Let $\pi(x) = e^{f(x)}$; then

$$\log \frac{\pi(x)w_e}{\pi(y)w_{\bar{e}}} = f(x) + A(e) - f(y) = 0.$$

Thus (i) implies (iii), and (iii) implies vanishing of every cycle product by telescoping. If every cycle integral vanishes, fix a base vertex x_0 and define $f(x)$ as the integral of A along any path from x_0 to x . Cycle vanishing makes this path independent, and then $A(e) = f(y) - f(x)$, so $[A] = 0$. $\square$

Theorem 3.37 (Unique affinity–conductance factorization). *Let $q_e > 0$ be positive oriented edge capacities on a bidirected graph. Set*

$$c_e := \sqrt{q_e q_{\bar{e}}}, \quad A_q(e) := \log \frac{q_e}{q_{\bar{e}}}.$$

Then $c_{\bar{e}} = c_e$, $A_q(\bar{e}) = -A_q(e)$, and

$$q_e = c_e e^{A_q(e)/2}.$$

This factorization is unique among a symmetric positive capacity c and an antisymmetric real one-cochain. In particular, when $q_e = w_e$ is the resolved branch capacity of [Theorem 3.35](#), one has $A_q = A$.

Proof. The symmetry and antisymmetry are immediate. Moreover

$$\sqrt{q_e q_{\bar{e}}} \exp\left(\frac{1}{2} \log \frac{q_e}{q_{\bar{e}}}\right) = q_e.$$

If $q_e = \tilde{c}_e e^{\tilde{A}(e)/2}$ with $\tilde{c}_e = \tilde{c}_{\bar{e}} > 0$ and $\tilde{A}(\bar{e}) = -\tilde{A}(e)$, then $q_e q_{\bar{e}} = \tilde{c}_e^2$ and $q_e/q_{\bar{e}} = e^{\tilde{A}(e)}$. Positivity and injectivity of the real exponential give $\tilde{c} = c$ and $\tilde{A} = A_q$. $\square$

Remark 3.38 (Affinity does not determine capacity). The cohomology class $[A]$ fixes only the oriented ratio of the two edge capacities. The symmetric conductance c_e is an independent unoriented renewal-capacity datum. The sign class introduced below does not enter the positive factorization.

The affinity class measures nonreciprocal weight. It is continuous and real. The Lorentzian sign class has a different origin.

Affinity has now been decomposed into an antisymmetric log ratio and an independent symmetric conductance. Neither datum supplies a sign-changing line or a two-sheeted cover. To obtain orientation, the theory needs stable resolved records and a rule for transporting their contrasts. The next subsection first explains how such records can arise canonically from specified microscopic interactions.

3.6 Microscopic extraction of a stable resolved instrument

The record quotient below begins once a finite raw outcome family is operationally resolved. A channel by itself does not determine such an outcome decomposition. This subsection gives a conditional microscopic bridge: a specified system–environment interaction, together with an interaction-complete monitoring phase, canonically determines a stable classical pointer algebra and hence a resolved instrument. The construction is finite dimensional and basis independent.

This is a conditional bridge rather than a reconstruction from the averaged channel alone. The interaction family determines which environment observables are repeatedly monitored; their commutant contains the nondemolished information, and the centre of that commutant is the classical part that can label stable outcomes.

Let

$$V : H_{\text{in}} \longrightarrow H_{\text{out}} \otimes E$$

be a minimal Stinespring isometry for a channel

$$\Phi_*(\rho) = \text{Tr}_E(V\rho V^*).$$

Let P be a finite-dimensional probe interface and let

$$\mathcal{H} = \{H_1, \dots, H_q\}, \quad H_\nu = H_\nu^* \in \mathcal{B}(P \otimes E),$$

be a specified family of microscopic probe–environment interactions.

Definition 3.39 (Environment coefficient algebra). Define

$$\mathcal{C}_E(\mathcal{H}) := C^*(\{(\omega \otimes \text{id}_E)(H_\nu) : \omega \in \mathcal{B}(P)_*, 1 \leq \nu \leq q\}) \subseteq \mathcal{B}(E).$$

Its commutant

$$\mathcal{F}_E := \mathcal{C}_E(\mathcal{H})'$$

is called the *nondemolition stable algebra*, and

$$\mathcal{Z}_E := Z(\mathcal{F}_E)$$

is its stable classical pointer algebra.

Proposition 3.40 (Minimality and nondemolition characterization). *The algebra $\mathcal{C}_E(\mathcal{H})$ is the unique smallest unital C^* -subalgebra $\mathcal{C} \subseteq \mathcal{B}(E)$ satisfying*

$$H_\nu \in \mathcal{B}(P) \otimes \mathcal{C} \quad (1 \leq \nu \leq q).$$

In particular it is independent of every probe basis, operator-Schmidt decomposition, and generating list. Moreover, for $X \in \mathcal{B}(E)$,

$$\boxed{X \in \mathcal{F}_E \iff [\mathbf{1}_P \otimes X, H_\nu] = 0 \text{ for every } \nu.}$$

Proof. Choose matrix units (e_{ab}) of $\mathcal{B}(P)$ and write

$$H_\nu = \sum_{a,b} e_{ab} \otimes h_{ab}^{(\nu)}.$$

Each coefficient is a slice, $h_{ab}^{(\nu)} = (\omega_{ba} \otimes \text{id}_E)(H_\nu)$, so it belongs to $\mathcal{C}_E(\mathcal{H})$, and therefore $H_\nu \in \mathcal{B}(P) \otimes \mathcal{C}_E(\mathcal{H})$. Conversely, if $H_\nu \in \mathcal{B}(P) \otimes \mathcal{C}$, every slice of H_ν belongs to $\mathcal{C}$. Hence $\mathcal{C}_E(\mathcal{H}) \subseteq \mathcal{C}$, proving minimality and basis independence.

Finally,

$$[\mathbf{1}_P \otimes X, H_\nu] = \sum_{a,b} e_{ab} \otimes [X, h_{ab}^{(\nu)}].$$

This vanishes for every ν exactly when X commutes with every slice coefficient, equivalently with the C^* -algebra generated by them. $\square$

Equip $\mathcal{B}(E)$ with the normalized Hilbert–Schmidt inner product

$$\langle X, Y \rangle_2 := \tau_E(X^*Y), \quad \tau_E := \frac{1}{\dim E} \text{Tr}.$$

Let $\mathcal{C}_{0,\text{sa}}$ be the orthogonal complement of $\mathbb{R}\mathbf{1}_E$ in $\mathcal{C}_E(\mathcal{H})_{\text{sa}}$, and choose any real orthonormal basis $A_1, \dots, A_m$ of that space.

Theorem 3.41 (Interaction-complete stable-pointer selection). *Define*

$$\mathcal{L}_{\mathcal{H}}(X) := -\frac{1}{2} \sum_{j=1}^m [A_j, [A_j, X]].$$

Then:

- (i) $\mathcal{L}_{\mathcal{H}}$ is independent of the chosen real orthonormal basis;
- (ii) $D_t := e^{t\mathcal{L}_{\mathcal{H}}}$ is a norm-continuous semigroup of unital, trace-preserving, Hilbert–Schmidt self-adjoint completely positive maps;
- (iii) its fixed algebra is exactly

$$\text{Fix}(D) = \mathcal{F}_E = \mathcal{C}_E(\mathcal{H})';$$

- (iv) as $t \rightarrow \infty$, D_t converges in operator norm to the Hilbert–Schmidt orthogonal projection

$$\Delta_{\mathcal{H}} : \mathcal{B}(E) \longrightarrow \mathcal{F}_E;$$

this limit is the unique trace-preserving conditional expectation onto $\mathcal{F}_E$.

Proof. If $B_k = \sum_j O_{kj} A_j$ is another real orthonormal basis, then O is orthogonal and

$$\sum_k [B_k, [B_k, X]] = \sum_{i,j} \left(\sum_k O_{ki} O_{kj} \right) [A_i, [A_j, X]] = \sum_i [A_i, [A_i, X]],$$

proving (i).

For each self-adjoint A_j , the map $-\frac{1}{2}[A_j, [A_j, \cdot]]$ is the bounded Lindblad generator

$$A_j X A_j - \frac{1}{2}(A_j^2 X + X A_j^2).$$

Their sum therefore generates a norm-continuous UCP semigroup. Each summand annihilates $\mathbf{1}_E$, and Hilbert–Schmidt cyclicity makes the generator self-adjoint. Trace preservation follows either from the Lindblad form or from $\text{Tr}[A_j, Y] = 0$, proving (ii).

For every X ,

$$-\langle X, \mathcal{L}_{\mathcal{H}} X \rangle_2 = \frac{1}{2} \sum_j \|[A_j, X]\|_2^2.$$

Hence $\mathcal{L}_{\mathcal{H}} X = 0$ if and only if $[A_j, X] = 0$ for every j . Since the A_j , together with the identity, linearly span the self-adjoint part of $\mathcal{C}_E(\mathcal{H})$, this is equivalent to $X \in \mathcal{C}_E(\mathcal{H})'$, proving (iii).

The generator is self-adjoint and negative semidefinite on the finite Hilbert space $\mathcal{B}(E)$. Its spectrum is therefore real and nonpositive, and zero is isolated from the remaining finite spectrum. The spectral theorem gives norm convergence of $e^{t\mathcal{L}_{\mathcal{H}}}$ to the orthogonal projection onto its kernel, namely $\Delta_{\mathcal{H}}$. A norm limit of UCP maps is UCP, and the limit is unital, trace preserving, idempotent, and has range $\mathcal{F}_E$. For $A, B \in \mathcal{F}_E$, every D_t is an $\mathcal{F}_E$ -bimodule map because the A_j commute with A, B ; passing to the limit gives

$$\Delta_{\mathcal{H}}(AXB) = A\Delta_{\mathcal{H}}(X)B.$$

Thus $\Delta_{\mathcal{H}}$ is a conditional expectation. If E_1, E_2 are trace-preserving conditional expectations onto the same finite-dimensional subalgebra, then for $Y \in \mathcal{F}_E$,

$$\langle Y, X - E_i X \rangle_2 = \tau_E(Y^* X) - \tau_E(E_i(Y^* X)) = 0,$$

using trace preservation and the bimodule property. Both are therefore the orthogonal projection onto $\mathcal{F}_E$, proving uniqueness. $\square$

Theorem 3.42 (Canonical pointer instrument). *Let $z_1, \dots, z_r$ be the minimal central projections of $\mathcal{F}_E$. They are uniquely determined as an unordered family. Define CP trace-nonincreasing maps*

$$\mathcal{I}_\alpha(\rho) := \text{Tr}_E[(\mathbf{1} \otimes z_\alpha)V\rho V^*(\mathbf{1} \otimes z_\alpha)].$$

Then

$$\sum_{\alpha=1}^r \mathcal{I}_\alpha = \Phi_*,$$

so $\{\mathcal{I}_\alpha\}$ is a basis-independent resolved instrument. Its raw classical outcome algebra is canonically

$$\mathcal{Z}_E \cong C(\{1, \dots, r\}; \mathbb{C}).$$

Applying conditional future equivalence to these pointer outcomes produces the unique reduced future-distinct record algebra of Theorem 3.45. Reversible monodromy of the resulting record sectors therefore feeds canonically into the determinant-line construction.

Proof. The minimal central projections of a finite-dimensional C^* -algebra are intrinsic and unique up to permutation. Each $\mathcal{I}_\alpha$ is CP, being the composition of the isometric dilation, compression by a projection, and partial trace. Its trace is at most $\text{Tr} \rho$, so it is trace-nonincreasing.

Since the z_α are pairwise orthogonal and sum to $\mathbf{1}_E$,

$$V\rho V^* = \sum_{\alpha, \beta} (\mathbf{1} \otimes z_\alpha)V\rho V^*(\mathbf{1} \otimes z_\beta).$$

For $\alpha \neq \beta$, the partial trace of the off-diagonal block vanishes: in an orthonormal basis adapted to $E = \bigoplus_\alpha z_\alpha E$, every diagonal matrix coefficient of that block is zero. Therefore

$$\Phi_*(\rho) = \text{Tr}_E(V\rho V^*) = \sum_\alpha \mathcal{I}_\alpha(\rho).$$

The centre is the span of the z_α , hence is canonically the algebra of functions on their unordered spectrum. The final assertions follow by applying Theorems 3.44 and 3.45 and then the determinant transport results below. $\square$

Remark 3.43 (Sharp boundary of pointer extraction). The construction is canonical relative to the specified microscopic interaction family and the interaction-complete monitoring phase. It is not canonical relative to the averaged system channel alone. Different interaction algebras may have different commutants and hence different stable pointer centres, even when the reduced channel agrees. Pure Hamiltonian motion also need not converge to $\Delta_{\mathcal{H}}$; actual monitoring, phase randomization, or an equivalent irreversible coarse graining is load-bearing.

The stable-pointer theorem and canonical instrument theorem supply physically resolved raw outcomes, but they need not yet be minimal for prediction. Some pointer labels may have identical conditional futures. The next subsection quotients precisely that redundancy and constructs the canonical algebra on which record orientation is defined.

3.7 Canonical predictive record algebra

The determinant construction should not begin from arbitrarily named record coordinates. Once a finite family of raw outcomes is physically resolved, its minimal future-distinct algebra is fixed by prediction.

The construction mirrors the rooted predictive quotient of [Section 2](#): raw record names are irrelevant, and only distinctions that can affect future experiments are retained. This prevents the later determinant sign from depending on an arbitrary over-refinement of the outcome alphabet.

Definition 3.44 (Conditional future equivalence of records). At a predictive vertex x , let $\mathcal{O}_x$ be a finite set of physically resolved raw outcome records. For every admissible future experiment F , let $p_F^x(o) \in [0, 1]$ be its conditional probability after observing $o \in \mathcal{O}_x$. Set

$$o \equiv_x o' \iff p_F^x(o) = p_F^x(o') \text{ for every admissible } F,$$

and write $\Omega_x := \mathcal{O}_x / \equiv_x$ with quotient map q_x .

Define the real unital algebra generated by conditional future predictions,

$$\mathcal{R}_x^{\text{pred}} := \text{alg}_{\mathbb{R}}(\mathbf{1}, \{p_F^x : F \text{ admissible}\}) \subseteq C(\mathcal{O}_x; \mathbb{R}).$$

Theorem 3.45 (Canonical future-distinct record algebra). *For every finite resolved record set,*

$$\boxed{\mathcal{R}_x^{\text{pred}} = q_x^* C(\Omega_x; \mathbb{R}) = \{f \in C(\mathcal{O}_x; \mathbb{R}) : o \equiv_x o' \Rightarrow f(o) = f(o')\}.$$

Consequently $C(\Omega_x; \mathbb{R})$ is the unique reduced minimal finite commutative record algebra through which all conditional future predictions factor.

Proof. Every generator p_F^x is constant on $\equiv_x$ -classes, giving one inclusion. For the reverse inclusion, fix a class $C \in \Omega_x$. For every $D \neq C$, some future experiment separates C and D . Affinely rescale its classwise prediction function to obtain $g_D \in \mathcal{R}_x^{\text{pred}}$ with $g_D|_C = 1$ and $g_D|_D = 0$. Then

$$e_C := \prod_{D \neq C} g_D$$

is the indicator of C . The class indicators span $C(\Omega_x; \mathbb{R})$, proving the algebra identity.

If $r : \mathcal{O}_x \rightarrow Y$ is any future-sufficient realization, equality $r(o) = r(o')$ forces $o \equiv_x o'$, so q_x factors uniquely through $r(\mathcal{O}_x)$. If r is reduced, this factor is bijective. This is the claimed minimality and uniqueness. $\square$

Proposition 3.46 (Descent of reversible record transport). *Suppose a reversible predictive process $u : x \rightarrow y$ transports raw records by a bijection $\tau_u : \mathcal{O}_x \rightarrow \mathcal{O}_y$ and pulls future experiments back so that*

$$p_F^y(\tau_u(o)) = p_{u^*F}^x(o).$$

Then τ_u descends uniquely to a bijection $\bar{\tau}_u : \Omega_x \rightarrow \Omega_y$. Pullback gives a unital order isomorphism of the canonical record algebras, and composition and inversion descend functorially.

Proof. If $o \equiv_x o'$, then for every F at y ,

$$p_F^y(\tau_u(o)) = p_{u^*F}^x(o) = p_{u^*F}^x(o') = p_F^y(\tau_u(o')),$$

so $\tau_u(o) \equiv_y \tau_u(o')$. The same argument for u^{-1} proves that the descended map is bijective. Functoriality follows from uniqueness of descent. $\square$

Theorem 3.47 (Uniqueness of resolved-record parity). *Let $k \geq 2$. Every group homomorphism*

$$\zeta : S_k \longrightarrow \{\pm 1\}$$

is either the trivial character or the permutation sign. Consequently, for reversible relabellings of k future-distinct records, determinant parity is the unique nontrivial one-dimensional real functorial invariant.

Proof. All transpositions in S_k are conjugate, so a homomorphism to the abelian group $\{\pm 1\}$ takes one common value on every transposition. Transpositions generate S_k . If their common value is $+1$, the homomorphism is trivial. If it is -1 , the value on any permutation is $(-1)^m$, where m is the number of transpositions in a decomposition. This is well defined and equals the usual sign character. The determinant of the permutation representation is exactly that character; the invariant constant line has determinant $+1$, so the same character acts on the contrast module. $\square$

For a finite set Ω_x , its canonical contrast module is

$$R_x^0 := \left\{ f \in C(\Omega_x; \mathbb{R}) : \sum_{\omega \in \Omega_x} f(\omega) = 0 \right\}.$$

Every descended record bijection preserves R_x^0 . Thus the determinant line of the contrast module is obtained from the conditional predictive quotient, not from a choice of record names.

Remark 3.48 (The remaining instrument boundary). The theorem reconstructs the minimal record algebra after raw outcomes have been physically resolved. An averaged CP channel does not in general select a preferred outcome decomposition. The remaining input is therefore the operational availability of resolved outcomes, not the choice of their future-sufficient quotient.

For reversible relabellings, the canonical record quotient carries one unique nontrivial real one-dimensional invariant: permutation parity on the contrast module. Real records, however, may be transported noisily rather than by exact permutations. We therefore extend the construction to every positive mass-preserving correspondence that remains invertible on future-distinct contrasts.

3.8 Determinant orientation for noisy record correspondences

Reversible permutations are not the largest class carrying record orientation. A noisy channel can blur record values while still preserving the dimension and orientation of their contrast space. The determinant detects precisely this coarser topological information: its magnitude may contract, but its sign remains defined until a contrast direction is actually lost. Let

$$M_x := \mathbb{R}^{\Omega_x}, \quad \mathbf{u}_x(m) := \sum_{\omega \in \Omega_x} m_\omega, \quad M_x^0 := \ker \mathbf{u}_x.$$

A positive mass-preserving record correspondence along a chosen orientation of a geometric edge is a linear map

$$K_e : M_x \longrightarrow M_y, \quad \mathbf{u}_y K_e = \mathbf{u}_x.$$

It restricts to

$$Q_e := K_e|_{M_x^0} : M_x^0 \longrightarrow M_y^0.$$

Definition 3.49 (Determinant-regular noisy record edge). A noisy record edge is *determinant regular* when $|\Omega_x| = |\Omega_y|$ and Q_e is a linear isomorphism. The reverse transport of the associated real local system is defined algebraically as Q_e^{-1} ; it is not required to be positive or to coincide with a physical reverse stochastic channel.

Theorem 3.50 (Signed cover from determinant-regular noisy records). *Let G be a connected graph of constant finite record rank, choose one orientation of each geometric edge, and assign a determinant-regular positive mass-preserving record correspondence K_e to that orientation. Then the lines*

$$\mathcal{D}_x := \det(M_x^0)$$

with transports $\det Q_e$ form a real line local system. In local oriented generators $d_x \in \mathcal{D}_x \setminus \{0\}$, write

$$(\det Q_e)d_x = a_e d_y, \quad a_e \in \mathbb{R}^\times,$$

and put $\chi_{\det}(e) = \text{sgn}(a_e)$. Then:

- (i) *signs multiply under path composition;*
- (ii) *changing d_x by $\epsilon_x d_x$, $\epsilon_x \in \{\pm 1\}$, changes $\chi_{\det}$ by a vertex coboundary;*
- (iii) *the gauge-invariant obstruction is*

$$\boxed{[\chi_{\det}] = w_1(\mathcal{D}) \in H^1(G; \mathbb{Z}/2);}$$

- (iv) *for deterministic record permutations this class agrees with ordinary permutation parity on the contrast module.*

Thus operational reversibility of the noisy channel is unnecessary; only linear nondegeneracy of its action on future-distinct record contrasts is required.

Proof. Mass preservation gives $K_e(M_x^0) \subseteq M_y^0$. Determinant regularity makes Q_e invertible, so its top exterior power is a nonzero map $\det Q_e : \mathcal{D}_x \rightarrow \mathcal{D}_y$. Functoriality of exterior powers gives

$$\det(Q_f Q_e) = (\det Q_f)(\det Q_e),$$

and therefore multiplicativity of the scalar signs along paths. Replacing d_x by $\epsilon_x d_x$ changes the edge scalar to $\epsilon_y^{-1} a_e \epsilon_x$; over $\{\pm 1\}$ this is precisely multiplication by a coboundary. Hence the sign cocycle represents the first Stiefel–Whitney class of the determinant-line local system.

For a permutation correspondence, Q_e is the augmentation representation of the permutation. The constant line has determinant +1, so the determinant on M_x^0 equals the determinant of the full permutation matrix, namely its ordinary parity. $\square$

Example 3.51 (Noisy orientation reversal without a positive inverse). Let $\Omega = \{0, 1\}$ and

$$K_p = \begin{pmatrix} p & 1-p \\ 1-p & p \end{pmatrix}, \quad 0 \leq p \leq 1.$$

This is a positive bistochastic record channel. On $M^0 = \mathbb{R}(1, -1)$,

$$Q_p(1, -1) = (2p - 1)(1, -1).$$

Hence

$$\chi_{\det}(K_p) = \begin{cases} +1, & p > 1/2, \\ -1, & p < 1/2. \end{cases}$$

For $0 < p < 1$ and $p \neq 1/2$, the channel is not a permutation and its linear inverse is not positive, yet it carries a well-defined orientation sign. At $p = 1/2$ all record contrast is erased and determinant transport becomes singular.

Proposition 3.52 (Orientation is stable away from rank loss). *In a continuous family of determinant-regular record correspondences, $\text{sgn det } Q_e$ is locally constant. It can change only at a parameter value where*

$$\det Q_e = 0,$$

equivalently where at least one future-distinct record contrast is lost.

Proof. The determinant is continuous and a nonzero real number cannot change sign without passing through zero. $\square$

The noisy-record theorem shows that orientation survives dissipation whenever contrast transport remains nonsingular, and that its sign can change only at a rank-loss wall. The next subsection packages this local determinant data as a line system over the graph and identifies the resulting principal double cover.

3.9 Stable records and determinant orientation

Definition 3.53 (Future-distinct record system). A future-distinct record system over G consists of finite-dimensional real record spaces R_x and invertible linear transports

$$L_e : R_x \longrightarrow R_y, \quad e : x \rightarrow y,$$

with $L_{\bar{e}} = L_e^{-1}$. The spaces are reduced: two record vectors are identified when all future record tests agree. The determinant line is

$$\det R_x := \bigwedge^{\dim R_x} R_x.$$

The canonical construction above supplies this datum in the reversible resolved-record phase by taking $R_x = R_x^0$ and letting L_e be the descended record transport. The abstract definition also permits other stable real record carriers and determinant-regular noisy transports.

At this level the construction no longer depends on a particular pointer model. Any reduced real record carrier with invertible transport has a determinant line, and the signs of its transition maps define the same cohomological obstruction.

Definition 3.54 (Record-orientation cochain). Assume the record rank is constant on the connected component. Define

$$\chi(e) := \text{sgn det } L_e \in \{\pm 1\} \cong \mathbb{Z}/2.$$

Changing the orientation of R_x by $g(x) \in \{\pm 1\}$ acts by

$$\chi^g(e : x \rightarrow y) = g(y)\chi(e)g(x).$$

The gauge-invariant class is

$$[\chi] \in H^1(G; \mathbb{Z}/2).$$

Theorem 3.55 (Determinant line and principal signed cover). *The determinant transports $\det L_e$ define a real line local system on G . Its first Stiefel–Whitney class is $[\chi] \in H^1(G; \mathbb{Z}/2)$. The associated orientation cover*

$$G^\chi = (V \times \{\pm 1\}, E_G^\chi), \quad (x, \eta) \xrightarrow{e} (y, \eta\chi(e)),$$

is a principal $\mathbb{Z}/2$ -cover. It is trivial if and only if $[\chi] = 0$, and it is connected for every nonzero class.

Proof. The top exterior power is functorial, so the maps $\det L_e$ compose along paths and define a rank-one real local system. Choosing an orientation in each fibre identifies its transition functions with their signs $\chi(e)$. A change of local orientation multiplies the edge signs by the displayed coboundary, so the isomorphism class is $[\chi]$, the first Stiefel–Whitney class of the line system.

The displayed graph is the standard cover associated with the homomorphism $\pi_1(G) \rightarrow \mathbb{Z}/2$ obtained by multiplying χ around loops. The deck action $(x, \eta) \mapsto (x, -\eta)$ is free and transitive on each fibre. A global section exists exactly when the transition signs are a coboundary, equivalently $[\chi] = 0$. For a connected base, a nonzero homomorphism to $\mathbb{Z}/2$ is surjective, so the corresponding two-sheeted cover is connected. $\square$

Corollary 3.56 (Minimal real signed enrichment). *Among local enrichments that preserve the positive record cone, recover the unsigned base by a deck quotient, and have both positive and negative local lines, the orientation cover has minimal fibre rank two. Its canonical fundamental symmetry is*

$$J_{e(x,\eta)} = \eta e_{(x,\eta)}.$$

Every objectwise-minimal such enrichment is classified, up to orientation gauge, by $H^1(G; \mathbb{Z}/2)$.

Proof. Indefiniteness requires at least one positive and one negative local line, so the fibre rank is at least two. The orientation cover realizes rank two. Cone-preserving unitaries permute the two extremal rays, leaving only the identity and sheet exchange. Their edgewise choices are precisely $\mathbb{Z}/2$ -valued transition functions, modulo vertex reorientations. This gives the stated cohomology classification and the diagonal sheet-sign operator. $\square$

The determinant line therefore produces the minimal signed enrichment of the positive predictive graph. It remains to compare this mod-two obstruction with the real affinity class derived earlier. Although both are associated with cycles, they encode different questions and must not be identified.

3.10 Affinity and orientation are independent

Affinity asks whether opposite path weights balance; orientation asks whether a record frame returns with reversed determinant sign. The following two theorems prove independence both constructively, by realizing all four combinations, and functorially, by ruling out any coefficient homomorphism from real affinity to mod-two parity.

Theorem 3.57 (Independence theorem). *The real affinity class $[A] \in H^1(G; \mathbb{R})$ and the determinant orientation class $[\chi] \in H^1(G; \mathbb{Z}/2)$ are logically independent. More precisely, on every graph with a cycle there exist resolved models realizing each of the four possibilities*

$$([A] = 0 \text{ or } [A] \neq 0) \quad \times \quad ([\chi] = 0 \text{ or } [\chi] \neq 0).$$

Consequently detailed balance neither implies nor is implied by orientability of the record local system.

Proof. It suffices to work on one oriented cycle. Choose arbitrary positive edge weights. Taking $w_e = w_{\bar{e}}$ gives $A = 0$; multiplying one forward weight by e^a with $a \neq 0$ gives a nonzero cycle affinity. Independently choose one-dimensional real record transports $L_e = \lambda_e$ with $\lambda_e \neq 0$. If every $\lambda_e > 0$, then $[\chi] = 0$. If exactly one edge transport is negative, the product of signs around the cycle is -1 , so $[\chi] \neq 0$. The scalar weights and determinant signs can be chosen independently, yielding all four cases. $\square$

Remark 3.58 (Why the distinction matters). A real affinity records how strongly forward and backward path weights differ. A determinant sign records whether a future-distinct record frame returns with its orientation reversed. The former can vary continuously through zero; the latter is locally constant on every determinant-regular stratum and changes only across a rank-loss wall. The Lorentzian cover is controlled by the parity class, not by entropy production or nonequilibrium affinity.

Theorem 3.59 (No canonical mod-two reduction of real affinity). *Every additive group homomorphism*

$$f : (\mathbb{R}, +) \longrightarrow \mathbb{Z}/2$$

is zero. Hence no nontrivial map on cohomology induced by a coefficient homomorphism can send

$$H^1(G; \mathbb{R}) \longrightarrow H^1(G; \mathbb{Z}/2)$$

and convert the real affinity class into the orientation class. The magnitude and parity invariants must arise from distinct components of the operational data.

Proof. For every $x \in \mathbb{R}$ there is $y \in \mathbb{R}$ with $x = 2y$. Therefore

$$f(x) = f(2y) = 2f(y) = 0$$

in the exponent-two group $\mathbb{Z}/2$. Thus f is identically zero, and functorial cohomology applied to a coefficient homomorphism produces only the zero map. $\square$

The separation is now complete: entropy production cannot generate the signed cover, and orientability cannot enforce detailed balance. The real cover is already sufficient for an indefinite fundamental symmetry. A further choice is needed only if its sign holonomy is to act as a phase on complex amplitudes.

3.11 Amplitude square roots

The real signed cover is sufficient for a Krein fundamental symmetry. If the same sign holonomy is retained as a phase acting on amplitudes, one must choose a square root.

Theorem 3.60 (Classification and minimality of amplitude lifts). *Let G be connected. Identify $\mathbb{Z}/4$ and $\mathbb{Z}/2$ with the fourth and second roots of unity when discussing amplitude holonomy, and let $q : \mathbb{Z}/4 \rightarrow \mathbb{Z}/2$ be reduction modulo two. Then*

$$q_* : H^1(G; \mathbb{Z}/4) \rightarrow H^1(G; \mathbb{Z}/2)$$

is surjective. For every $[\chi] \in H^1(G; \mathbb{Z}/2)$, the fibre $q_^{-1}([\chi])$ is a torsor for $H^1(G; \mathbb{Z}/2)$. If $b_1(G) = r$, there are exactly 2^r gauge-inequivalent lifts.*

If a loop has real holonomy -1 , every amplitude lift has order-four holonomy on that loop. Hence $\mathbb{Z}/4$ is the minimal finite coefficient group that can contain an amplitude square root of a nontrivial signed sector. No group homomorphism $s : \mathbb{Z}/2 \rightarrow \mathbb{Z}/4$ satisfies $q \circ s = \text{id}$; therefore the signed class does not canonically select one of its amplitude lifts.

Proof. Use the short exact sequence

$$0 \longrightarrow \mathbb{Z}/2 \xrightarrow{\iota} \mathbb{Z}/4 \xrightarrow{q} \mathbb{Z}/2 \longrightarrow 0, \quad \iota(1) = 2.$$

The associated cohomology sequence contains

$$H^1(G; \mathbb{Z}/2) \xrightarrow{\iota_*} H^1(G; \mathbb{Z}/4) \xrightarrow{q_*} H^1(G; \mathbb{Z}/2) \longrightarrow H^2(G; \mathbb{Z}/2).$$

A graph is a one-dimensional CW complex, so $H^2(G; \mathbb{Z}/2) = 0$ and q_* is surjective. Since the map on H^0 induced by q is surjective, exactness also makes ι_* injective. Thus two lifts of the same class differ freely and transitively by $H^1(G; \mathbb{Z}/2)$. For a connected graph homotopy equivalent to a wedge of r circles, this group has 2^r elements.

If $u(\gamma)^2 = -1$, then $u(\gamma)^4 = 1$ while $u(\gamma)^2 \neq 1$, so $u(\gamma)$ has order four. Conversely the order-four phase i squares to -1 . Finally, if a section s existed, the image $s(1)$ of the order-two element $1 \in \mathbb{Z}/2$ would have order dividing two. But $q(s(1)) = 1$, so $s(1)$ is an odd class in $\mathbb{Z}/4$ and therefore has order four, a contradiction. $\square$

Remark 3.61 (The second pillar). The retrodictive dual, the affinity class, and the signed cover are now separated. The KMS–Petz dual is selected by a faithful stationary state. The real affinity is selected by resolved forward/backward branch weights. The orientation class is selected by stable-record monodromy. Only the last produces the principal double cover and its fundamental symmetry.

Section efsec:retrodictive has therefore produced two independent cycle invariants and assigned them distinct roles. The KMS–Petz construction and resolved branch weights give the real affinity class, whereas stable record transport gives the determinant class and its signed cover; amplitude square roots exist but are not canonically selected. The next section asks what further phase properties turn these finite ingredients into modular and Clifford data of Lorentzian type.

4 The signed modular Lorentzian phase

The previous sections reconstruct the finite positive memory and its possible signed cover. Lorentzian geometry requires a further recurrent phase: some predictive transformations must be reversible, renewal depth must acquire a modular scale, spatial records must have a nondegenerate second moment, and the translational diamond defect must lie in its exchange-symmetric phase. These are phase properties, not universal consequences of complete positivity.

The construction now changes character. Sections efsec:predictive and efsec:retrodictive identified canonical structures once their operational domains were specified. Here we select recurrent components and impose checkable phase criteria. Reversible units yield a projective Clifford factor, the signed cover supplies its temporal row, pressure selects a modular scale, and record moments supply metric and torsion data. The closing hierarchy states exactly which conclusions are structural, phase dependent, or delegated to the companion continuum theorem.

4.1 Canonical reductions on recurrent and Perron components

Proposition 4.1 (Component reduction for stochastic and weighted transfers). *Let G be a finite directed graph and let $C_1, \dots, C_m$ be its strongly connected components.*

- (i) *The condensation graph of the C_j is finite and acyclic.*

- (ii) Every matrix B supported on the edges of G is block upper triangular after a topological ordering of the components, and

$$\rho(B) = \max_{1 \leq j \leq m} \rho(B|_{C_j}).$$

- (iii) If P is stochastic, every stationary probability measure is supported on the union of the closed communicating classes, equivalently the terminal components of the positive-probability support graph.
- (iv) Under a stationary Markov law, almost every trajectory lies in one closed recurrent class after a finite random time. This statement is not true for arbitrary deterministic infinite paths in the underlying graph.

Consequently pressure analysis is performed on an irreducible component block attaining the maximal Perron radius, whereas a stationary stochastic phase is reduced to a selected closed recurrent class.

Proof. Collapsing each strongly connected component gives a directed acyclic graph; otherwise a directed cycle among components would make their union strongly connected. A topological order therefore makes every supported matrix block upper triangular. The spectrum of a block upper triangular matrix is the union of the spectra of its diagonal blocks, proving (ii).

Let P be stochastic and let $\pi P = \pi$. In the finite condensation graph, a nonterminal component has a directed path to a terminal component. Summing the stationarity equation over a downward-closed set and iterating along the acyclic order shows that a stationary measure assigns zero mass to every transient component; equivalently, finite-state Markov-chain recurrence theory places all stationary mass on closed communicating classes [54]. Starting from a stationary law, the chain is in the recurrent support almost surely and visits transient states only finitely often, which proves (iv). A deterministic path may ignore an available exit forever, so no pathwise statement holds without probabilistic qualification. $\square$

Global strong connectivity is therefore unnecessary. The relevant reduction must, however, match the theorem being used: closed recurrent classes for stationary Markov dynamics and Perron-maximizing irreducible blocks for general weighted pressure transfers.

The component reduction prevents two distinct notions of recurrence from being conflated. Stationary stochastic arguments belong on closed communicating classes, whereas pressure belongs on irreducible blocks attaining the maximal Perron radius. After making the appropriate reduction, we can examine the reversible part of the renewal dynamics and the algebra generated by its projective implementers.

4.2 Reversible units and the primitive Clifford factor

Let K_{rev} be a finite elementary abelian two-group of realized reversible labels acting on a finite recurrent predictive algebra. By Theorem 2.37, every label acts by a $*$ -automorphism. After selecting one invariant central orbit, quotienting the action kernel, and choosing a simple block, we obtain a faithful action on a matrix factor $M_N(\mathbb{C})$.

Automorphisms of a matrix factor are inner, but their unitary implementers are defined only up to phase. This unavoidable phase ambiguity is not a defect: its commutator records the alternating binary form from which the Clifford algebra will emerge.

Theorem 4.2 (Automatic projective lift). *Choose unitary implementers R_α of the automorphisms,*

$$\Phi_\alpha = \text{Ad}(R_\alpha), \quad \alpha \in K_{\text{rev}}, \quad R_0 = \mathbf{1}.$$

Then there is a normalized scalar two-cocycle $\sigma \in Z^2(K_{\text{rev}}, U(1))$ such that

$$R_\alpha R_\beta = \sigma(\alpha, \beta) R_{\alpha+\beta}.$$

Its antisymmetric ratio is a gauge-invariant alternating form

$$\sigma(\alpha, \beta) \sigma(\beta, \alpha)^{-1} = (-1)^{\omega_{\text{sp}}(\alpha, \beta)}, \quad \omega_{\text{sp}} \in \text{Alt}^2(K_{\text{rev}}, \mathbb{F}_2).$$

Proof. Every automorphism of a complex matrix factor is inner. The unitaries $R_\alpha R_\beta$ and $R_{\alpha+\beta}$ implement the same automorphism, so their ratio lies in the centre and is a scalar phase. Associativity of the unitary product is exactly the two-cocycle identity for σ . Replacing R_α by $\mu(\alpha) R_\alpha$ changes σ by the coboundary $\delta\mu$.

Because K_{rev} is abelian, the antisymmetric ratio is a bicharacter. Because $2\alpha = 0$, its square is one, so it takes values in $\{\pm 1\}$ and defines an $\mathbb{F}_2$ -valued alternating form. Coboundaries are symmetric on an abelian group and cancel from the ratio, proving gauge invariance. $\square$

The automatic projective lift extracts the spatial commutator data from the reversible predictive action alone. It has not yet identified which generator should be timelike. That information comes from the signed cover: sheet holonomy determines how the fundamental symmetry commutes with every reversible cycle.

The signed cover supplies a temporal generator. Let J be the sheet-sign fundamental symmetry and suppose a reversible cycle class α shifts the signed sheet by the holonomy pairing with $[\chi]$.

Theorem 4.3 (Canonical temporal row). *For every realized reversible cycle class $\alpha \in K_{\text{rev}}$,*

$$J R_\alpha = (-1)^{\langle [\chi], \alpha \rangle} R_\alpha J.$$

Adjoin a temporal label t with $R_t = J$ and set $K_{\text{full}} = K_{\text{rev}} \oplus \langle t \rangle$. The full commutator form is

$$\omega((\alpha, \epsilon), (\beta, \delta)) = \omega_{\text{sp}}(\alpha, \beta) + \epsilon \langle [\chi], \beta \rangle + \delta \langle [\chi], \alpha \rangle.$$

Hence the temporal row is fixed by the signed cover and is not an independent projective input.

Proof. A lifted cycle beginning at (x, η) ends at $(x, \eta(-1)^{\langle [\chi], \alpha \rangle})$. Since J multiplies a basis state by its sheet sign, moving J through the lifted revision produces the displayed scalar. Define $R_{\alpha+\epsilon t} := R_\alpha J^\epsilon$ and move J^ϵ through R_β . Combining the resulting sign with the spatial projective cocycle of [Theorem 4.2](#) gives the formula for the full antisymmetric commutator. $\square$

The temporal-row theorem joins the spatial projective form and the determinant holonomy into one alternating form on the enlarged label group. Some labels may still commute with everything and therefore contribute only central redundancy. Quotienting this radical leaves the primitive nondegenerate sector.

Theorem 4.4 (Radical reduction and primitive Clifford factor). *Let*

$$\text{rad } \omega := \{a \in K_{\text{full}} : \omega(a, b) = 0 \text{ for all } b \in K_{\text{full}}\}$$

and let $K_{\text{prim}} = K_{\text{full}} / \text{rad } \omega$. If $\dim_{\mathbb{F}_2} K_{\text{prim}} = 2m$, then the noncentral factor of the twisted group algebra is

$$\mathbb{C}_\sigma[K_{\text{prim}}] \cong M_{2^m}(\mathbb{C}) \cong \text{Cl}_{2m}(\mathbb{C}).$$

The faithful irreducible projective representation is unique up to unitary equivalence.

Proof. The induced alternating form on K_{prim} is nondegenerate by construction and therefore has even dimension. Choose a symplectic basis $(p_1, q_1, \dots, p_m, q_m)$. After scalar rephasing, the corresponding implementers satisfy

$$R_{p_j} R_{q_j} = -R_{q_j} R_{p_j},$$

while generators belonging to different pairs commute. Each pair generates $M_2(\mathbb{C})$, and the full twisted algebra is their tensor product $M_2(\mathbb{C})^{\otimes m} = M_{2^m}(\mathbb{C})$. This is the complex Clifford algebra on $2m$ generators. The finite Stone–von Neumann theorem gives uniqueness of the irreducible representation of the nondegenerate finite Heisenberg extension. $\square$

The radical quotient identifies the finite matrix factor generated by the noncommuting revisions. When the determinant class pairs nontrivially with a reversible cycle, the abstract temporal row also produces a concrete operator of negative square, as the following corollary shows.

Corollary 4.5 (Intrinsic timelike generator). *If some $\alpha_0 \in K_{\text{rev}}$ satisfies $\langle [\chi], \alpha_0 \rangle = 1$, normalize J and R_{α_0} as Hermitian involutions. Then*

$$\gamma^0 := JR_{\alpha_0}$$

is anti-Hermitian and

$$(\gamma^0)^2 = -\mathbf{1}.$$

Proof. The temporal-row relation gives $JR_{\alpha_0} = -R_{\alpha_0}J$. Therefore

$$(JR_{\alpha_0})^* = R_{\alpha_0}J = -JR_{\alpha_0}, \quad (JR_{\alpha_0})^2 = -J^2 R_{\alpha_0}^2 = -\mathbf{1}.$$

$\square$

A nontrivial signed cover and a timelike Clifford generator establish the *possibility* of temporal orientation. They do not choose one of the two sheets, nor do they guarantee that future experiments can distinguish the choice. The next subsection separates these three logically different levels.

4.3 Selection and predictive visibility of temporal orientation

Let ϑ denote the deck automorphism on the signed observable algebra; thus $\vartheta(J) = -J$. An observable a is deck odd when $\vartheta(a) = -a$.

The first result is a no-go theorem: finite primitive dynamics that respects deck symmetry selects the symmetric state, not an orientation. The second result asks a different question—whether, once an oriented state has been selected by some additional mechanism, the predictive test algebra is rich enough to observe it.

Theorem 4.6 (Finite primitive symmetric phases cannot select an orientation). *Let T be a primitive finite-dimensional renewal transfer on the signed state space and assume deck covariance,*

$$T \circ \vartheta_* = \vartheta_* \circ T.$$

Then its unique stationary state ρ_ is deck invariant and*

$$\text{Tr}(\rho_* a) = 0$$

for every deck-odd observable a , in particular for $a = J$.

Proof. By [Theorem 2.42](#), T has a unique stationary density. Deck covariance makes $\vartheta_*(\rho_*)$ stationary as well, so uniqueness gives $\vartheta_*(\rho_*) = \rho_*$. For deck-odd a ,

$$\mathrm{Tr}(\rho_* a) = \mathrm{Tr}(\vartheta_*(\rho_*) a) = \mathrm{Tr}(\rho_* \vartheta(a)) = -\mathrm{Tr}(\rho_* a),$$

and the expectation vanishes. $\square$

Corollary 4.7 (Possible orientation-selection mechanisms). *A nonzero stationary deck-odd order parameter requires at least one departure from a finite primitive deck-symmetric phase: explicit deck-symmetry breaking, stationary-state degeneracy, boundary or superselection data, or a singular infinite-volume/infinite-depth limit in which uniqueness fails.*

Theorem 4.8 (Deck-odd observability criterion). *Let $\omega_- := \omega_+ \circ \vartheta$. On the deck-even algebra $\mathcal{A}^\vartheta = \{a : \vartheta(a) = a\}$ one has*

$$\omega_-|_{\mathcal{A}^\vartheta} = \omega_+|_{\mathcal{A}^\vartheta}.$$

Hence a predictive quotient using only deck-even future tests identifies the two orientations. Conversely, if an admissible deck-odd observable a satisfies $\omega_+(a) \neq 0$, then

$$\omega_-(a) = -\omega_+(a),$$

so the two phases are predictively distinguishable.

Proof. For deck-even a , $\omega_-(a) = \omega_+(\vartheta(a)) = \omega_+(a)$. For deck-odd a , the same identity gives $\omega_-(a) = -\omega_+(a)$. A single admissible test with different expectations separates the predictive classes. $\square$

Remark 4.9 (Three distinct levels). A nontrivial class $[\chi]$ gives the possibility of temporal orientation; a state-level phase chooses one sheet; a deck-odd admissible test makes that choice operationally visible. None of these statements implies either of the other two.

We have now distinguished topology, state selection, and observability. The signed cover supplies a binary temporal direction, but not a rate at which renewal depth should be read as time. The next subsection derives a dimensionless modular scale from Perron–Frobenius pressure and realizes it on an unbounded graded path space.

4.4 Perron pressure and the affine modular clock

Let the selected recurrent component carry positive branch capacities $q_e > 0$ and positive depths $\ell_e \geq \ell_0 > 0$. Define the weighted transfer matrix

$$B(s)_{xy} := \sum_{e: x \rightarrow y} q_e e^{-s\ell_e}.$$

When these capacities come from the resolved bidirected branch law, write $q_e = c_e e^{A(e)/2}$ as in [Theorem 3.37](#).

There are four steps. Gauge covariance first shows that the pressure is independent of local normalizations. Supercriticality then selects a unique root β . The associated Doob transform gives a stationary edge process whose mean depth and entropy production admit an exact identity. Finally, a graded path cover turns the scalar root into an affine modular flow without contradicting the finite-cycle obstruction.

Proposition 4.10 (Gauge covariance of pressure and the Doob law). *Let a vertex gauge act by $A^f(e : x \rightarrow y) = A(e) + f(x) - f(y)$ and put $q_e^f = c_e e^{A^f(e)/2}$. With $D_f = \text{diag}(e^{f(x)/2})$,*

$$B^f(s) = D_f B(s) D_f^{-1}.$$

Consequently the pressure $\log \rho(B(s))$, its zero β , and its derivatives are gauge invariant. If h, ν are Perron vectors at the root, then choosing $h^f = D_f h$ and $(\nu^f)^T = \nu^T D_f^{-1}$ leaves the Doob edge law

$$p_e = q_e e^{-\beta \ell_e} \frac{h_y}{h_x}$$

and the stationary vertex law $\pi_x = \nu_x h_x$ unchanged.

Proof. For an edge $e : x \rightarrow y$, $q_e^f = e^{f(x)/2} q_e e^{-f(y)/2}$, which gives the diagonal similarity after summing parallel edges. Similarity preserves the spectrum and hence the pressure data. Substitution of the transformed Perron vectors gives $p_e^f = p_e$ and $\pi_x^f = \pi_x$. $\square$

The similarity relation shows that pressure depends only on gauge-invariant cycle data, even though individual oriented capacities depend on local choices. We can therefore use its spectral radius to select an intrinsic exponent.

Theorem 4.11 (Pressure-root selection). *Assume the recurrent component is strongly connected and let $r(s) = \rho(B(s))$. Then r is continuous and strictly decreasing on $[0, \infty)$, with $r(s) \rightarrow 0$ as $s \rightarrow \infty$. Consequently:*

- (i) *if $r(0) > 1$, there is a unique $\beta > 0$ with $r(\beta) = 1$;*
- (ii) *if $r(0) = 1$, the unique nonnegative root is $\beta = 0$;*
- (iii) *if $r(0) < 1$, no nonnegative root exists.*

At the root the Perron eigenvalue is simple and has strictly positive left and right eigenvectors.

Proof. For $s_2 > s_1$, every nonzero matrix entry satisfies $B(s_2)_{xy} < B(s_1)_{xy}$. Irreducibility and the strict monotonicity part of Perron–Frobenius imply $r(s_2) < r(s_1)$. Continuity follows from continuity of the finite matrix entries and of the spectral radius. The positive lower bound on ℓ_e makes every entry tend to zero, so $r(s) \rightarrow 0$. The three alternatives follow from the intermediate value theorem and strict monotonicity. Simplicity and positivity of the Perron eigenvectors are the irreducible Perron–Frobenius theorem. $\square$

The pressure-root theorem identifies exactly when a positive modular exponent exists: it is a supercritical phase property, not a universal consequence of recurrence. At the root, the Perron vectors canonically normalize the weighted transfer into a stationary stochastic law.

Theorem 4.12 (Stationary pressure law and mean depth). *Let $h > 0$ and $\nu > 0$ be right and left Perron eigenvectors at β , normalized by $\nu^T h = 1$. For an edge $e : x \rightarrow y$, define*

$$p_e := q_e e^{-\beta \ell_e} \frac{h_y}{h_x}, \quad \pi_x := \nu_x h_x.$$

Then p is a stochastic edge law and π is stationary. If $P(s) := \log r(s)$, its stationary mean depth satisfies

$$\mu_\ell := \sum_e \pi_{s(e)} p_e \ell_e = -P'(\beta) > 0.$$

Proof. The row normalization is

$$\sum_{e:s(e)=x} p_e = h_x^{-1}(B(\beta)h)_x = 1.$$

For a vertex y ,

$$\sum_{e:t(e)=y} \pi_{s(e)} p_e = h_y \sum_x \nu_x B(\beta)_{xy} = \nu_y h_y = \pi_y.$$

The Perron root is simple, so the standard simple-eigenvalue perturbation formula gives $r'(\beta) = \nu^T B'(\beta)h$; see, for example, [54]. Since $r(\beta) = 1$,

$$-P'(\beta) = -r'(\beta) = \sum_e \nu_{s(e)} q_e e^{-\beta \ell_e} h_{t(e)} \ell_e = \sum_e \pi_{s(e)} p_e \ell_e.$$

Positivity follows from $\ell_e > 0$. □

The Doob law gives the probabilistic process selected by the pressure root, and the derivative of pressure gives its average renewal depth. To interpret β , we now decompose each capacity into symmetric conductance and affinity and compare the resulting stationary edge fluxes.

The pressure root also relates the modular scale to the information carried by the stationary edge process. Assume in the remainder of this subsection that the selected edge set is bidirected, $\ell_{\bar{e}} = \ell_e$, and

$$q_e = c_e e^{A(e)/2}, \quad c_{\bar{e}} = c_e > 0, \quad A(\bar{e}) = -A(e),$$

as in [Theorem 3.37](#).

Definition 4.13 (Stationary entropy production and entropy rate). Let

$$F_e := \pi_{s(e)} p_e.$$

Then $(F_e)_e$ and $(F_{\bar{e}})_e$ are probability distributions on the oriented edge set. Define

$$\mathcal{A}(e) := \log \frac{F_e}{F_{\bar{e}}}, \quad \sigma := \sum_e F_e \mathcal{A}(e),$$

and

$$h_{\text{rate}} := - \sum_e F_e \log p_e, \quad \kappa_c := \sum_e F_e \log c_e.$$

The quantity σ is the relative entropy $D_{\text{KL}}(F \| F \circ -)$ and is therefore nonnegative.

Lemma 4.14 (Input affinity and stationary affinity are cohomologous). *Let h, ν be the Perron vectors of [Theorem 4.12](#), and set*

$$v(x) := \log \frac{h_x}{\nu_x}.$$

Then

$$\mathcal{A}(e) = A(e) + v(t(e)) - v(s(e)).$$

Consequently

$$\boxed{\sum_e F_e A(e) = \sum_e F_e \mathcal{A}(e) = \sigma \geq 0.}$$

Proof. For $e : x \rightarrow y$,

$$F_e = \pi_x q_e e^{-\beta \ell_e} \frac{h_y}{h_x} = \nu_x q_e e^{-\beta \ell_e} h_y.$$

Using $\ell_{\bar{e}} = \ell_e$ gives

$$\log \frac{F_e}{F_{\bar{e}}} = \log \frac{q_e}{q_{\bar{e}}} + \log \frac{\nu_x h_y}{\nu_y h_x} = A(e) + v(y) - v(x).$$

Stationarity implies that the expectation of every vertex coboundary vanishes:

$$\sum_e F_e (v(t(e)) - v(s(e))) = \sum_y \pi_y v(y) - \sum_x \pi_x v(x) = 0.$$

Hence the two stationary means agree. Finally,

$$\sigma = \sum_e F_e \log \frac{F_e}{F_{\bar{e}}} = D_{\text{KL}}(F \| F \circ \cdot) \geq 0$$

by the log-sum inequality. $\square$

The preceding lemma shows that the input affinity and the affinity measured by the stationary edge process differ only by a vertex coboundary. Their stationary means therefore agree and equal the nonnegative entropy production. Substituting this fact into the Doob normalization yields the central balance law.

Theorem 4.15 (Entropy–affinity–depth identity). *Under the preceding bidirected pressure hypotheses,*

$$\beta \mu_\ell = h_{\text{rate}} + \kappa_c + \frac{1}{2} \sigma.$$

Equivalently,

$$\beta = \frac{h_{\text{rate}} + \mathbb{E}_F[\log c_e] + \frac{1}{2} \mathbb{E}_F[A(e)]}{\mu_\ell}.$$

In the neutral-conductance normalization $c_e \equiv 1$,

$$\beta = \frac{h_{\text{rate}} + \sigma/2}{\mu_\ell}.$$

Thus nonequilibrium affinity contributes a positive one-half correction to the entropy-per-depth law, while the symmetric conductance remains an independent unoriented capacity datum.

Proof. Taking logarithms of the Doob law gives

$$\log p_e = \log q_e - \beta \ell_e + \log h_{t(e)} - \log h_{s(e)}.$$

Average with respect to F . By stationarity, the final coboundary has zero mean, so

$$\beta \mu_\ell = - \sum_e F_e \log p_e + \sum_e F_e \log q_e = h_{\text{rate}} + \sum_e F_e \log q_e.$$

The unique affinity–conductance factorization yields

$$\log q_e = \log c_e + \frac{1}{2} A(e).$$

Substitution and [Theorem 4.14](#) prove the first identity. Division by $\mu_\ell > 0$ gives the second, and $c_e \equiv 1$ gives the final specialization. $\square$

Remark 4.16 (Why the factor one half is forced). The affinity is the full oriented log ratio $A(e) = \log(q_e/q_{\bar{e}})$. A single oriented pressure capacity equals the geometric mean c_e times the square root of that ratio. The coefficient $1/2$ is therefore algebraic, not conventional.

The identity explains what fixes the modular exponent: predictive entropy, symmetric renewal capacity, and nonequilibrium circulation per unit mean depth. It still remains to realize that exponent as a consistent flow. A finite return cycle cannot carry the same nonzero eigenvalue on every edge, so the clock must remember accumulated path depth.

Construction 4.17 (Graded path-clock cover). Let $\mathcal{P}_{\text{fin}}(G)$ be the set of finite directed paths in the selected recurrent graph and put

$$\mathcal{H}_{\text{clk}} := \ell^2(\mathcal{P}_{\text{fin}}(G)).$$

For a path $w = e_1 \cdots e_k$, set

$$L(w) := \sum_{j=1}^k \ell_{e_j}, \quad N\delta_w = L(w)\delta_w.$$

If appending e to w is composable, define $S_e\delta_w = \delta_{we}$, and set $S_e\delta_w = 0$ otherwise. Then $[N, S_e] = \ell_e S_e$ on the finite-path core.

Theorem 4.18 (Pressure-selected affine path clock). *Let β be the pressure root of [Theorem 4.11](#) and define the positive self-adjoint operator*

$$\Delta_\beta := e^{\beta N}$$

on the graded path-clock cover. Then every path-creation operator is a modular eigenoperator:

$$\boxed{\Delta_\beta^{it} S_e \Delta_\beta^{-it} = e^{it\beta\ell_e} S_e.}$$

Thus the pressure root determines an affine modular clock without assigning a nonzero common exponent directly to every edge of the finite recurrent base.

Proof. For every basis vector for which we is defined,

$$\Delta_\beta^{it} S_e \Delta_\beta^{-it} \delta_w = e^{-it\beta L(w)} e^{it\beta L(we)} \delta_{we} = e^{it\beta\ell_e} S_e \delta_w.$$

Both sides vanish when we is not composable. The finite-path span is a core, so the identity extends to the bounded creation operators. $\square$

The graded cover resolves the finite-cycle obstruction by separating the finite recurrent base from the unbounded depth coordinate. Every edge creation operator then has the pressure-selected modular frequency. The next theorem shows that, in an irreducible realization, no different compatible Hamiltonian is possible except for an additive scalar.

Theorem 4.19 (Rigidity of a compatible modular Hamiltonian). *Let N and S_e satisfy $[N, S_e] = \ell_e S_e$. If a faithful modular flow on a selected factor obeys*

$$\sigma_t(S_e) = e^{it\beta\ell_e} S_e$$

and the joint commutant of the elementary resets is scalar, then

$$H_{\text{mod}} = \beta N + c\mathbf{1}.$$

Thus every compatible irreducible realization of the pressure-selected clock is affine in renewal depth, up to an additive scalar.

Proof. Writing $\sigma_t = \text{Ad}(e^{itH_{\text{mod}}})$ and differentiating at zero gives $[H_{\text{mod}} - \beta N, S_e] = 0$ for every e . The scalar-commutant hypothesis gives $H_{\text{mod}} - \beta N = c\mathbf{1}$. $\square$

Proposition 4.20 (Finite-cycle obstruction). *Suppose $e_1, \dots, e_m$ form a closed reset cycle, each S_{e_j} is a modular eigenoperator with exponent β_j , and*

$$S_{e_m} \cdots S_{e_1} = cP, \quad c \neq 0,$$

where P is a modular-invariant one-dimensional corner. Then $\sum_j \beta_j = 0$. In particular, a common exponent on such a cycle must be zero.

Proof. The product is a modular eigenoperator with exponent $\sum_j \beta_j$, whereas cP is modular invariant. Since the product is nonzero, $e^{it\sum_j \beta_j} = 1$ for every t , so the sum vanishes. $\square$

Remark 4.21 (Critical state versus trace density). At $r(\beta) = 1$, the finite-path partition series is critical and need not define a trace-class density on $\mathcal{H}_{\text{clk}}$. The natural normalized object is the Perron boundary Markov state of [Theorem 4.12](#), or a faithful semifinite modular weight. For $s > \beta$, the weighted path series is subcritical and convergent.

The pressure theorem selects a dimensionless modular exponent and the graded cover realizes its flow. Converting the resulting parameter into seconds requires a physical unit calibration and is not part of the dimensionless reconstruction.

Pressure has supplied an intrinsic dimensionless clock, while the signed and projective structures supplied its temporal algebraic direction. Lorentzian data also require a nondegenerate spatial quadratic form and compatible comparison between neighbouring frames. These arise next from second moments of stable direction records.

4.5 Spatial second moments and metric comparison

Let $V_{\text{sp}} \cong \mathbb{R}^d$ be the real span of direction records on the selected recurrent component. A local stationary direction law ν_x has second moment

$$M_x := \int_{V_{\text{sp}}} vv^\top d\nu_x(v).$$

The second moment is the lowest-order statistic that can define a positive quadratic geometry without assuming coordinates. Symmetry may force it to be isotropic, but only positive definiteness is needed; covariance under transport then determines the orthogonal comparison maps.

Proposition 4.22 (Symmetry forces ellipticity). *Let a compact group K act orthogonally and irreducibly on V_{sp} . If ν is a finite positive K -invariant measure, not supported at the origin, and has finite second moment, then*

$$M = \int vv^\top d\nu(v) = cI$$

for a scalar $c > 0$.

Proof. Invariance gives $kMk^{-1} = M$ for every $k \in K$. The matrix M is self-adjoint and positive. By real Schur theory, the self-adjoint part of the commutant of an irreducible orthogonal representation is scalar: any additional complex or quaternionic generators are skew-adjoint. Thus $M = cI$. Since $\text{Tr } M = \int \|v\|^2 d\nu(v) > 0$, one has $c > 0$. $\square$

Literal isotropy is not required. Positive definiteness of M_x is enough to define a coframe; constant anisotropy can be absorbed into a spatial frame. The next proposition derives the isometric comparison used downstream.

Proposition 4.23 (Second-moment covariance derives orthogonal comparison). *Let $M_x, M_y \succ 0$ be neighbouring second moments and let $T_e : V_{\text{sp},x} \rightarrow V_{\text{sp},y}$ be an invertible record transport satisfying*

$$T_e M_x T_e^\top = M_y.$$

Then

$$O_e := M_y^{-1/2} T_e M_x^{1/2}$$

is orthogonal. If the determinant orientation of the record system is preserved, $O_e \in \text{SO}(d)$. Thus metric compatibility of neighbouring renewal frames follows from covariance of their quadratic statistics.

Proof. Directly,

$$O_e O_e^\top = M_y^{-1/2} T_e M_x T_e^\top M_y^{-1/2} = I.$$

Hence $O_e \in O(d)$. Its determinant sign is the transported determinant-line orientation, so orientation preservation gives determinant $+1$. $\square$

The second-moment results supply the spatial coframe and metric-compatible comparison transport. They do not determine whether elementary translations close around a plaquette. That first-moment question is independent of metric compatibility and is addressed by the exchange symmetry of two-step diamond records.

4.6 Exchange symmetry and translational plaquette closure

The last upstream input required for the curved Cartan theorem is a first-moment closure law. It does not follow from associativity or ordinary one-step detailed balance. It follows from a stronger symmetry of one common microscopic two-step law.

The guiding distinction is parity under exchange of the two path orders. A translational closure defect is exchange odd and must vanish in an exchange-invariant stationary state; a rotational defect is exchange even and may remain nonzero. This is the finite precursor of the separation between torsion and curvature.

Fix two elementary directions $i \neq j$. Let Ω_{ij} be the finite space of resolved microscopic diamond records, including the two ordered histories ij and ji , their terminal-location records, and any internal environment labels retained by future tests. Let $\vartheta : \Omega_{ij} \rightarrow \Omega_{ij}$ exchange the two path orders.

Definition 4.24 (Exchange-covariant diamond process). A Markov kernel K_{ij} on Ω_{ij} is exchange covariant when

$$K_{ij}(\vartheta\omega, \vartheta A) = K_{ij}(\omega, A)$$

for every state ω and event A . A translational defect $d_{ij}^{(h)} : \Omega_{ij} \rightarrow V_{\text{sp}}$ is exchange odd when

$$d_{ij}^{(h)}(\vartheta\omega) = -d_{ij}^{(h)}(\omega).$$

Theorem 4.25 (Stationary exchange closure). *Assume K_{ij} is primitive and exchange covariant. Let π_{ij} be its unique stationary law. Then π_{ij} is exchange invariant and every exchange-odd translational defect has zero stationary mean:*

$$\int d_{ij}^{(h)} d\pi_{ij} = 0.$$

If the developed first-moment defect has the expansion

$$\int d_{ij}^{(h)} d\pi_{ij} = h^2 T_{ij} + O(h^3),$$

then $T_{ij} = O(h)$ and, in the continuum limit, $T_{ij} \rightarrow 0$. If the expansion is exact through second order with an h -independent coefficient, then $T_{ij} = 0$ identically. No exchange-even rotational defect is constrained.

Proof. Push the stationary law forward by ϑ . Exchange covariance gives

$$(\vartheta_* \pi_{ij}) K_{ij} = \vartheta_*(\pi_{ij} K_{ij}) = \vartheta_* \pi_{ij},$$

so $\vartheta_* \pi_{ij}$ is stationary. Primitivity gives uniqueness, hence $\vartheta_* \pi_{ij} = \pi_{ij}$. Therefore

$$\int d_{ij}^{(h)} d\pi_{ij} = \int d_{ij}^{(h)} \circ \vartheta d\pi_{ij} = - \int d_{ij}^{(h)} d\pi_{ij},$$

and the integral vanishes.

Substituting the asymptotic expansion gives $0 = h^2 T_{ij} + O(h^3)$, hence $T_{ij} = O(h)$. If T_{ij} is the fixed second-order coefficient, division by h^2 and passage to $h \downarrow 0$ gives $T_{ij} = 0$. The argument uses oddness and says nothing about an even rotational observable. $\square$

The stationary exchange theorem gives a direct symmetry argument for closure. The same condition can also be characterized variationally. Introducing a source conjugate to the exchange-odd defect turns its expectation into the gradient of a convex pressure, so zero source becomes the Euler–Lagrange point.

Definition 4.26 (Exchange-source pressure). Let Ω_{ij} be a finite resolved diamond space with exchange involution ϑ . Let $S_h : \Omega_{ij} \rightarrow \mathbb{R}$ be exchange even and let $D_h : \Omega_{ij} \rightarrow V_{\text{sp}}$ be exchange odd:

$$S_h \circ \vartheta = S_h, \quad D_h \circ \vartheta = -D_h.$$

For $\lambda \in V_{\text{sp}}^*$ define

$$\Psi_h(\lambda) := \log \sum_{\omega \in \Omega_{ij}} \exp(-S_h(\omega) + \langle \lambda, D_h(\omega) \rangle),$$

and let $\pi_{h,\lambda}$ be the associated Gibbs law.

Theorem 4.27 (Zero exchange source is the torsion-free stationary point). *For the pressure of [Theorem 4.26](#),*

$$\nabla \Psi_h(\lambda) = \mathbb{E}_{\pi_{h,\lambda}} D_h, \quad \nabla^2 \Psi_h(\lambda) = \text{Cov}_{\pi_{h,\lambda}}(D_h) \succeq 0.$$

Moreover $\Psi_h(-\lambda) = \Psi_h(\lambda)$. Hence $\lambda = 0$ is a global minimum and

$$\nabla \Psi_h(0) = 0 \quad \Longleftrightarrow \quad \mathbb{E}_{\pi_{h,0}} D_h = 0.$$

The minimum is unique modulo the annihilator of

$$\text{span}\{D_h(\omega) - D_h(\omega') : \omega, \omega' \in \Omega_{ij}\}.$$

When D_h is the developed translational diamond defect, zero source is therefore exactly the Euler–Lagrange condition for first-moment plaquette closure.

Proof. Because the sum is finite, differentiation under the sum is legitimate. The first derivative of the log partition function is the Gibbs expectation of D_h , and the second derivative is its covariance bilinear form. The latter is positive semidefinite, so Ψ_h is convex.

The change of variables $\omega \mapsto \vartheta\omega$, together with the parities of S_h and D_h , gives

$$\Psi_h(-\lambda) = \log \sum_{\omega} e^{-S_h(\vartheta\omega) + \langle \lambda, D_h(\vartheta\omega) \rangle} = \Psi_h(\lambda).$$

Thus the convex function is even and has vanishing gradient at the origin, which is consequently a global minimum. Its Hessian vanishes on a covector λ exactly when $\langle \lambda, D_h \rangle$ is constant on the support; these covectors form the stated annihilator. Outside that subspace the pressure is strictly convex. Finally, taking $D_h = d_{ij}^{(h)}$ identifies the gradient with the stationary translational first moment, so the zero-source stationarity equation is precisely plaquette closure. $\square$

Remark 4.28 (Why ordinary detailed balance is insufficient). A reversible one-step edge law may coexist with a stationary distribution that selects one of two exchange-related diamond orders. Such a phase has nonzero exchange-odd first moment and therefore torsion. The exact upstream input is not generic stationarity or detailed balance but selection of the exchange-invariant, zero-source translational phase. This condition is falsifiable and admits torsional counterphases.

Exchange covariance therefore supplies two equivalent descriptions of the torsion-free phase: invariance of the stationary diamond law and stationarity at zero exchange source. What remains to be shown is that weaker conditions, such as ordinary stationarity or one-step detailed balance, do not already force this choice. The following countermodel establishes that sharp boundary.

4.7 A sharp torsion phase countermodel

The exchange theorem gives exact closure in an exchange-invariant stationary state. The next model proves that ordinary stationarity and one-step detailed balance do not force that state to be selected.

The model is intentionally elementary. Its Ising order parameter records the choice between the two diamond orders, while an independent isotropic direction record keeps the spatial second moment nondegenerate. This isolates torsion selection from both reversibility and metric regularity.

Construction 4.29 (Local torsion-record heat-bath field). On $\mathbb{Z}^2$ let $\tau_x \in \{\pm 1\}$ and consider the nearest-neighbour zero-field Ising specification with coupling $\theta \geq 0$. Its single-site conditional law is

$$q_x(s \mid \tau) = \frac{\exp\left(\theta s \sum_{y \sim x} \tau_y\right)}{2 \cosh\left(\theta \sum_{y \sim x} \tau_y\right)}.$$

Let global sign reversal $\vartheta\tau = -\tau$ exchange the two orders of one microscopic diamond. Fix $v \neq 0$ and $\kappa > 0$, and define the exchange-odd translational defect

$$d_{ij}^{(h)}(\tau) = \kappa h^2 \tau_0 v.$$

For every stationary Gibbs state μ ,

$$\Pi_{\text{tr}} \mathcal{P}_{ij}^{(h)}(\mu) = \kappa h^2 \mu(\tau_0) v.$$

Independently attach to each update a deck-even direction record taking the six values $\{\pm e_1, \pm e_2, \pm e_3\}$ with equal probability. Its second moment is $I_3/3$, so the model has a fixed nondegenerate spatial second moment in every stationary phase without changing the torsion defect.

Theorem 4.30 (Torsion-free and torsional stationary phases coexist). *The heat-bath law of Theorem 4.29 is local, finite range, and satisfies one-step detailed balance in every Gibbs state. Moreover:*

- (i) *if $4 \tanh \theta < 1$, the infinite-volume Gibbs state is unique, exchange invariant, and $\Pi_{\text{tr}} \mathcal{P}_{ij}^{(h)} = 0$;*
- (ii) *if $\theta \geq \frac{1}{2} \log 12$, there are two extremal deck-related Gibbs states $\mu_\theta^\pm$ satisfying*

$$\pm \mu_\theta^\pm(\tau_0) \geq \frac{203}{216},$$

and hence

$$\pm v \cdot \Pi_{\text{tr}} \mathcal{P}_{ij}^{(h)}(\mu_\theta^\pm) \geq \kappa h^2 \frac{203}{216} \|v\|^2 > 0;$$

- (iii) *the symmetric mixture $\bar{\mu}_\theta = (\mu_\theta^+ + \mu_\theta^-)/2$ is stationary and detailed balanced at the same coupling but has zero translational defect.*

Thus one microscopic detailed-balanced law admits both torsion-free and torsional stationary phases.

Proof. Let c_{xy} be the maximal total-variation change of the conditional law at x when only the neighbouring spin at y is changed. The plus probability in local field h is $(1 + \tanh(\theta h))/2$; changing one neighbour changes h by two, and the maximal difference occurs between fields -1 and 1 . Hence $c_{xy} \leq \tanh \theta$ for $x \sim y$ and $c_{xy} = 0$ otherwise. Thus $\alpha := \sup_x \sum_y c_{xy} \leq 4 \tanh \theta$. In a finite box, update the same random site in two heat-bath chains and use a maximal coupling of their conditional Bernoulli laws. If D is their Hamming distance, then

$$\mathbb{E}[D' \mid \tau, \tau'] \leq \left(1 - \frac{1 - \alpha}{|\Lambda|}\right) D.$$

When $\alpha < 1$, iteration and exhaustion of the lattice show that the influence of arbitrary boundary conditions on every fixed cylinder event tends to zero; there is therefore a unique infinite-volume Gibbs state (the standard Dobrushin argument; see [55]). The unique state is invariant under $\tau \mapsto -\tau$, so its one-point function and the displayed defect vanish.

For the low-temperature statement, impose plus boundary conditions in a finite box. If $\tau_0 = -1$, the minus cluster containing the origin has an outer dual contour γ . Flipping all spins inside γ changes every crossing bond from disagreement to agreement and increases the Gibbs weight by $e^{2\theta|\gamma|}$. The number N_n of simple dual contours of length n that surround the origin is bounded by $4n3^{n-1}$. Hence, with $r = 3e^{-2\theta}$,

$$\mu_\theta^+(\tau_0 = -1) \leq \sum_{n \geq 4} 4n3^{n-1} e^{-2\theta n} = \frac{4}{3} \frac{r^4(4 - 3r)}{(1 - r)^2}.$$

If $\theta \geq \frac{1}{2} \log 12$, then $r \leq 1/4$ and the right-hand side is at most $13/432$. Therefore

$$\mu_{\theta}^{+}(\tau_0) = 1 - 2\mu_{\theta}^{+}(\tau_0 = -1) \geq \frac{203}{216}.$$

Monotone boundary limits give an extremal plus state; its global flip is the extremal minus state with opposite magnetization. Every Gibbs state is stationary and detailed balanced for the heat-bath dynamics because the update uses its exact single-site conditional distribution. Substitution in the defect formula proves (ii). Convex combinations preserve stationarity and detailed balance, while the opposite magnetizations cancel in the symmetric mixture, proving (iii). $\square$

Corollary 4.31 (Torsion freedom is a genuine selection principle). *No theorem using only local finite-range UCP or Markov dynamics, stationarity, one-step detailed balance, microscopic exchange covariance, bounded terminal records, and a nondegenerate spatial second moment can imply translational plaquette closure in every stationary phase. Closure is selected, for example, by uniqueness plus exchange covariance, restriction to exchange-invariant states, or zero exchange source.*

Proof. The two extremal low-temperature states in [Theorem 4.30](#) satisfy all the listed properties and have nonzero $O(h^2)$ translational defect, while their symmetric mixture satisfies the same microscopic laws and has zero defect. $\square$

The coexistence theorem proves that torsion freedom is a genuine phase selection: the same detailed-balanced microscopic law can support symmetric torsion-free and exchange-broken torsional stationary states. We can now summarize the entire dependency chain without hiding model choices or continuum assumptions inside the foundational reconstruction.

4.8 Logical hierarchy and the upstream-to-downstream bridge

Every premise of a theorem is a hypothesis, but not every hypothesis is a foundational axiom. We use the following distinctions.

This vocabulary is needed to read the final theorems correctly. The first theorem collects structures reconstructed uniformly from the operational package. The phase assumption then lists additional checkable properties of a chosen model. The second theorem derives the finite signed modular Clifford data in that phase, and the third invokes the separate analytic continuum theorem only after those finite inputs have been established.

Definition 4.32 (Axioms, models, phases, and selections). (i) A *foundational axiom* is imposed uniformly on every admissible operational theory; here these are the clauses of [Assumption 2.1](#) together with the specified finite domain and ancillary access.

- (ii) A *microscopic model specification* chooses a particular resolved graph, branch instrument, capacities, depths, reversible actions, direction records, diamond law, and parameter values inside that theory.
- (iii) A *phase criterion* is a checkable predicate on a specified model or parameter region, such as nontrivial determinant holonomy, supercriticality, projective rank, or ellipticity.
- (iv) A *physical selection* chooses one recurrent component, extremal state, boundary condition, zero-source state, or continuum scaling net when the model admits several possibilities.

Theorem 4.33 (I. Structural reconstruction). *Assume the finite operational reconstruction package and complete ancillary stability. Then every admissible model canonically determines:*

- (i) *the rooted future quotient Q_+ and its unique reduced predictive renewal memory;*
- (ii) *the complete monoidal process quotient $Q_{||}$, equal to the ordinary quotient precisely in the parallel-complete theory;*
- (iii) *a finite oriented complex predictive C^* -algebra and a finite alphabet of UCP renewal maps;*
- (iv) *on every selected recurrent support for which a primitive weighted transfer and channelwise stationary faithful state exist, the canonical KMS–Petz retrodictive family.*

No signed, supercritical, orientation-broken, elliptic, or torsion-free phase is asserted by this theorem.

Proof. Items (i)–(ii) are [Theorems 2.3, 2.7, 2.9](#) and [2.13](#). The complex algebra and UCP maps are [Theorems 2.33](#) and [2.36](#). A primitive weighted transfer selects the unique faithful state by [Theorem 2.42](#); channelwise stationarity and [Theorem 3.3](#) give the common UCP retrodictive family. $\square$

The structural theorem deliberately stops before any Lorentzian claim. It provides the common arena in which signed and unsigned, critical and subcritical, reversible and dissipative models can all live. The following criteria select the particular recurrent phase whose finite data match the input requirements of the companion construction.

Assumption 4.1 (Checkable signed Lorentzian phase criteria). A specified microscopic model M has a selected irreducible recurrent block satisfying:

- (C1) its resolved outcomes are either supplied operationally or extracted from an interaction-complete stable-pointer phase as in [Theorem 3.42](#); their future-distinct record transport is determinant regular and has nonzero $[\chi] \in H^1(G; \mathbb{Z}/2)$, with an admissible deck-odd test;
- (C2) its realized reversible subgroup has, after kernel and radical reduction, a nonzero temporal row and the required spatial projective rank;
- (C3) after the canonical successor refinement of [Theorem 3.21](#), its resolved branches are represented by their conditional predictive successors in a reduced future-tomographic realization, and the Petz-reindexed opposite branches satisfy the corresponding reverse-renewal condition; the resulting reciprocal scalar weights have positive depths and define a supercritical Perron block, so the unique pressure root $\beta > 0$ exists;
- (C4) it admits two deck-related extremal orientation states, and one of them is selected rather than their symmetric mixture;
- (C5) its spatial second moment is positive definite and covariant between neighbouring cells;
- (C6) its diamond state is exchange invariant, or equivalently selected at zero exchange source, and has the regularity required by the intended continuum theorem.

Theorem 4.34 (II. Signed modular Clifford phase). *Let M be a microscopic model supplied by [Theorem 4.33](#) and satisfying [Assumption 4.1](#). Then M canonically supplies:*

- (i) a real affinity class $[A] \in H^1(G; \mathbb{R})$ and the unique factorization $q_e = c_e e^{A(e)/2}$ of each reciprocal capacity;
- (ii) the independent determinant class $[\chi] \in H^1(G; \mathbb{Z}/2)$, its principal signed cover, and the classified noncanonical $\mathbb{Z}/4$ amplitude lifts;
- (iii) a projective reversible action whose radical quotient is a primitive complex Clifford factor, together with an intrinsic anti-Hermitian generator γ^0 satisfying $(\gamma^0)^2 = -\mathbf{1}$;
- (iv) the unique pressure-selected graded modular clock $\Delta_\beta = e^{\beta N}$;
- (v) a positive spatial coframe with orthogonal comparison transport;
- (vi) exact stationary first-moment plaquette closure in the selected exchange-invariant phase, with no constraint on the exchange-even rotational defect.

Thus M lies in a signed modular Clifford–Lorentzian universality class. The finite criterion set is nonempty by [Theorem E.6](#).

Proof. Conditional predictive unifilarity, canonical successor refinement, and future tomography give branchwise reference renewal by [Theorems 3.16](#) and [3.22](#); hence scalar path weights are multiplicative by [Theorems 3.17](#) and [3.28](#). Branchwise Petz reversal gives reciprocal scalar fluxes by [Theorems 3.18](#) and [3.19](#); the reverse-renewal clause of [Assumption 4.1](#) makes the reindexed Petz branches ray homogeneous. Then [Theorems 3.36](#) and [3.37](#) gives (i). The canonical record quotient, uniqueness of parity, determinant line, signed cover, amplitude lifts, and independence from affinity are [Theorems 3.45](#) to [3.47](#), [3.50](#), [3.55](#), [3.57](#), [3.59](#) and [3.60](#). Projective lifting, temporal exchange, radical reduction, and the timelike generator are [Theorems 4.2](#) to [4.5](#). Supercriticality gives the pressure root by [Theorem 4.11](#); the stationary law and its entropy–affinity–depth identity are [Theorems 4.12](#) and [4.15](#), and the graded clock and its rigidity are [Theorems 4.18](#) and [4.19](#). Positive second moments and covariance give (v) by [Theorem 4.23](#). Finally (vi) is [Theorems 4.25](#) and [4.27](#). The deck-odd test retains the selected temporal sign by [Theorem 4.8](#). $\square$

The second theorem is the finite culmination of this paper. It assembles the independent ingredients derived throughout Sections [efsec:predictive–](#)

[efsec:phase](#) into a signed modular Clifford–Lorentzian phase, while still remaining entirely finite. A smooth curved spacetime sector requires one final, explicitly separate step: selection of a scaling net satisfying the analytic hypotheses of the companion theorem.

Theorem 4.35 (III. Realized curved sector, conditional on the companion theorem). *Let M satisfy [Theorem 4.34](#). Choose one recurrent component, one deck-oriented extremal state, an exchange-invariant or zero-source diamond state, and a continuum scaling net satisfying the analytic hypotheses of the companion Lorentzian spectral-geometry theorem [\[1\]](#). Then that theorem applies to the finite datum reconstructed here and yields the corresponding oriented torsion-free Lorentzian Dirac continuum sector. When the specified model admits additional components or stationary phases, alternative selections may instead realize the opposite temporal orientation, a symmetric unoriented mixture, or a torsional sector; the existence of such alternatives is model dependent.*

Proof. [Theorem 4.34](#) supplies exactly the finite inputs used by the companion theorem: UCP predictive memory, signed/Krein cover, primitive projective Clifford sector, graded affine modular clock, positive second-moment coframe, orthogonal comparison, and first-moment closure. The

selected scaling net supplies the remaining analytic domain assumptions. Opposite temporal sectors, when both exist, are related by deck reversal. The countermodel of [Theorem 4.30](#) separately proves that the structural framework contains microscopic laws for which different stationary selections have zero or nonzero translational defect; it is not used to assert this multiplicity for every model M . $\square$

Corollary 4.36 (Exact logical strength). *The paper does not prove that the operational axioms force every model to become Lorentzian spacetime. It proves:*

- (a) *the axioms reconstruct a broad finite complex UCP predictive framework;*
- (b) *within that framework, explicit checkable criteria define a nonempty signed modular Clifford phase;*
- (c) *physical component, state, and scaling selections inside that phase allow the companion curved-sector theorem to be applied.*

The unsigned, subcritical, nonreversible, unoriented, spatially degenerate, torsional, and nonsmooth counterphases remain admissible models of the same structural theory.

The corollary is the practical reading guide for the paper’s logical strength. The operational axioms reconstruct the finite framework; phase criteria identify a nonempty Lorentzian-type universality class; and additional physical and analytic selections realize a particular continuum sector. Counterphases are not failures of the construction but part of its stated scope.

Remark 4.37 (No circularity). No Lorentzian metric, causal cone, signed cover, Clifford algebra, or physical time coordinate is used to construct the rooted quotient, the complex algebra, the UCP maps, the KMS dual, the record determinant, or the pressure root. The continuum theorem is invoked only after all finite signed modular data have been reconstructed. Conversely, the phase and state selections are not relabelled as universal consequences.

5 Interpretation and outlook

5.1 From Predictive Relations to Temporal and Geometric Structure

Prediction as an operational primitive The reconstruction begins below the level at which spacetime, quantum states, or even a preferred notion of history have been specified. Its primitive data are composable operations, closed-experiment weights, and a family of admissible future continuations and tests. Two pasts are identified when no such future experiment distinguishes them. The resulting quotient therefore retains neither a microscopic trajectory nor a complete record of what has happened, but exactly the information that can still affect what happens next. In this precise sense, the basic object of the theory is an effective difference under continuation.

This viewpoint is related to the predictive equivalence underlying causal-state constructions in computational mechanics [\[17, 18\]](#), but the present setting is both operational and compositional. The objects being compared are generated processes rather than sample paths of a fixed stochastic model, and the quotient must remain compatible with admissible sequential and parallel contexts. Moreover, outcome unifilarity is not imposed on a hidden-state presentation. It appears only after successful histories have been quotiented by their conditional future profiles. In a future-tomographic operator realization, that quotient-level determinism forces each resolved branch to return the reference state of its unique predictive successor. The scalar renewal law is therefore

a consequence of predictive reduction, rather than a modelling assumption about microscopic dynamics.

The universality of the quotient gives a limited but mathematically definite sense in which the retained predictive content is representation independent. Every future-sufficient realization factors through it, whereas the choice of a particular microscopic representative does not follow from the quotient alone. Thus the construction does not assign ontological priority to one hidden history. It assigns priority to those distinctions between histories that remain operationally effective.

The quantum operator structure enters only at a later stage. Under the finite sharp-purification and interference hypotheses used in the reconstruction programme [3, 13, 14, 15], the reduced predictive state spaces acquire Euclidean Jordan structure. The two-path phase law then selects complex matrix type and an associative orientation, while compatibility with arbitrary ancillary extensions upgrades positive deterministic dynamics to UCP dynamics. In this hierarchy, neither Hilbert space nor complete positivity is primitive. They are the finite operator form taken by a predictive theory satisfying specific purification, interference, and compositional stability conditions. This is also why the complex selector and parallel completeness must be kept logically distinct; tensor-product results such as [16] concern a different interface from the single-system phase law.

Retrodiction, orientation, and temporal structure Retrodiction adds a second structure without undoing the first. Relative to a faithful stationary reference state, the KMS–Petz map is the canonical adjoint of forward prediction [30, 31, 32]. It reverses compositional order, but it is not generally an inverse channel and does not by itself define a microscopic reversal of time. Forward and retrodictive branch weights may nevertheless fail to balance in the same path orientation. Their logarithmic ratio defines the real affinity class familiar from network thermodynamics [39, 40, 41].

The signed Lorentzian structure has a different origin. It records whether a frame of future-distinct records returns with reversed orientation after transport around a recurrent cycle. The corresponding determinant line is a real local system, with its first Stiefel–Whitney class in $H^1(G; \mathbb{Z}/2)$ [42]. This parity is invisible to positive scalar weights alone. Conversely, a nonzero affinity need not alter the determinant class. The strict independence of these two invariants is therefore more than a difference of notation: nonequilibrium circulation and record orientation belong to different coefficient systems and classify different operational effects.

The two sheets of the resulting cover should not be read as two universes, nor as matter and antimatter sectors. They are the minimal real carrier of an orientation distinction forgotten by the unsigned predictive quotient. The positive theory records what the two orientations have in common; the signed extension records how they differ. When a reversible predictive cycle exchanges the sheets, its implementer anticommutes with the fundamental symmetry. Their product then has negative square and supplies the intrinsic timelike Clifford generator. Temporal sign is therefore not introduced as an additional positive coordinate. It arises as an algebraic opposition between two ways of carrying the same unsigned predictive content.

A physical arrow of time requires a further state selection. A finite primitive transfer that is symmetric under sheet exchange has a unique sheet-symmetric stationary state, so it cannot select a nonzero deck-odd order parameter. An oriented phase must instead involve explicit symmetry breaking, stationary-state degeneracy, boundary or superselection data, or a singular large-system or large-depth limit. The selected sign is operationally retained only if a deck-odd future test remains admissible. The reconstruction therefore locates the logical ingredients from

which an arrow could arise, but it does not identify the thermodynamic or cosmological arrow with the existence of the signed cover itself.

The pressure root supplies another temporal ingredient. Its role is to turn renewal depth into an intrinsic dimensionless modular scale. The exact identity

$$\beta\mu_\ell = h_{\text{rate}} + \mathbb{E}[\log c_e] + \frac{1}{2}\sigma$$

separates three contributions: predictive entropy, symmetric renewal capacity, and nonequilibrium circulation. The corresponding affine action is realized on the graded path cover, where unbounded depth is separated from the finite recurrent base. This realizes, in a finite renewal setting, the relation between modular flow and intrinsic time advocated in the thermal-time programme [46]. It does not, however, fix an absolute second or energy unit. Dimensionful calibration remains external to the pressure theorem.

Spatial geometry and the status of torsion freedom The spatial metric is likewise reconstructed from operational statistics rather than imposed at the primitive level. A positive second moment of stable direction records defines a quadratic frame, and covariance of this moment under transport forces neighbouring normalized frames to be compared orthogonally. Metric compatibility is therefore encoded by preservation of quadratic predictive data.

The microscopic diamond law then separates translation from rotation. The translational defect is odd under exchange of the two path orders, whereas the rotational defect is even. An exchange-invariant stationary phase cancels the former but leaves the latter unconstrained. In continuum language, this is the required distinction between torsion and curvature [47]: the Levi-Civita sector is selected by vanishing translational closure defect, not by suppressing rotational holonomy.

The Ising countermodel shows why this selection cannot be hidden inside ordinary stationarity or one-step detailed balance. One and the same local reversible law can admit a symmetric torsion-free stationary state and extremal exchange-broken states with nonzero translational defect. Torsion freedom is therefore a genuine phase or state-selection condition. This strengthens the logical interpretation of the curved reconstruction: the theory does not define torsional sectors away, but identifies the additional microscopic symmetry that selects the torsion-free branch.

Scope of the reconstruction The result is a conditional sufficiency theorem, not a uniqueness theorem for fundamental physics. It shows that a stated operational package together with a stated recurrent phase supplies the finite signed, modular, Clifford, and metric data required by the companion Lorentzian construction [1]. It does not show that every operational theory enters that phase, that the phase is dynamically inevitable, or that its continuum limit is unique.

Several limitations are structural rather than merely technical. An averaged channel does not select its resolved instrument without additional interaction or monitoring data. A nontrivial signed cover does not select one of its orientations. A positive pressure root fixes a dimensionless exponent but not a laboratory unit. A nondegenerate record moment does not by itself derive three spatial dimensions, and finite signed modular data do not guarantee a smooth Dirac continuum. The unsigned, subcritical, nonreversible, spatially degenerate, unoriented, and torsional regimes remain consistent alternatives. Their presence is important: it makes each phase condition falsifiable rather than definitional.

The conceptual claim should therefore be read narrowly. What lies beneath the geometric starting object of the companion paper need not be another geometry. It can be a compositional

system of distinctions whose future consequences can be compressed, compared with their retrodictive adjoints, transported with or without orientation reversal, and organized by recurrent statistical laws. In the selected phase studied here, these operations produce the finite algebraic precursors of Lorentzian propagation.

5.2 Outlook and testable interfaces

The intermediate constructions have uses that do not depend on taking a continuum spacetime limit. They suggest several concrete directions in which the framework can be tested, refined, or applied.

Predictive reconstruction from experimental data. The conditional quotient and the tomographic frame estimate turn predictive model reduction into a quantitative inference problem. Experimental branch data can be grouped by approximate equality of conditional future statistics, and the resulting successor states can be tested against the accumulated trace-norm error bound. This extends causal-state and quantum-memory reconstruction [18, 19, 20] from model fitting toward certification of a minimal predictive instrument. A natural next step is a finite-sample theory that combines statistical confidence regions with the deterministic perturbation bounds proved here.

Independent control of affinity and record orientation. The cohomological independence of $[A] \in H^1(G; \mathbb{R})$ and $[\chi] \in H^1(G; \mathbb{Z}/2)$ yields a direct experimental prediction: continuous changes in nonequilibrium driving can alter entropy production without changing the record-orientation sector, while a determinant crossing can flip the orientation class without requiring a singularity in the scalar affinity. Engineered Markov, photonic, or measurement-feedback networks provide natural settings in which edge fluxes and record-transport determinants could be measured simultaneously. Such experiments would test the separation between thermodynamic and signed information at the finite-network level.

Pointer and record engineering. The stable-pointer construction is canonical relative to a specified interaction family. Read in reverse, it provides a design principle: choose system–environment couplings whose interaction algebra has a commutant with the desired stable centre. This connects the reconstruction to einselection, decoherence-free subsystems, and redundant record formation [21, 22, 23, 24, 25, 26, 27, 28, 29]. Determinant regularity then quantifies the noise margin within which the orientation carried by those records survives.

Thermal quantum dynamics. The KMS–Petz branch reversal and cancellation current are directly relevant to KMS-detailed-balanced quantum Markov dynamics. In particular, the entropy–affinity–depth identity gives an exact stationarity diagnostic for resolved implementations of thermal samplers [36, 56]. Deviations from the identity would distinguish imperfect sampling, incorrect branch resolution, and genuine nonstationarity. It would be useful to formulate corresponding finite-time and finite-sample inequalities.

The geometric continuation. For the spacetime programme, the next mathematical tasks are more sharply localized by the present reconstruction. One must classify microscopic models that enter the required nontrivial signed and projective phase, derive rather than postulate the relevant spatial rank when possible, control the scaling of the exchange-closure error, and prove

the analytic compactness and strong-resolvent hypotheses needed by the companion continuum theorem. A separate metrological bridge is also required to convert the dimensionless modular parameter into physical units. These are downstream questions; they no longer concern the existence or canonical meaning of the finite renewal object itself.

Taken together, the results suggest a research programme rather than a closed ontology. Prediction supplies the canonical reduction, retrodiction supplies a stationary comparison, stable records supply orientation, recurrence supplies projective and modular structure, and coarse-grained record statistics supply metric and connection data. The principal open question is how broadly these structures arise in microscopic models without being imposed. The present paper makes that question mathematically accessible by isolating the exact interfaces at which each additional structure enters and the counterphases in which it fails.

A Finite-dimensional operator-system facts

This appendix records the elementary finite-dimensional facts used in [Section 2](#).

Lemma A.1 (Choi criterion in the present setting). *Let $\Phi : M_n(\mathbb{C}) \rightarrow M_m(\mathbb{C})$ be linear. Define its Choi matrix*

$$C_\Phi := \sum_{i,j=1}^n E_{ij} \otimes \Phi(E_{ij}) \in M_n(\mathbb{C}) \otimes M_m(\mathbb{C}).$$

Then Φ is completely positive if and only if $C_\Phi \geq 0$.

Proof. If Φ is completely positive, then

$$C_\Phi = (\text{id}_n \otimes \Phi)(|\Omega\rangle\langle\Omega|) \geq 0, \quad |\Omega\rangle = \sum_i e_i \otimes e_i.$$

Conversely suppose $C_\Phi \geq 0$ and write $C_\Phi = \sum_a |v_a\rangle\langle v_a|$. Reshape each $v_a \in \mathbb{C}^n \otimes \mathbb{C}^m$ into a matrix $V_a : \mathbb{C}^n \rightarrow \mathbb{C}^m$. Comparing matrix entries gives

$$\Phi(x) = \sum_a V_a x V_a^*.$$

This Kraus form is completely positive. □

Lemma A.2 (Unitality and stationarity). *For a linear map $\Phi : \mathcal{A} \rightarrow \mathcal{A}$ on a finite-dimensional C^* -algebra, the trace adjoint satisfies*

$$\Phi(\mathbf{1}) = \mathbf{1} \iff \text{Tr}(\Phi_*(x)) = \text{Tr}(x)$$

for all x . More generally, $\Phi_(\rho) = \rho$ if and only if the state $\varphi_\rho(a) = \text{Tr}(\rho a)$ is invariant under Φ .*

Proof. By definition,

$$\text{Tr}(\Phi_*(x)) = \text{Tr}(\Phi_*(x)\mathbf{1}) = \text{Tr}(x\Phi(\mathbf{1})).$$

This equals $\text{Tr}(x)$ for all x exactly when $\Phi(\mathbf{1}) = \mathbf{1}$. The second statement is

$$\varphi_\rho(\Phi(a)) = \text{Tr}(\rho\Phi(a)) = \text{Tr}(\Phi_*(\rho)a),$$

which equals $\varphi_\rho(a)$ for all a exactly when $\Phi_*(\rho) = \rho$. □

B Determinant walls and parity

The orientation class is locally constant only while the record transport is invertible. At rank-loss walls its continuation requires path data.

Proposition B.1 (Transverse determinant crossing). *Let $L(t)$ be a C^1 path of real $r \times r$ matrices with $\det L(t_0) = 0$ and*

$$\left. \frac{d}{dt} \det L(t) \right|_{t=t_0} \neq 0.$$

Then $\operatorname{sgn} \det L(t)$ changes sign across t_0 . The parity of a path with finitely many transverse crossings is

$$(-1)^{N_{\text{cross}}},$$

independent of sufficiently small transverse perturbations that fix the endpoints.

Proof. A nonzero derivative makes $\det L(t)$ a scalar function with a simple zero, so it has opposite signs on the two sides of t_0 . Under a sufficiently small endpoint-fixed transverse perturbation, crossings can be created or annihilated only in pairs; therefore their number modulo two is invariant. This is the finite-dimensional parity principle of [57, 58]. $\square$

Proposition B.2 (First nonzero determinant jet). *Let $L(t)$ be C^k near t_0 and suppose*

$$\det L(t_0) = \cdots = \frac{d^{m-1}}{dt^{m-1}} \det L(t_0) = 0, \quad \frac{d^m}{dt^m} \det L(t_0) \neq 0.$$

Then the determinant sign changes across t_0 if and only if m is odd.

Proof. Taylor's theorem gives

$$\det L(t) = c(t - t_0)^m + o(|t - t_0|^m), \quad c \neq 0.$$

For sufficiently small $|t - t_0|$, the leading term controls the sign. It has opposite signs on the two sides exactly when m is odd. $\square$

Definition B.3 (Endpoint path parity). Let V, W be real vector spaces of the same finite dimension and let

$$Q : [a, b] \longrightarrow \operatorname{Hom}(V, W)$$

be continuous with Q_a, Q_b invertible. Choose arbitrary orientations of V and W and define

$$\operatorname{par}(Q) := \operatorname{sgn} \det Q_b \operatorname{sgn} \det Q_a \in \{\pm 1\}.$$

The product is independent of the two orientation choices.

Theorem B.4 (Perturbative parity classification). *The parity of [Theorem B.3](#) has the following properties.*

- (i) *it is invariant under homotopies through paths with invertible endpoints;*
- (ii) *it is multiplicative under path concatenation;*
- (iii) *every C^1 path admits an arbitrarily small endpoint-fixed perturbation $\tilde{Q}$ with finitely many regular crossings, and for every such perturbation*

$$\boxed{\operatorname{par}(Q) = (-1)^{\sum_{t_*} \dim \ker \tilde{Q}_{t_*}};}$$

- (iv) *for an isolated infinitely flat crossing, parity gives a canonical relation between the signs of the two regular sides even though no finite determinant jet does;*
- (v) *for a nonisolated singular interval, parity canonically relates the two regular endpoint phases but does not canonically decompose the interval into individual signed walls.*

Thus endpoint-fixed perturbative parity is the maximal path-level mod-two datum available without additional local regularity.

Proof. Changing the orientation of either V or W reverses both endpoint determinant signs, leaving their product unchanged. During an endpoint-invertible homotopy, neither endpoint determinant can cross zero, so the product is constant. Under concatenation, the determinant sign at the common endpoint appears twice and cancels.

For the crossing formula, let t_* be a regular crossing of a C^1 path. Write

$$K = \ker Q_{t_*}, \quad C = \operatorname{coker} Q_{t_*}, \quad m = \dim K = \dim C.$$

The derivative induces the crossing map

$$\Gamma_{t_*} : K \longrightarrow C, \quad \Gamma_{t_*} = \pi_C \dot{Q}_{t_*}|_K,$$

and regularity means that Γ_{t_*} is invertible. Choose splittings $V = K \oplus V_1$ and $W = C \oplus W_1$ such that $Q_{t_*}|_{V_1} : V_1 \rightarrow W_1$ is an isomorphism. The Schur-complement expansion then gives, in oriented coordinates,

$$\det Q_t = c(t - t_*)^m + o(|t - t_*|^m), \quad c \neq 0.$$

Hence passage through the crossing changes the determinant sign by $(-1)^m$.

The singular matrices form a stratified algebraic hypersurface whose top stratum consists of corank-one matrices. Finite-dimensional relative transversality gives an arbitrarily small endpoint-fixed perturbation meeting the regular strata transversely and in finitely many points; higher-corank coincidences may be separated by a further arbitrarily small perturbation. Multiplying the local factors $(-1)^{\dim \ker}$ gives the ratio of endpoint signs and hence the displayed formula. Its independence of the chosen perturbation follows from the already proved homotopy invariance. This is the finite-dimensional parity principle; see [57, 58].

For flat or nonisolated singular sets, the endpoint expression remains defined whenever the endpoints are invertible. It supplies only their total relative parity; without a distinguished local crossing or nonzero jet there is no canonical finer decomposition. $\square$

Remark B.5 (Sharp singular boundary). At a singular map itself, no nonzero determinant transport exists. A pointwise sign can therefore be continued only after adding local jet data, a regular crossing structure, a path parity, or an independent orientation. Rank loss is a genuine boundary of the nondegenerate signed phase rather than a removable coordinate singularity.

C Approximate exchange covariance

The exact closure theorem has a quantitative perturbative version.

Theorem C.1 (Mixing-gap stability of exchange closure). *Represent probability laws as row vectors. Let K be a primitive stochastic matrix on a finite diamond space, let P be the permutation matrix of the exchange involution, and let $\pi K = \pi$. Assume*

$$\|PK - KP\|_{1 \rightarrow 1} \leq \varepsilon$$

and that every zero-mass row vector μ satisfies

$$\|\mu K\|_1 \leq (1 - \lambda)\|\mu\|_1$$

for some $\lambda > 0$. Then

$$\|\pi P - \pi\|_1 \leq \frac{\varepsilon}{\lambda}.$$

For every bounded exchange-odd column vector d , meaning $Pd = -d$,

$$|\pi d| \leq \frac{\varepsilon}{2\lambda}\|d\|_\infty.$$

Proof. Set $\mu = \pi P - \pi$, which has total mass zero. Using $\pi K = \pi$,

$$\mu K - \mu = \pi PK - \pi K - \pi P + \pi = \pi(PK - KP).$$

Hence $\|\mu K - \mu\|_1 \leq \varepsilon$. The reverse triangle inequality and the mixing-gap assumption give

$$\varepsilon \geq \|\mu - \mu K\|_1 \geq \|\mu\|_1 - \|\mu K\|_1 \geq \lambda\|\mu\|_1,$$

which proves the first estimate. Since $Pd = -d$,

$$\mu d = \pi Pd - \pi d = -2\pi d.$$

Therefore

$$2|\pi d| = |\mu d| \leq \|\mu\|_1\|d\|_\infty \leq \frac{\varepsilon}{\lambda}\|d\|_\infty.$$

□

D A worked finite signed renewal model

We close with a minimal example displaying the independence of the positive, retrodictive, signed, and projective structures.

Example D.1 (Three-cycle with a nontrivial signed cover). Let G be the directed three-cycle with both orientations on every edge. Take $\mathcal{A} = M_2(\mathbb{C})$ and the faithful state $\rho = \mathbf{1}/2$. Let the three forward renewal channels be unitary conjugations

$$\Phi_1 = \text{Ad}(\sigma_x), \quad \Phi_2 = \text{Ad}(\sigma_z), \quad \Phi_3 = \text{id},$$

with the corresponding reverse channels equal to themselves. Then every channel is its KMS–Petz dual because the Hilbert–Schmidt and symmetric KMS pairings coincide at the tracial state.

Assign positive branch weights with forward/backward ratios $e^{a_1}, e^{a_2}, e^{a_3}$. The cycle affinity is $a_1 + a_2 + a_3$ and may be chosen independently of the operator channels. Let the stable record fibre be $R_x = \mathbb{R}^2$ at every vertex, with two orientation preserving transports and one reflection. The product determinant around the cycle is negative, so $[\chi] \neq 0$ and the orientation cover is connected, regardless of whether the cycle affinity vanishes.

The automorphisms $\text{Ad}(\sigma_x)$ and $\text{Ad}(\sigma_z)$ commute even though their implementers anticommute. Their projective commutator is $-\mathbf{1}$, so the nondegenerate spatial revision algebra is $M_2(\mathbb{C})$. An odd sign-holonomy cycle then supplies the canonical temporal exchange relation with the sheet operator J . A full $3+1$ primitive block requires one additional independent spatial revision label, as in the rank-four construction of the companion paper.

Verification. At $\rho = \mathbf{1}/2$, Γ_ρ is a scalar multiple of the identity map and the trace adjoint of a unitary conjugation is its inverse. Pauli conjugations are involutive, so $\Phi_j^\sharp = \Phi_j$. The affinity statement is [Theorem 3.36](#). The reflection contributes one negative determinant and therefore one nontrivial cycle sign. Finally $\sigma_x \sigma_z = -\sigma_z \sigma_x$, while their central sign disappears under conjugation; this is exactly the projective mechanism of [Theorem 4.2](#). $\square$

E A common-origin local UCP–Clifford phase

The preceding bridge is conditional on explicit phase criteria. This appendix constructs one finite-range microscopic instrument satisfying them jointly. The construction is model specific; its purpose is to prove nonemptiness and compatibility, not universal uniqueness.

Let Λ be a finite periodic box in $\mathbb{Z}^2$ and $\Omega_\Lambda = \{\pm 1\}^\Lambda$. For $i \in \Lambda$ and $s \in \{\pm 1\}$, write $\eta^{i,s}$ for the configuration obtained by replacing η_i by s . Fix $\theta \geq 0$ and define the heat-bath probability

$$q_i(s \mid \eta) = \frac{\exp\left(\theta s \sum_{j \sim i} \eta_j\right)}{2 \cosh\left(\theta \sum_{j \sim i} \eta_j\right)}.$$

Choose site weights $\nu_i > 0$ with $\sum_i \nu_i = 1$. Let the spatial record rank be $d = 3$, choose $\lambda_{i,a} > 0$, $b_a \in [-1, 1]$, and $\varepsilon \in \mathbb{R}$ with

$$\delta := |\varepsilon| \max_a |b_a| < 1.$$

Set

$$r_{i,a}(u) := \frac{\lambda_{i,a}(1 + \varepsilon b_a u)}{\sum_{c=1}^3 \lambda_{i,c}(1 + \varepsilon b_c u)}, \quad u \in \{\pm 1\}.$$

The argument $u = \eta_i s$ couples the direction record to whether the local orientation is preserved or flipped.

Let $S = \mathbb{C}^4$ carry Hermitian Clifford generators $\Gamma_0, \Gamma_1, \Gamma_2, \Gamma_3$ satisfying

$$\Gamma_\mu \Gamma_\nu + \Gamma_\nu \Gamma_\mu = 2\delta_{\mu\nu} \mathbf{1}.$$

Put $J = \Gamma_0$, $R_a = \Gamma_a$, and let $\tau_S = \text{Tr}/4$. Choose $\kappa_b > 0$ for $b \in \{0, 1, 2\}$ with $\sum_b \kappa_b = 1$, and define UCP maps

$$\Psi_{a,0} = \text{Ad}(R_a), \quad \Psi_{a,1} = \text{Ad}(JR_a), \quad \Psi_{a,2}(X) = \tau_S(X) \mathbf{1}.$$

On $\hat{\mathcal{A}}_\Lambda = C(\Omega_\Lambda) \otimes M_4(\mathbb{C})$ define the resolved Heisenberg branch

$$(\hat{K}_{i,s,a,b} F)(\eta) = \nu_i q_i(s \mid \eta) r_{i,a}(\eta_i s) \kappa_b \Psi_{a,b}(F(\eta^{i,s})), \quad (1)$$

and set $\hat{K}_\Lambda = \sum_{i,s,a,b} \hat{K}_{i,s,a,b}$.

Proposition E.1 (UCP law, marginal, and deck covariance). *Every resolved branch in (1) is CP and $\hat{K}_\Lambda$ is UCP. On scalar observables $F = f \otimes \mathbf{1}$,*

$$\hat{K}_\Lambda(f \otimes \mathbf{1}) = (K_\Lambda f) \otimes \mathbf{1},$$

where K_Λ is the ordinary random-scan Ising heat-bath channel. Under deck reversal $(\Theta F)(\eta) = F(-\eta)$, the record transforms as $(s, a, b) \mapsto (-s, a, b)$ and

$$\Theta \hat{K}_{i,s,a,b} = \hat{K}_{i,-s,a,b} \Theta.$$

Proof. Each $\Psi_{a,b}$ is UCP. Pullback along $\eta \mapsto \eta^{i,s}$, multiplication by a nonnegative scalar coefficient, and composition with a CP map preserve complete positivity. Since $\sum_s q_i(s \mid \eta) = 1$, $\sum_a r_{i,a}(u) = 1$, and $\sum_b \kappa_b = 1$, the sum maps $\mathbf{1}$ to $\mathbf{1}$. Every internal map fixes scalar matrices, giving the classical marginal. Finally $q_i(-s \mid -\eta) = q_i(s \mid \eta)$ and $(-\eta_i)(-s) = \eta_i s$, so the direction weight is unchanged and deck covariance follows. $\square$

Let

$$\mu_{\Lambda,\theta}(\eta) = Z^{-1} \exp \left(\theta \sum_{\langle i,j \rangle} \eta_i \eta_j \right)$$

and define the faithful state

$$\hat{\mu}_{\Lambda,\theta}(F) = \sum_{\eta} \mu_{\Lambda,\theta}(\eta) \tau_S(F(\eta)).$$

Theorem E.2 (Resolved transition balance and primitivity). *For every configuration η , site i , redraw value s , direction a , and internal mode b , put $\xi = \eta^{i,s}$ and define the scalar transition weight*

$$w_{i,s,a,b}(\eta) := \nu_i q_i(s \mid \eta) r_{i,a}(\eta_i s) \kappa_b.$$

Then the reverse elementary transition $\xi \rightarrow \eta$ has fixed redraw value η_i and satisfies

$$\boxed{\mu_{\Lambda,\theta}(\eta) w_{i,s,a,b}(\eta) = \mu_{\Lambda,\theta}(\xi) w_{i,\eta_i,a,b}(\xi).}$$

Each internal map $\Psi_{a,b}$ is self-adjoint for the normalized Hilbert–Schmidt pairing. Consequently the complete channel $\hat{K}_{\Lambda}$ is self-adjoint for the symmetric KMS pairing of $\hat{\mu}_{\Lambda,\theta}$ and hence satisfies KMS detailed balance. If $\kappa_2 > 0$, its Schrödinger predual is primitive; therefore $\hat{\mu}_{\Lambda,\theta}$ is the unique faithful finite-volume stationary state and the dynamics mixes exponentially.

Proof. The heat-bath identity gives

$$\mu(\eta) q_i(s \mid \eta) = \mu(\xi) q_i(\eta_i \mid \xi).$$

Moreover $\xi_i \eta_i = s \eta_i = \eta_i s$, so the direction factor is identical on the two orientations of the same elementary transition. Multiplication by $\nu_i \kappa_b$ proves the boxed scalar identity.

For $b = 0$, R_a is a Hermitian involution. For $b = 1$, JR_a is unitary and $(JR_a)^{-1} = -(JR_a)$; the central sign disappears under adjoint action, so $\text{Ad}(JR_a)$ is an involution. Both inner automorphisms are therefore self-adjoint on $L^2(M_4(\mathbb{C}), \tau_S)$. The depolarizing map is the orthogonal projection onto $\mathbb{C}\mathbf{1}$ and is self-adjoint as well. Expanding the KMS inner product of F with $\hat{K}_{\Lambda}G$, changing variables $\eta \leftrightarrow \xi$ transition by transition, and applying the scalar and internal self-adjointness identities gives

$$\langle F, \hat{K}_{\Lambda}G \rangle_{\hat{\mu}} = \langle \hat{K}_{\Lambda}F, G \rangle_{\hat{\mu}}.$$

Thus the full channel is its KMS adjoint.

For primitivity, choose a classical block carrying nonzero positive mass. A single transition with $b = 2$ replaces its internal density by a strictly positive multiple of $\mathbf{1}$. All heat-bath, direction, and internal-mode probabilities are strictly positive. A finite sequence of single-site redraws reaches every target classical configuration, while unitary conjugations preserve strict positivity. Since the state space is finite, one common power sends every nonzero positive block state to a strictly positive density in every block. The finite-dimensional Perron theorem for primitive CP maps gives uniqueness and exponential mixing. $\square$

Proposition E.3 (Pressure, sign, and spatial moment). *Fix an activity $\mathfrak{a} > 1$ and define the depth-one pressure transfer*

$$\widehat{\mathcal{B}}_z = \mathfrak{a}e^{-z}\widehat{K}_\Lambda.$$

Then its unique pressure zero is

$$\beta = \log \mathfrak{a},$$

and its Doob-normalized channel is $\widehat{K}_\Lambda$. Let $\widehat{G}_\Lambda$ be the positive-support graph of the resolved finite-volume transitions and quotient it by the free global deck involution $\eta \mapsto -\eta$. Primitivity makes $\widehat{G}_\Lambda$ connected, so

$$\widehat{G}_\Lambda \longrightarrow G_\Lambda := \widehat{G}_\Lambda / \langle \Theta \rangle$$

is a connected, hence nontrivial, principal $\mathbb{Z}/2$ -cover. If the full resolved record (s, a, b) is immediately readable, its deck action is the product of 9 disjoint transpositions and therefore has determinant -1 on the record contrast module. The associated determinant character represents the nonzero classifying this cover.

Attach vectors $v_{i,a} \in \mathbb{R}^3$ and define

$$M_i(\eta) = \sum_{s,a} q_i(s \mid \eta) r_{i,a}(\eta_i s) v_{i,a} v_{i,a}^\top.$$

Assume

$$0 < \lambda_- \leq \lambda_{i,a} \leq \lambda_+, \quad \sum_{a=1}^3 v_{i,a} v_{i,a}^\top \succeq g_* I_3.$$

Then, with

$$r_* := \frac{\lambda_-(1-\delta)}{3\lambda_+(1+\delta)},$$

one has

$$M_i(\eta) \succeq r_* g_* I_3, \quad M_i(-\eta) = M_i(\eta).$$

Thus deck-related orientation phases have the same positive rank-three spatial second moment.

Proof. A unital finite-dimensional channel has spectral radius one, so $\rho(\widehat{\mathcal{B}}_z) = \mathfrak{a}e^{-z}$ and the root is $\log \mathfrak{a}$. Every heat-bath coefficient is positive, hence the support graph is connected; the global flip has no fixed configuration and acts freely. A trivial double cover of a connected graph has disconnected total space, so the connected quotient cover is nontrivial. Deck reversal fixes (a, b) and exchanges the two values of s ; there are $3 \cdot 3 = 9$ transpositions. The constant line has determinant $+1$, so the contrast module has the same determinant -1 , realizing the nontrivial cover character.

For every a, u ,

$$r_{i,a}(u) \geq \frac{\lambda_-(1-\delta)}{3\lambda_+(1+\delta)} = r_*.$$

Hence for $\xi \in \mathbb{R}^3$,

$$\xi^\top M_i(\eta) \xi \geq r_* \sum_a (\xi \cdot v_{i,a})^2 \geq r_* g_* \|\xi\|^2.$$

Deck reversal sends (η, s) to $(-\eta, -s)$ while preserving both $q_i(s \mid \eta)$ and $r_{i,a}(\eta_i s)$, so relabelling the sum proves deck evenness. $\square$

Theorem E.4 (Clifford factor generated by coherent branches). *Conditioning on $b = 0, 1$ leaves the predictive units*

$$\alpha_a = \text{Ad}(\Gamma_a), \quad \beta_a = \text{Ad}(\Gamma_0 \Gamma_a).$$

They generate $\text{Ad}(\Gamma_0)$ because $\beta_a \alpha_a = \text{Ad}(\Gamma_0)$. Let $H = \mathbb{F}_2^4$ with basis e_0, e_1, e_2, e_3 and choose implementing monomials $R_x = \Gamma_0^{x_0} \Gamma_1^{x_1} \Gamma_2^{x_2} \Gamma_3^{x_3}$. Their commutator form has matrix

$$\Omega_{\mu\nu} = \begin{cases} 0, & \mu = \nu, \\ 1, & \mu \neq \nu, \end{cases}$$

over $\mathbb{F}_2$. The form is nondegenerate and

$$\mathbb{C}_\sigma[H] \cong \text{Cl}_4(\mathbb{C}) \cong M_4(\mathbb{C}).$$

Proof. The coherent branches contain $\text{Ad}(\Gamma_a)$ and $\text{Ad}(\Gamma_0 \Gamma_a)$, so their products contain every generator. Reordering Clifford monomials produces only a scalar sign, defining a projective co-cycle; distinct generators anticommute and equal generators commute, giving Ω . If $\Omega x = 0$ and $S = \sum_j x_j$, the i th equation is $S + x_i = 0$, so every $x_i = S$. Summing gives $S = 4S = 0$ over $\mathbb{F}_2$, hence $x = 0$. Thus the alternating form is nondegenerate. The 16 Clifford monomials are linearly independent and span the 16-dimensional algebra $M_4(\mathbb{C})$, proving the isomorphisms. $\square$

Theorem E.5 (Unified common-origin phase). *Assume*

$$\theta \geq \frac{1}{2} \log 12, \quad \mathfrak{a} > 1, \quad \kappa_b > 0,$$

and the uniform spatial nondegeneracy hypotheses above. Then the single resolved branch family (1) simultaneously has:

- (i) *primitive finite-volume UCP dynamics and a unique faithful stationary state;*
- (ii) *exact resolved transition balance and full-channel KMS detailed balance;*
- (iii) *in the thermodynamic limit, two extremal low-temperature deck-related orientation phases, separated by the deck-odd local spin test;*
- (iv) *pressure root $\beta = \log \mathfrak{a}$;*
- (v) *a nontrivial connected signed cover and determinant character;*
- (vi) *a uniformly positive rank-three deck-even spatial second moment;*
- (vii) *a primitive projective revision factor $\text{Cl}_4(\mathbb{C}) \cong M_4(\mathbb{C})$ generated by the coherent resolved branches.*

Therefore the finite algebraic signed modular Clifford criteria—through the coframe stage—are jointly realizable by one local microscopic edge instrument.

Proof. Items (i)–(ii) are [Theorems E.1](#) and [E.2](#). Summing over direction and internal records gives the two-dimensional Ising heat-bath marginal, whose plus and minus boundary limits are distinct by the Peierls estimate in the proof of [Theorem 4.30](#); this gives (iii). Item (iv), the sign, and the spatial moment are [Theorem E.3](#). The final item is [Theorem E.4](#). Every property belongs to the same branch family, rather than to commuting factors appended after the fact. $\square$

Corollary E.6 (Nonemptiness of the finite phase criteria). *Choose the common-origin model with coefficients independent of i , so its phase-averaged spatial moment is translation invariant and neighbouring comparison may be taken to be the identity. Form its local product with the two-state diamond kernel*

$$\Omega_{\diamond} = \{+, -\}, \quad K_{\diamond}(\zeta, \zeta') = \frac{1}{2}, \quad \vartheta_{\diamond}(\zeta) = -\zeta,$$

and any bounded exchange-odd defect $d_{\diamond}(-\zeta) = -d_{\diamond}(\zeta)$. The product kernel is primitive and exchange covariant; its stationary diamond law is uniform and has zero mean defect. The pre-existing determinant line of the common-origin record sector embeds as an invariant factor and retains its nonzero class. Consequently all finite criteria of [Assumption 4.1](#), including exchange-invariant first-moment closure, are simultaneously satisfied by an explicit local product model. A smooth or quantitative continuum conclusion still requires the separate scaling regularity stipulated in that assumption.

Proof. The common-origin factor supplies sign holonomy, the projective Clifford rank, pressure supercriticality, broken temporal orientation, and a positive rank-three moment by [Theorem E.5](#). Site-independent coefficients make the averaged moment constant, hence covariant under identity comparison. The displayed diamond kernel has strictly positive entries, so it is primitive, and it commutes with the exchange involution. Its unique stationary law is $(1/2, 1/2)$, which annihilates every exchange-odd defect. Tensor products of UCP channels are UCP, tensor products of primitive finite channels are primitive after a common power, and the determinant local system from one invariant record factor is unchanged by adjoining an independent factor. Thus the combined finite model satisfies all stated criteria. $\square$

Remark E.7 (Scope of the rank-three statement). The equality between record determinant parity and spatial-rank parity is a property of this common record alphabet, not a universal proof that every operational theory has three spatial dimensions. In this model class, nontrivial determinant parity requires odd rank and a nonzero spatial bivector requires rank at least two, so rank three is the first admissible value.

F Logical dependency ledger

Datum	Logical status	Origin in this paper
Rooted future quotient	canonical reduced memory	Future-profile quotient and minimality, Theorems 2.3 and 2.13 ; no natural raw-history selector is determined without extra structure, Theorem 2.15 .
Approximate predictive compression	quantitatively stable under rank and transversality control	A near-idempotent, near-sufficient map lies near an exact minimal projection; Theorems 2.20 and 2.21 .
Complete process quotient	canonical monoidal completion	Largest monoidal congruence below ordinary equivalence, Theorems 2.7 and 2.9 .
Parallel completeness	primitive no-holism axiom	Equality $\mathbb{Q}_{\parallel} \cong \mathbb{Q}_1$; Theorems 2.8 and 2.11 .

Datum	Logical status	Origin in this paper
Process tomography	interface condition	Identifies operational profiles with full accumulated channels; Theorem 2.39 . A single rooted orbit is insufficient by Theorem 2.40 .
Finite Euclidean Jordan structure	derived from sharp-reversible realisation	Theorems 2.24, 2.27 and 2.28 .
Pure sharpness boundary	irreducible effect-side input	Reversible dilation alone does not determine pure sharpness; Theorem 2.25 .
Finite complex C^* -algebra	reconstructed from Jordan structure and two-path phases	Theorem 2.33 .
Complete positivity	ancillary-stability theorem	Theorem 2.36 .
Faithful stationary state	selected by a primitive weighted transfer	Theorem 2.42 .
KMS–Petz reverse	canonical for stationary channels and reference-renewing resolved branches	Theorems 2.43, 3.3 and 3.18 .
Scalar path weights	derived on the conditional predictive orbit	Outcome-unifilar prediction, canonical successor refinement, and future tomography force reference renewal, Theorems 3.12, 3.16, 3.17 and 3.22 .
Instrument gauge boundary	Kraus-invariant within a fixed resolved branch	Theorems 3.23 and 3.30 .
Approximate scalar renewal	quantitative finite-history stability	Tomographic frame control and CP trace-norm contraction, Theorem 3.25 .
Petz path reversal	exact stationary adjointness identity	Reversed-path Petz comparison is identically balanced, forcing the same-orientation affinity convention; Theorems 3.33 and 3.34 .
Real affinity and conductance	resolved reciprocal branch data modulo gauge	Theorems 3.36 and 3.37 .
Stable raw instrument	conditional microscopic derivation	A specified interaction-complete monitoring phase selects the nondemolition commutant, its central pointer algebra, and a basis-independent resolved instrument; Theorems 3.40 to 3.42 .

Datum	Logical status	Origin in this paper
Canonical record algebra	conditional predictive quotient of resolved outcomes	Theorem 3.45.
Signed orientation	unique nontrivial one-dimensional record parity, robust under noisy regular transport	Theorems 3.47, 3.50, 3.52 and 3.55.
Singular orientation walls	jet and path-level parity classification	Finite-order crossings are controlled by determinant jets and general endpoint-invertible paths by perturbative parity; Theorems B.2 and B.4.
Affinity–parity separation	no canonical coefficient reduction	Theorems 3.57 and 3.59.
$\mathbb{Z}/4$ amplitude lifts	classified torsor, noncanonical	Theorem 3.60.
Projective multiplier	automatic on a finite factor	Theorem 4.2.
Primitive Clifford factor	canonical radical quotient	Theorem 4.4.
Selected temporal sign	phase plus deck-odd observability	Theorems 4.6 and 4.8.
Modular exponent β	pressure selected in a supercritical recurrent phase	Theorem 4.11.
Entropy–affinity–depth law	exact identity for the stationary pressure process	Theorems 4.14 and 4.15.
Affine modular clock	canonical on the graded path cover	Theorems 4.18 and 4.20.
Orthogonal frame comparison	derived from second-moment covariance	Theorem 4.23.
Translational closure	exchange-invariant phase or zero-source pressure stationarity	Theorems 4.25 and 4.27; torsional counterphases in Theorem 4.30.
Nonemptiness of the finite signed phase	explicit local product model	Theorem E.6.
Physical time unit	metrological calibration	Not fixed by the dimensionless pressure law.

Datum	Logical status	Origin in this paper
Lorentzian continuum	downstream theorem	Companion work [1].

G Necessity and no-go ledger

The following table records why the principal assumptions and phase conditions cannot be deleted wholesale. It distinguishes a missing proof from a genuine logical obstruction.

Input or condition	Role	Sharp boundary or counterexample
Arbitrary future continuations	Makes rooted equivalence a continuation congruence	Immediate tests alone can identify histories later separated by an allowed update; the future quantifier in Theorem 2.2 is load-bearing.
Choice of representative or normal form	Realizes the canonical quotient as a raw-history retract	The quotient admits no natural invariant selector when a profile-preserving automorphism exchanges representatives; Theorem 2.15 .
Parallel completeness	Eliminates process distinctions visible only with ancillas	The two-rebit construction gives processes ordinarily indistinguishable but ancilla-distinct; Theorem 2.10 .
Process tomography	Identifies rooted operational profiles with full accumulated channels	Two maps may agree on the entire available rooted orbit and differ on an unprepared direction; Theorem 2.40 .
Minimal spectral rank and transversality	Turn approximate compression into a nearby minimal exact retract	An exact sufficient projection may retain arbitrary redundant memory, while a correct-rank projection may erase the quotient; Theorem 2.22 .
Pure-process reversal	Supplies pure sharpness in the reconstruction package	Reversible dilation and essential uniqueness leave the admitted effect theory undetermined; the uniformly noisy effect model in Theorem 2.25 destroys pure sharpness.
Minimal continuous two-path interference	Selects complex rather than real, quaternionic, spin-factor, or exceptional Jordan type	Without (O4)–(O5), the other finite Euclidean Jordan factors remain compatible; Theorems 2.30 and 2.31 .
Complete ancillary stability	Upgrades positive deterministic maps to UCP maps	Single-system positivity alone permits positive non-CP transformations; complete positivity is precisely positivity at every matrix level, Theorem 2.36 .

Input or condition	Role	Sharp boundary or counterexample
Resolved instrument data	Makes branch weights and record outcomes operationally meaningful	An averaged channel has inequivalent Kraus/instrument decompositions. A canonical instrument follows only relative to a stable pointer phase, Theorems 3.42 and 3.43 .
Interaction-complete monitoring	Selects the maximal nondemolition pointer algebra	Monitoring a proper subalgebra may leave spurious stable records; pure Hamiltonian motion need not converge to a pointer expectation, Theorems 3.41 and 3.43 .
Conditional predictive successor resolution	Forces branchwise reference renewal and scalar path weights	A primitive stationary averaged channel can have branches that do not preserve the reference ray; Theorem 3.31 .
Determinant regularity	Defines signed transport of future-distinct record contrasts	The sign is locally constant only off the rank-loss wall; at $\det Q_e = 0$ a contrast is erased and no pointwise orientation transport exists, Theorem 3.52 .
Nonzero determinant class	Produces the principal signed cover	Positive scalar affinity does not produce a mod-two class, and no nonzero homomorphism $(\mathbb{R}, +) \rightarrow \mathbb{Z}/2$ exists; Theorems 3.57 and 3.59 .
Sufficient reversible projective rank	Produces a nontrivial Clifford factor	A purely dissipative predictive memory may possess no nontrivial units; radical reduction can also collapse the projective form, Theorem 4.4 .
Pressure supercriticality	Selects a positive modular exponent	If $r(0) \leq 1$, no positive pressure root exists; Theorem 4.11 .
Graded path cover	Carries a nonzero affine modular clock consistently	A common nonzero modular exponent on a finite scalar-return cycle is impossible; Theorem 4.20 .
Orientation-state selection and deck-odd tests	Produces and operationally retains one temporal sign	A finite primitive deck-symmetric transfer has a unique deck-even stationary state, and deck-even tests identify the two sheets; Theorems 4.6 and 4.8 .
Positive definite spatial second moment	Produces a nondegenerate spatial coframe	A degenerate direction law has no invertible quadratic frame and therefore no Lorentzian spatial block; Theorem 4.23 .
Exchange-invariant or zero-source diamond phase	Removes translational torsion while leaving rotation free	One-step detailed balance and stationarity admit torsional extremal phases; Theorems 4.30 and 4.31 .

Input or condition	Role	Sharp boundary or counterexample
Continuum scale separation and regularity	Permits the companion strong-resolvent Dirac limit	Finite signed modular data alone do not imply a smooth continuum; this remains an explicit domain condition in Theorem 4.35 .
Dimensionful calibration	Converts the dimensionless modular parameter to laboratory units	No dimensionless reconstruction can select an absolute second or joule; the pressure root fixes only an intrinsic dimensionless scale.

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